

Reinke Service Training

RST-02 Basic Pivot Electrical and Troubleshooting



Table of Contents

Safety and Theory of Operation	3
Safety Information.....	4
Pivot Theory of Operation	9
Electrical Components	10
Basic System Electrical Circuits.....	12
 Electrical Components, Electrical Properties, and Electrical Circuit Types	 15
Electrical Components on Reinke Systems.....	16
Electric Properties (Electricity Review)	24
Circuit Types (Series & Parallel Circuits).....	31
 Primary Electrical Circuits on a Reinke System	 33
Reinke Prints.....	34
Primary Electrical Circuits.....	35
 Common Options	 41
Isolation Relays	42
End Gun with Booster Pump.....	42
Low Pressure Shutdown	43
Auto Stop	43
Auto Reverse	43
Main Control Panel Options.....	46
Wrap Span	48
Single Phase.....	49
 Advanced / PAC III	 51
RPM Advanced.....	52
Main Control Panel Layout.....	52
PAC III Board	56
PAC III Normal Operation	58
Starting the System.....	63
PAC III Setup	63
PAC III Programming.....	70
PAC III Troubleshooting	77
 Troubleshooting	 81
Troubleshooting Theory.....	82
Service Bulletins and Alerts	83
Reference Material.....	84

This Page Intentionally Left Blank

Safety and Theory of Operation

CHAPTER 1

SAFETY INFORMATION

Reinke irrigation systems are designed with many electrical and mechanical safety features. However, each operator must read and understand this and all other accompanying owners manuals for the safe and efficient operation of your Reinke irrigation system. If this system is operated incorrectly, it can pose a safety threat to the operator and others, including serious injury or death. This system is powered by high voltage electricity. Any individual that will be engaging with the system in any way must read and fully understand the contents of this manual. This includes both the basic operation instructions (such as starting, stopping, adjusting water application amounts, and changing directions) as well as the safety obligations noted in this section.

In addition to the operational and safety information provided by this manual, you will also find safety signs and decals located throughout the system itself. For the safest experience possible, make sure you are able to locate and recognize every safety sign and decal located on the system. Throughout this manual, you will find safety information preceded by the words **Caution**, **Warning**, or **Danger**. These will be marked with a hazard symbol as shown below. Read each of these warnings closely, as they indicate potential hazards.



*The **Caution** sign indicates that a situation, or action, may result in minor damage to property or harm to individuals.*



*The **Warning** sign indicates that a situation, or action, may result in serious damage to property, injury, or death.*



*The **Danger** sign indicates that a situation, or action, poses an immediate threat that could result in severe injury or death.*



Make sure you read and understand the operator's manual BEFORE operating, or maintaining, the system. If anything seems unclear, contact your local Reinke dealer.

Ensure that anyone else who operates the system reads and understands the operator's manual.

Follow ALL safety guidelines on the system and in this manual. Please keep in mind that any unauthorized changes made to the system could cause damage to that system.



DO NOT attempt to perform any maintenance procedures until the Reinke main control panel disconnect switch and all pump and other disconnect switches are locked in the OFF position. Electrical component troubleshooting and replacement should be performed by a certified Reinke Service Technician to ensure built-in safety features remain intact. This also ensures the system remains compliant with the National Electric Code and Reinke Manufacturing specifications. Replace all protective guards and shields before restoring power to the system.

Danger

The movement of an electrically powered, gear-driven, irrigation system is relatively slow. Moving parts are exposed and may present a potential hazard. Therefore, keep all equipment, vehicles, people, livestock, etc. out of the system's path.

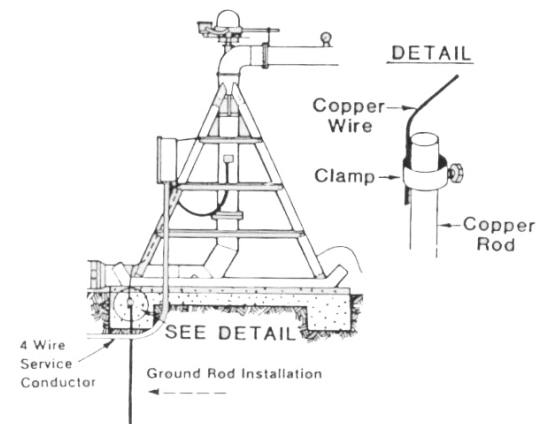
DO NOT allow anyone to ride, or climb, on the system unless they are qualified and required to do so for maintenance purposes.

The tower steps have been provided for access to the tower control boxes only. They are not intended for access to the span. For instance, should the sprinkler heads require service, use a ladder to reach them from the ground.

Danger

DO NOT start the system until all electrical equipment has been properly installed and grounded by a qualified electrician. Failing to do so could result in serious injury or death. Ensure that your contractor has grounded the system as required by the National Electrical Code and by all applicable local electrical codes.

When towing a system from field to field, avoid ditches, rough terrain, overhead power lines, etc. The ground wire **MUST** be re-attached to the ground rod, or concrete-encased electrode, and checked for electrical integrity each time the system is towed.



LOCKOUT / TAGOUT

Refer to U.S. Occupational Safety & Health Administration (OSHA) Regulations (Standards - 29 CFR) Typical minimal lockout procedures - 1910.147 App A, or applicable national, state or local regulations, for additional information.

On the following pages, you will find the typical minimal lockout procedures, as according to U.S. Occupational Safety & Health Administration (OSHA) Regulations. This procedure is used whenever maintenance or servicing is done on machines or equipment. It shall be used to ensure that the machine or equipment is stopped, isolated from all potentially hazardous energy sources, and locked out before anyone performs any servicing or maintenance where the unexpected start-up of the machine or equipment, or release of stored energy, could cause injury. When the energy isolating devices are not lockable, tagout may be used. Below is the sequence of Lockout:

1. Notify all affected employees that servicing or maintenance is required on a machine, or equipment, and that the machine, or equipment, must be shut down and locked out to perform the servicing or maintenance.
2. The authorized employee shall refer to the company procedure to identify the type and magnitude of the energy that the machine or equipment utilizes, understand the hazards of the energy, and know the methods to control the energy.
3. If the machine, or equipment, is operating, shut it down by the normal stopping procedure.

**Danger**

Be sure protective guards are installed on all belts and driveshafts of ancillary equipment such as combustion engines, electric motors, pumps, etc.

**Danger**

If you suspect a short circuit, or the system is not working correctly, DO NOT touch the system and keep others away from it. Call your Reinke service technician. Electrical component troubleshooting and replacement should be performed by a certified Reinke Service Technician to ensure built-in safety features remain intact. This also ensures the system remains compliant with the National Electric Code and Reinke Manufacturing Specifications.

**Danger**

Avoid any bodily contact with high pressure water streams from sprinklers and end guns.

**Caution**

Do not operate the system with water when temperatures are below 40°F (4.5°C). This can cause structural damage to the system. Damage from freeze-up is not covered under warranty.

**Warning**

Keep away from fields where the system is chemigating. Make sure the applied chemical and water does not blow or drift past the area of intended operation. A check valve must be installed between the pivot center and the pump to prevent the mixture of water and chemical from siphoning back into the irrigation water source. Comply with all local, state, and federal regulations.

**Caution**

Maintain adequate crop clearance. Allowing the systems trussing to drag in the crop, can cause structural damage to the system.

**Caution**

DO NOT oversize fuses. Fuses are sized for a specific circuit. It is very important to make sure you have the proper fuse size in place before initially starting the system and when replacing fuses.

**Caution**

In most states, it is unlawful to spray water on state and county roadways. This is a serious hazard and must not be allowed.

**Warning**

If your system is equipped with any auto-stop or auto-reverse mechanism, make sure they are working correctly and a tower barricade is properly installed. ***Reinke disclaims any and all liability (including any liability created pursuant to the Irrigation Systems Warranty) with regard to damage to the irrigation system, or to other property, or personal injury or death, caused by improper installation or maintenance of Reinke-supplied tower auto-reverse or auto-stop switches or tower barricades, or by use of customer-supplied barricades.***

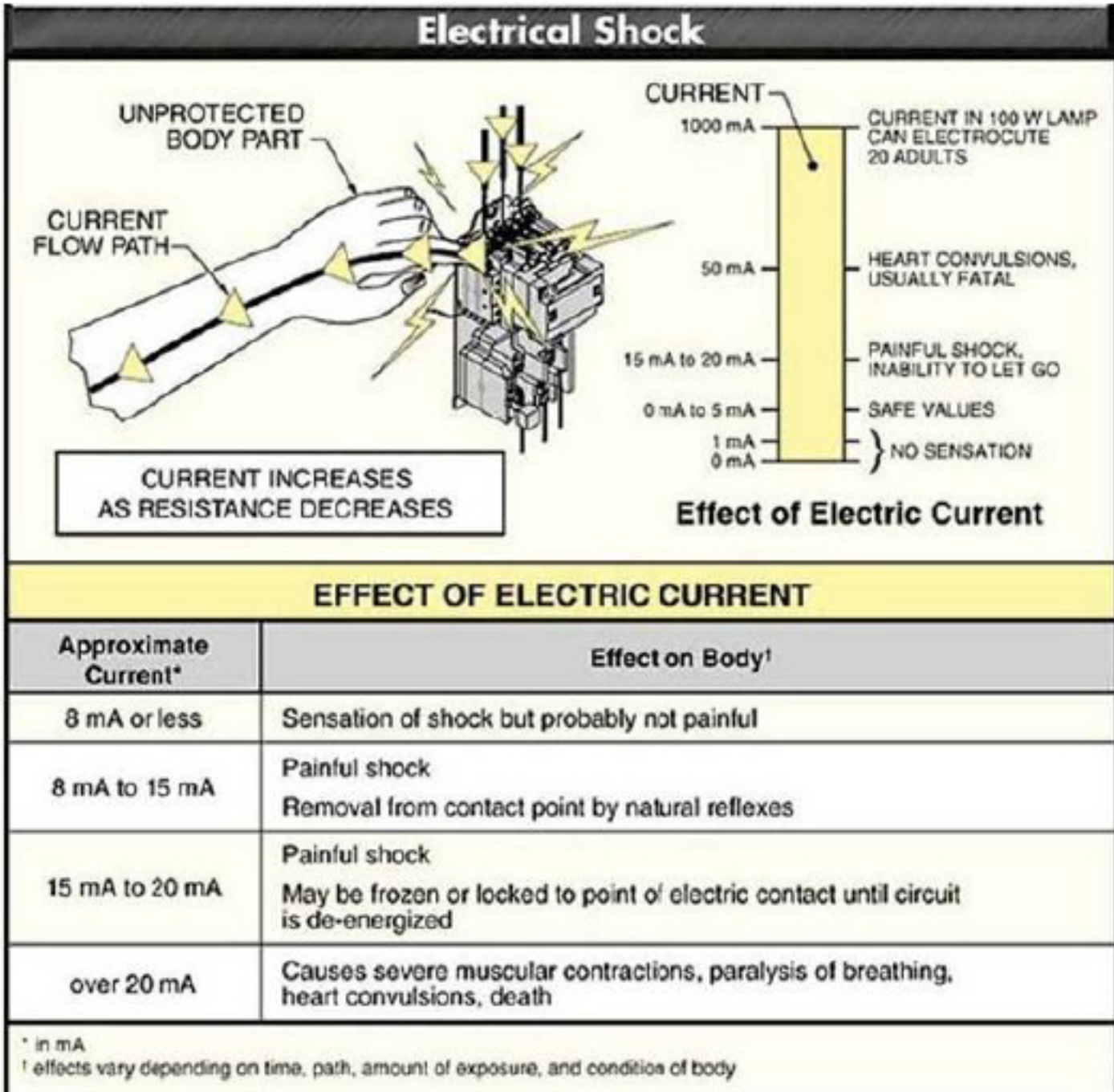
**Danger**

Drive shafts may start without warning. Keep away from drive shafts to prevent clothing or limbs from being entangled, resulting in severe injury.

**Caution**

Safe operation of any mechanized irrigation system is the responsibility of the operator, regardless of utilizing either on-system or remote control.

This manual was developed based on the information that was available at the time it was written. Reinke reserves the right to update this information, at any time, without providing prior notice.

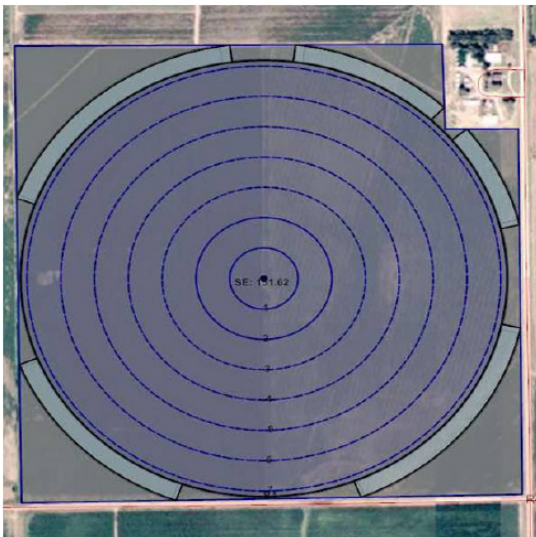


When dealing with electricity, it is important to understand the risks that are present. Electricity can be very dangerous to work with if not done so properly. The chart above shows the amount of current associated with the effect it can have on an individual's body. When looking at the chart, it is talking about mili-amps. This is a very small amount of current. Always make sure that when working with electricity, you know what is energized and what is not and where the dangerous areas are to avoid getting shocked. Factors that determine shock are discussed on the next page.

FACTORS THAT DETERMINE SHOCK

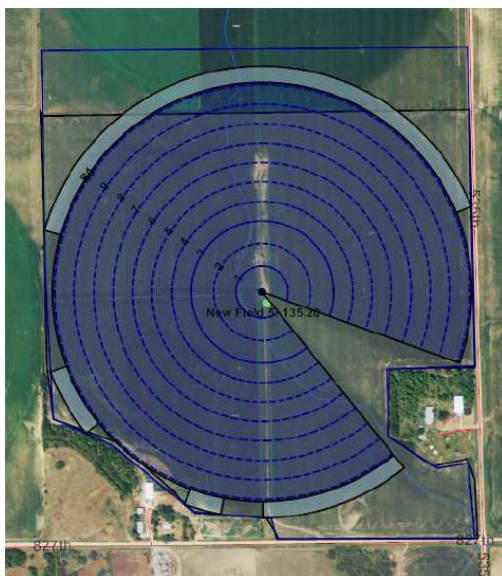
As seen in the previous chart, the amount of current that affects a body is very small. The time of exposure is also a determining factor in how severe a shock can be. The longer a person is exposed to electrical current, the more dangerous it is to the individual. The path through the body is very critical also. If one hand is grounded and the other hand is in contact with electrical current, the path that the current is going to take is from the hand through the heart and out the other hand to the ground. That is why it is important to try and do one handed checks if possible. Also, when approaching a main control panel, it is a good idea to tap the panel with the back of your hand with the other hand behind your back. The reason for this is to make sure that if something is wrong and the panel is energized then touching the panel with the back of the hand will cause your muscles to contract away from the panel and with the other hand behind the back then the path through the body will be from the hand and out the feet and not through the heart. The body area exposed is related to the path through the body. If the core of the body is electrocuted, it is much more severe than getting electrocuted on the extremity of the body. The size and condition of the body is also crucial. The bigger the body, the more resistance a body has. The smaller the body, the smaller the resistance. Electricity always follows the path of least resistance.

PIVOT THEORY OF OPERATION



FULL CIRCLE PIVOT

- A pivot is anchored at a central point.
- All spans can travel in a complete circle around the pivot center without any obstacles.
- The system can travel in forward (clockwise) or reverse (counter-clockwise).



PART CIRCLE PIVOT

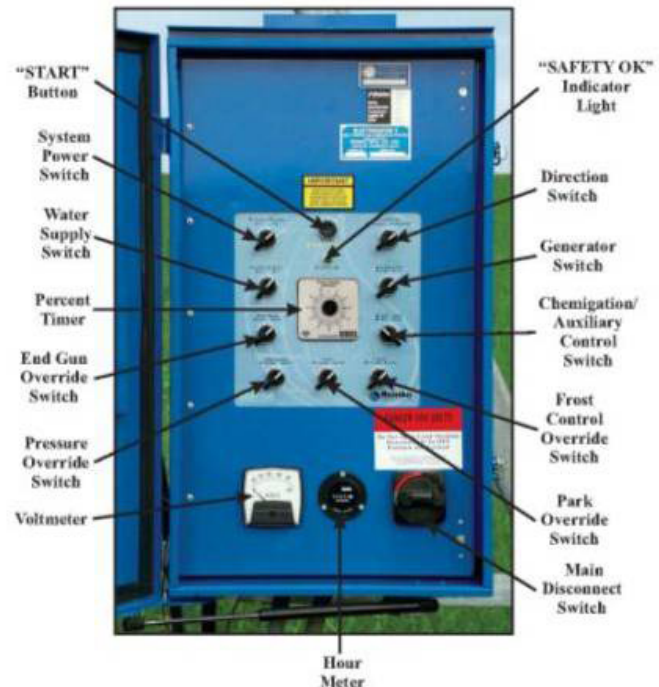
A part circle pivot is the same as a full circle pivot except that some obstacle prevents the spans from completing a full circle. The direction of system travel must switch to the opposite direction at the barricades placed at both sides of that obstacle.

ELECTRICAL COMPONENTS

MAIN CONTROL PANEL

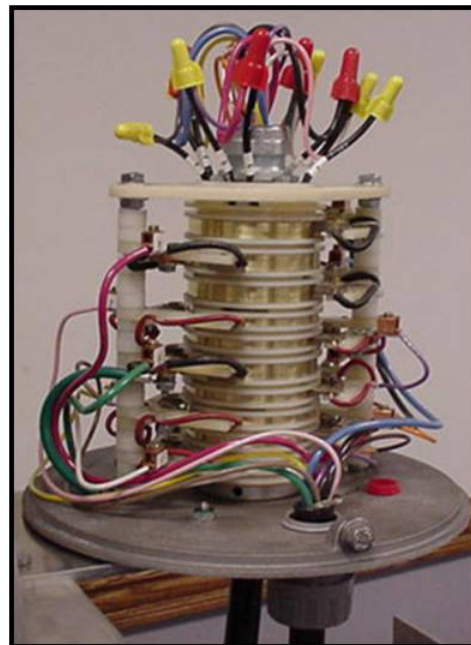
- control center of the system
- customizable as determined by the system options
- most commonly located at the pivot center but may be remotely located
- where the electrical service enters the system - this is most commonly 480 VAC electricity used by the drive motors to 120 volts for the control circuit

The panel shown below on the left is the basic panel. The standard panel is shown on the right.



COLLECTOR REEL

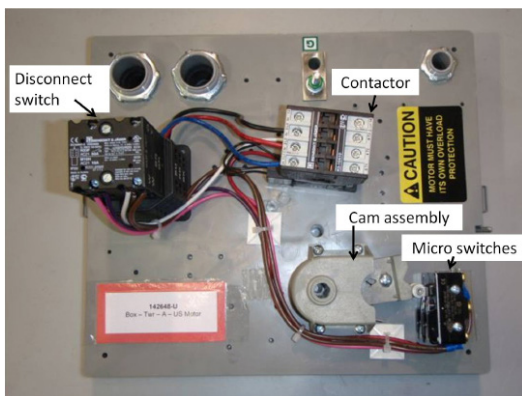
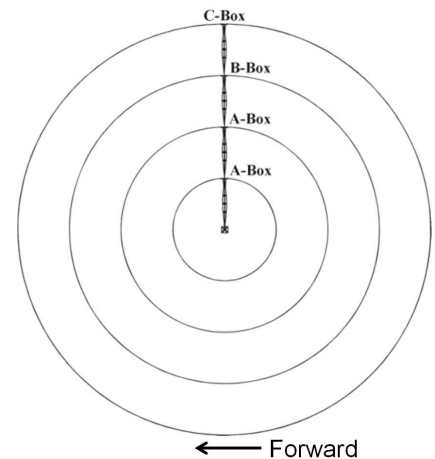
- is the point where the electrical cables make a transition from the rigid pivot point to the moveable system
- consists of stacked, insulated brass rings separated by insulators that remain stationary while the brushes rotate with the system around the brass rings, providing continuous flow of current without twisting the cable



TOWER BOXES

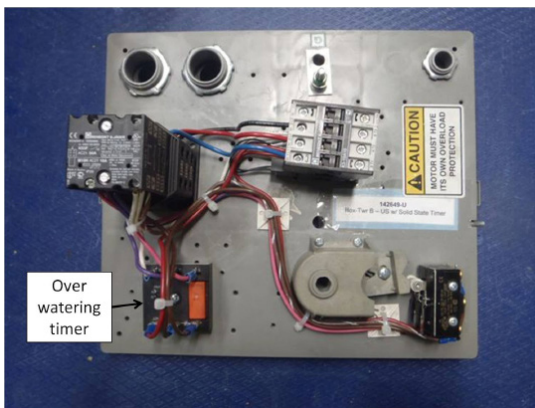
- Each tower box contains a disconnect switch, alignment, and safety controls for that tower.
- Serves as a junction box from one span cable to the next.
- "C" Box is used on the last tower.
- "B" Box is used on the next to last tower.
- "A" Box is used on all intermediate towers.

The picture shown here shows the locations of the A, B, and C boxes. Forward is clockwise and reverse is counter-clockwise.



"A" Box (intermediate tower box)

The "A" Box is installed on all the towers of a pivot except for the second to last, and the last tower. Included in the "A" box is the disconnect switch (allows power to come into the box or to turn it off), the contactor (supplies the 480V to the drive motor), and the micro-switches (the bottom switch tells the machine when to run and the top is the safety and will shut the machine down if tripped).



"B" Box (next to last tower)

The "B" box is installed on the second to last tower. It works just like an "A" box but also has an overwatering timer installed. If this tower does not move in a set amount of time the overwatering timer will time out and open the safety circuit shutting down the machine.



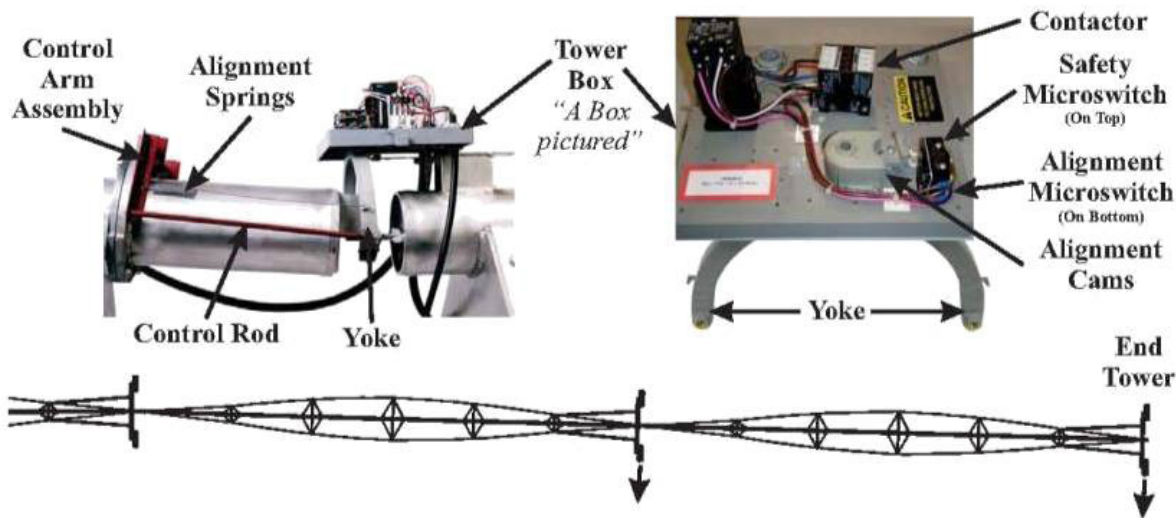
"C" Box (last tower box)

The last tower has nothing beyond it to align, therefore, there is no need for a cam assembly or micro switches. This box only contains the tower box disconnect and the contactor. The percent timer in the main control panel tells this tower when to run.

BASIC SYSTEM ELECTRICAL CIRCUITS

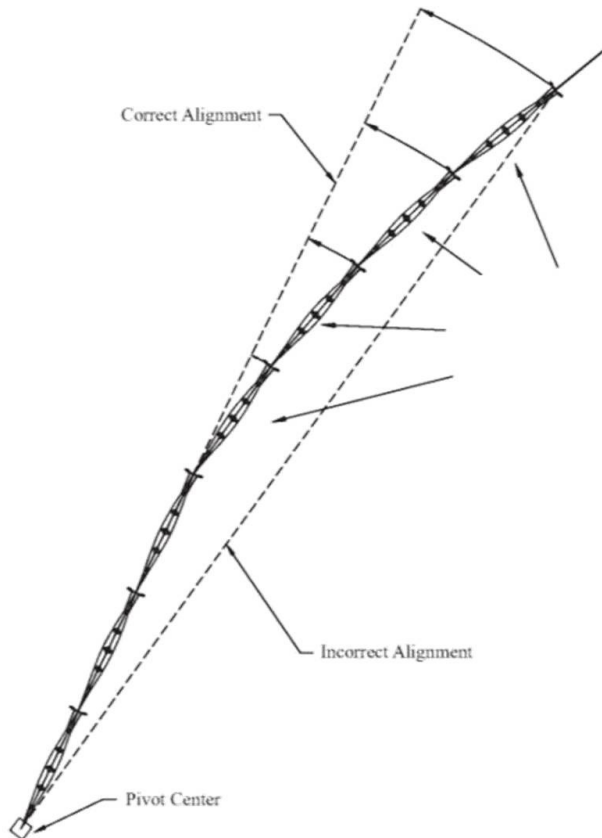
The **alignment circuit** is a parallel circuit. It determines when the tower is to move.

The percent timer in the main control panel controls the end tower, which determines the water application. All intermediate towers are controlled by micro switches in the tower boxes to keep the towers "in line" between the pivot point and the last tower.



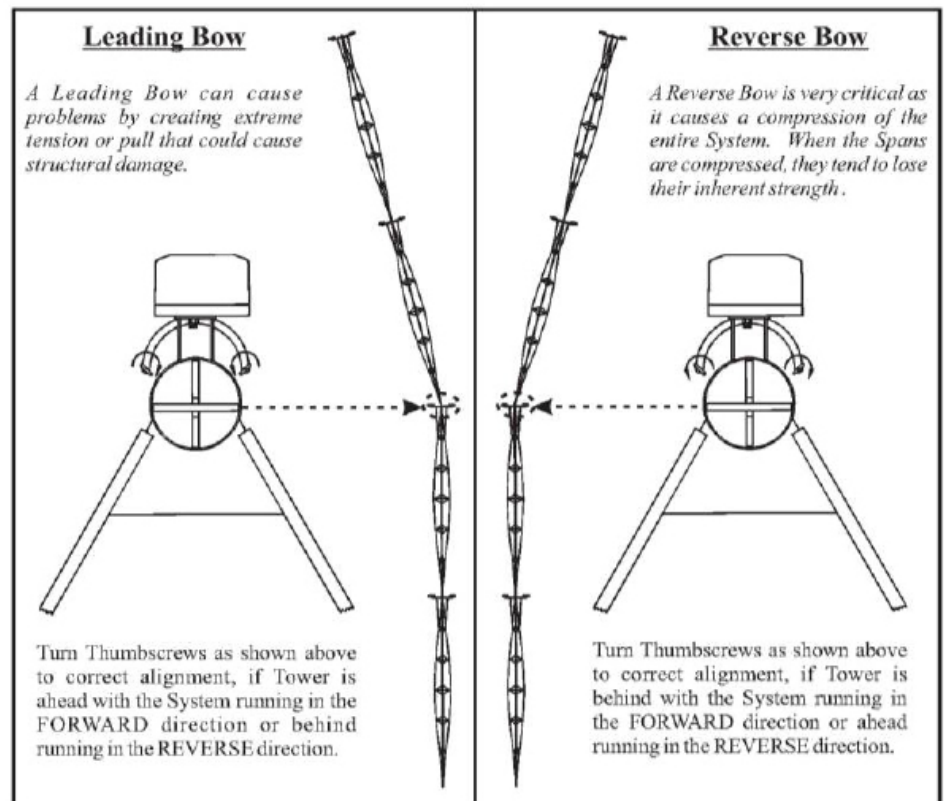
The alignment microswitch and safety microswitch are mechanically attached to the yoke via the alignment cams. These components are what tells the contactor when to pull in and make the tower move. The yoke rotates at the bottom of the box by sensing what the span out from it is doing. The yoke is connected to the control arm assembly by alignment springs and rods. The springs are in place to make sure that there is sufficient pressure on the rods to hold them into place to keep the system operational. To sum up how this works, the span ahead or out from the pivot center moves and rotates the yoke in turn tripping the alignment switch which then sends 120 volts to the coil of the contactor which will then send 480 volts to the center drive unit which makes the tower run. The tower will run until the microswitch de-activates removing the 120 volts from the coil of the contactor removing the 480 volts from the motor, stopping the tower from moving. If the span in front moves too far, too fast, the yoke will keep rotating and trip the safety cam which will shut down the system preventing structural damage to the system.

When a system is correctly aligned, it will have a slight bow with the center tower leading. When the direction of travel is changed, the system should bow the same amount only in the opposite direction. If a tower becomes stuck, or some other problem causes it to stop, the towers toward the end of the system will continue to run until the misalignment safety circuit shuts the system down. This results in a bend in the system at that tower. If the lagging tower is brought into alignment by continuing to run in the same direction, the outer towers will be forced outward to compensate. This causes considerable stress and possible damage to the system's structure. Therefore, after the problem has been corrected, run the system in the opposite direction so the outer towers run back even with the stalled tower. Then it is safe to operate the system in the desired direction.



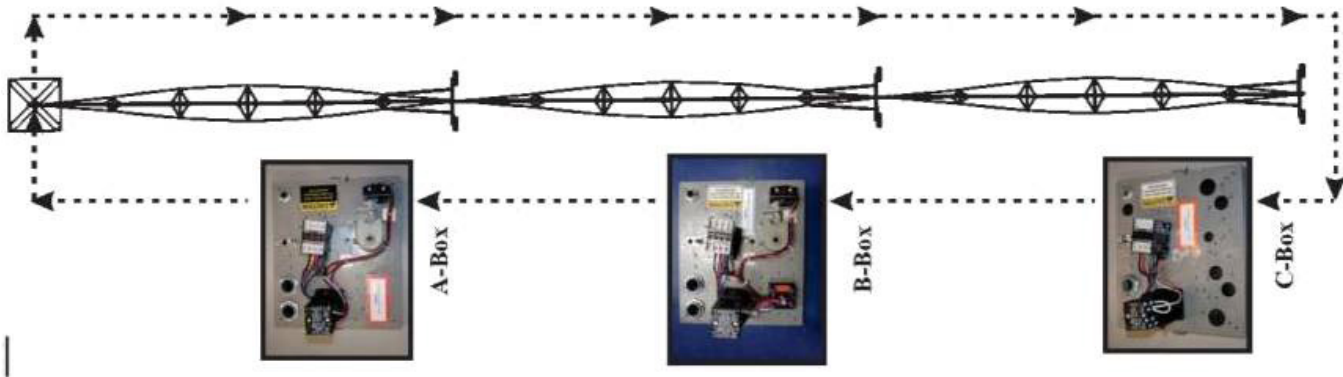
If the third tower from the pivot center is stuck or stalled, operating the system in the forward (clockwise) direction will cause the outer towers to be pushed outward, causing undue stress and possible damage to the system. Therefore, correct the tower problem, operate the system in the reverse (counter-clockwise) direction until the system is in alignment, and then operate the system in the forward (clockwise) direction.

The thumbscrews are turned in opposite directions in order to adjust the alignment of the system. When adjusting the thumbscrews, the direction that the right hand is rotating is the way that the tower is going to move. So if the tower needs to move ahead going forward, the thumbs would go out as seen in the reverse bow side.



The **safety circuit** is a series circuit. It carries no voltage when the safety circuit is complete.

The safety circuit supplies neutral to the safety relay, which supplies neutral to the reversing contactor in the main control panel. The circuit travels out to the last tower box and back through a series of tower box safety micro switches. Any break in circuit stops the system. When the safety circuit is complete, there is no voltage present on the brown or white wires. If the series circuit is broken, then there will be voltage present on the brown wire from the main control panel to the point where the circuit is broke. This is done by the 120 volts going through the coil of the relay because there is no more neutral there. This will be explained more in the safety circuit section of the main control panel.



The safety circuit goes out to the last tower box unimpeded then back through the safety switches back to the main control panel.

Electrical Components, Electrical Properties, and Electrical Circuit Types

CHAPTER 2

ELECTRICAL COMPONENTS ON REINKE SYSTEMS

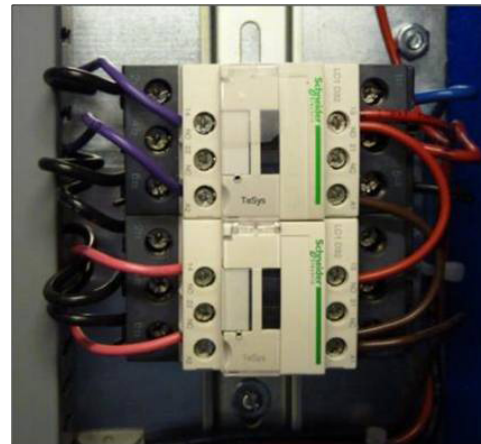
Arc Suppressor

The arc suppressor on the top can be installed wherever there is need; for instance, over a coil of a relay or contactor. The arc suppressor on the contactor on the bottom is pre-installed on tower box contactors and extends the life of the contactor by suppressing the amount of arc when engaging the contacts.



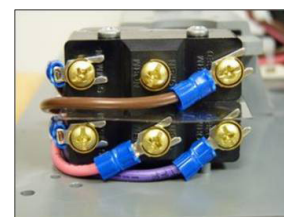
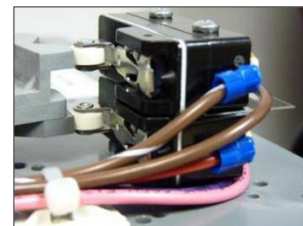
Auxiliary Contact

Auxiliary contacts are a set of contacts that are either attached to or accommodating an existing contactor. An auxiliary contact uses the same operation coil and armature as the existing or main contactor. The auxiliary contacts are either going to be normally open or normally closed which means that the normally open contacts close when the contactor is engaged and the normally closed contacts open when engaged.



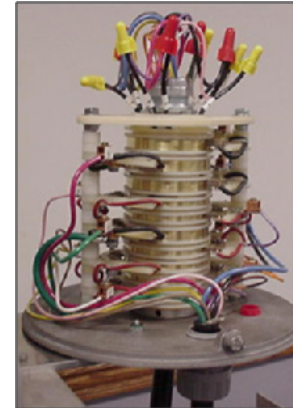
Basic Switch (Micro Switch)

The basic switch is a snap action switch that can be used as either a normally closed or normally open contact. The small-sized switch mechanism is enclosed in a case and is engaged by an actuator. The switch has a roller on the end of the actuator and can be found in "A", "B", "H", "S", and "G" tower boxes.



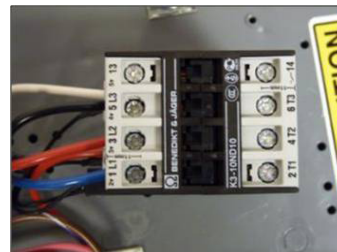
Collector Reel or Ring

The collector ring's purpose is to keep the wires from getting wound up during operation of the pivot. If there was just a junction box at the top, then the wires would eventually get wound so tight that they would pull out and cause an undesirable effect.



Contactor

(Starter) A set of three or four contacts magnetically engaged by an energized coil. It is secured to the base of tower control boxes and booster pump control boxes. An overload may be used with a contactor if the motor it operates doesn't have its own thermal protection.



Control Transformer

Contains two coils that transfer alternating current (AC) at the same frequency but at lower voltage and current values. On a Reinke system, a 4:1 step down transformer is used in the MCP to convert the 480 VAC incoming power supply to 120 VAC for the control circuit. This transformer is mounted on the inner mounting plate.



Direction Switch

Two or three position selector switch on the dead front panel in the MCP. The switch may have an “Auto” position between each direction change selection. This switch is used by the operator to set the direction that the system is to travel. The top picture is the front side. The bottom picture is the back side.



Electrical Wires and Cables

Electrical cables for a Reinke system are pre-engineered for proper wire size configuration. They are also cut to length for the application and have the outside sheathing trimmed back and the wires stripped for easy installation. Span cables come in a variety of sizes to meet the needs of different configurations of spans.



Electrical Motors

Electric motors used on a basic Reinke system are to serve as center drive motors for the towers. These are always coupled with a gear reduction assembly. There are three different speeds of center drives available on a Reinke system. There is a 25:1 which is a high speed motor, a 40:1 which is a standard speed motor, and a 60:1 which is a low speed motor.



Fuse

A fuse's purpose is to protect equipment from damage. It is sized so that if something goes wrong it will blow, cutting off electricity to the affected circuit. It is important that when replacing fuses not to put in a 20 amp fuse if a 10 amp fuse keeps blowing. Find out why the 10 amp fuse is blowing and correct that problem instead of just upsizing the fuses.



Large Service Wire Adapter

The large service wire adaptor is used when installing wires larger than 6 gauge into a main control panel. The large service wire adaptor will accept wires sized between 14 gauge to 2-0. The main disconnect on a Reinke panel will only allow for wires as large as #6.



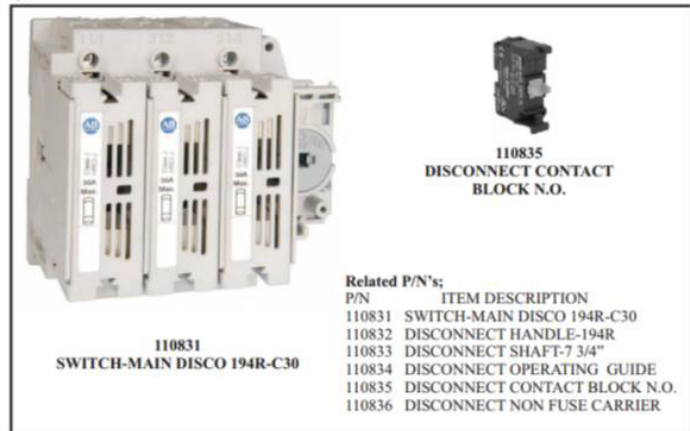
Main Control Panel (MCP)

A box enclosure mounted at the pivot or elsewhere (making it a remote panel) contains the system controls necessary for system operation. The main control panel is the brains of the system. The main control panel contains multiple parts to become one functioning unit. The main enclosure is the inside of the box. The outer door protects the electrical components from moisture and dust. The dead front panel is where the controls of the system are, which is also the RPM insert. The RPM insert can be "Standard", "Advanced", or "Preferred". The back plate is where the electrical components such as relays, contactors, disconnects, transformers, and terminal blocks are located.



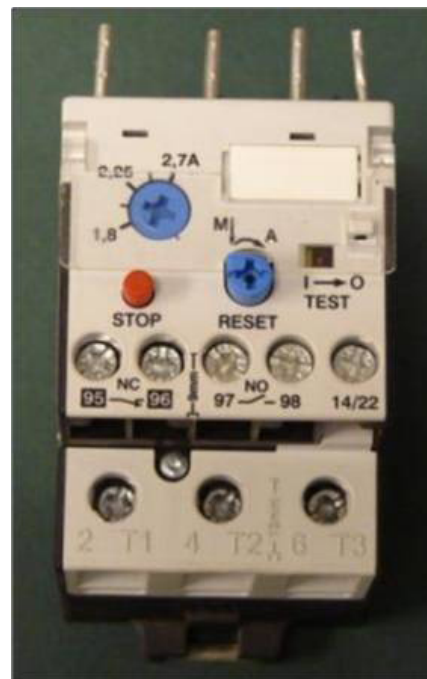
Main Disconnect Switch

The main disconnect is where the main power wires are landed to feed power to the panel. The incoming wires will always be hot on the panel unless they are disconnected at a previous place. The design of the disconnect has the screws and fuses covered to prevent any unnecessary contact.



Overload

The overload is attached to a contactor to protect against overloading of the tower motor. When excessive current is experienced, the overload disengages the contactor. The overload is adjustable between current levels. This overload's settings are from 1.8 amps to 2.7 amps. The stop button is to open the overload and keep the power from flowing through it. The normally closed and normally open contacts are what actually trip when there is too much current. Typically, the neutral to the contactor coil is ran through 95 to 96. When the overload senses too much current, it will open the contacts causing the neutral to open and drop out the contactor to prevent the motor from burning up.



Over-Watering (DO DO) Timer

The over-watering timer is a timer mounted in the next to last tower control box (B Box). The timer can be set from 0 – 20 minutes so that if the next to last tower does not move within this time, the safety circuit opens and shuts down the system. Forward and Reverse direction wires are connected to 1 and 2, 3 is the neutral, 4 is the initiate or reset that is wired to the coil of the contactor and is responsible for resetting the timer, and 5 and 6 are the contacts that "open" causing the safety to shut down the system.



Pressure Switch

The pressure switch is an adjustable pressure activated snap action switch that can be used as either a normally closed or normally open contact. This switch has two configurations: one open (shown in left picture) and one that comes in an enclosure (shown in right picture). This switch in the open configuration is used in the MCP and the booster pump box. The enclosed switch is used for any pressure sensing applications outside of the MCP or booster pump box. The pressure switch has three contacts on it: they are the common, normally open, and normally closed.



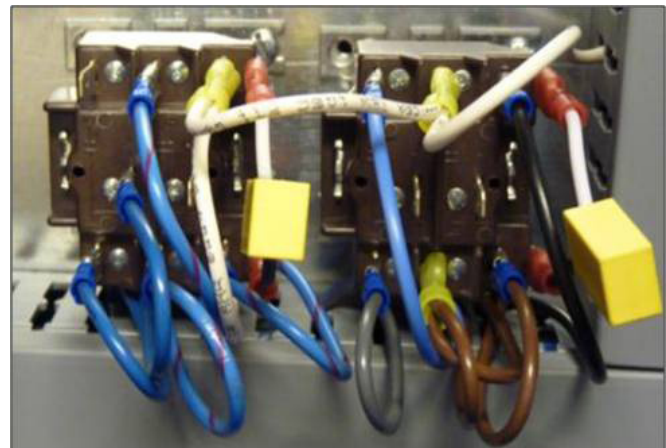
Percent Timer

The percent timer is a variable setting control mounted on the dead front panel in the MCP. It controls the amount of time (in percent) the last tower moves. The lower the percent timer setting, the more water is applied. The percent timer is a 60 second timer. If the timer was set at 50%, then the last tower would be on for 30 seconds and off for 30 seconds. At 75%, the last tower would be on for 45 seconds and off for 15 seconds.



Relay

A common relay on a Reinke system is referred to as a "Honeywell Relay". This relay is used in several applications. One example would be the safety relay in the main control panel. These relays have both normally open and normally closed contacts activated by one coil.



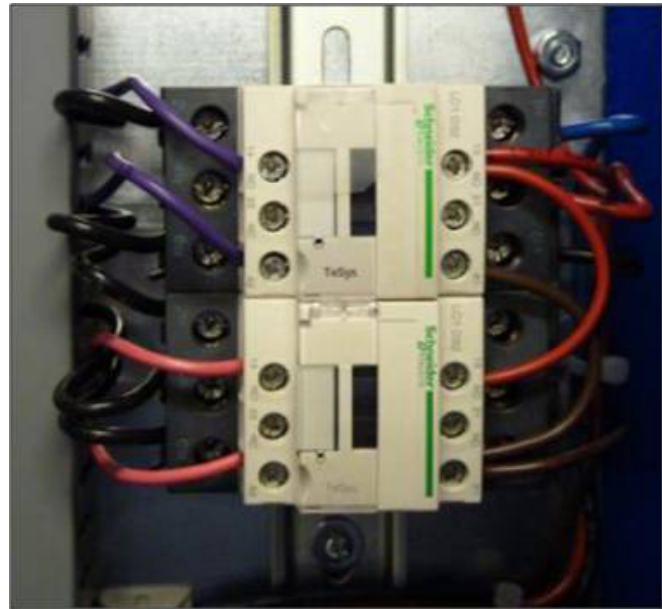
Resistor

Two common applications on Reinke systems are on strobe light installations and on remote panels. The resistors help to keep induced voltage off of components in which the voltage may cause problems.



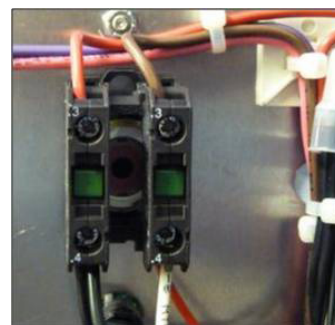
Reversing Contactor

The reversing contactor contains an interlocking magnetic forward contactor and one interlocking magnetic reverse contactor. Each contactor has three or four contacts attached to one armature, which are magnetically activated by an energized coil. The fourth contact is called an auxiliary contact. The reversing contactor is mounted on the inner mounting plate of the MCP. A reversing contactor is also used in the booster pump box if the system is equipped with a booster pump. The reversing contactor has internal jumpers that change the way the 3 phase voltage is distributed to the system. In forward, the purple side of the contactor pulls in and in turn sends the 3 phase out in a clockwise rotation. In reverse, the pink side of the contactor pulls in causing the rotation to be counter clockwise.



Start Button

A momentary activated push button switch on the dead front panel of the MCP. Once pushed, it sends power to the reversing contactor to start the system. Provided the main disconnect switch is on and safety circuit is closed, the latching circuit will continue to hold the reversing contactor in when the start button is released. The top picture is the front. The bottom picture is the back.



Strain Relief

A strain relief is a compression fitting that mechanically supports an electrical wire or cable through the wall of an enclosure or component. This is often referred to as a water-tight connector as it helps keep water out. On a Reinke system, they are also sometimes referred to as a “Woodhead” fitting. Woodhead is a common manufacturer of this fitting. There are multiple sizes of the strain relief to accommodate different sizes of cables.



Terminal Block

The terminal block is used when a group of wires need to be terminated. A common use of terminal blocks in the MCP is as the junction between the “Factory Wired” circuits and the “Field Wiring” that connects to them.



Tower Disconnect Switch

The tower disconnect switch is mounted on the base of a tower control box. It engages and disengages power to the components of the tower control boxes. The top picture shows the tower disconnect switch in the tower box. The bottom picture shows the switch handle and tower box lid interlock mechanism on the bottom of the box.



Voltmeter

The voltmeter is located on the dead front panel in the MCP and measures the incoming line voltage on the bottom side of the fuse clip at the main disconnect switch. The voltmeter is reading voltage off of two of the lines of the incoming 3 phase voltage. It is important to know what the voltage is, but be aware that the voltmeter may be reading correct but there is an issue with the third phase causing “single phasing” situation on the motors.



ELECTRIC PROPERTIES

(ELECTRICITY REVIEW)

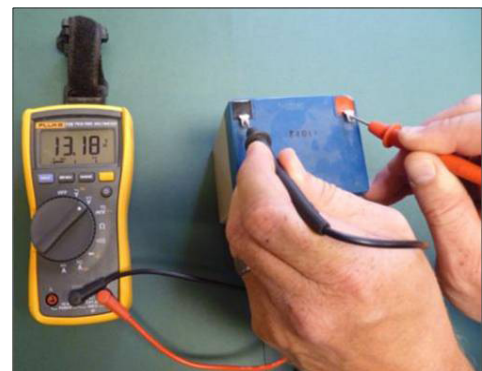
To perform service work on a Reinke irrigation system, a basic working knowledge of electricity is expected of a person. If you are unfamiliar with the essentials of electricity, it would be wise to gain some training on this subject. Troubleshooting or understanding the operation of a Reinke irrigation system requires the service person to have an idea of the working order of the items discussed in the manual. A service person must be knowledgeable enough to clearly define the problem at hand, quickly isolate it, and determine a possible solution. The service person must also be aware of the significance of the problem as well as the ramifications of the repair procedure. Those of you who have been away from irrigation systems for awhile or would like to refresh your memory, the following section will help define some of the basic electrical properties and provide a description of very basic Reinke irrigation system operation.

Voltage

When reading voltage on a voltmeter, it is important to make sure that the meter is on the proper setting. If measuring AC, make sure the meter is on Volts AC. A common example would be if you were reading the voltage on a direction wire: the reading on the meter should be approximately 120 Volts AC.

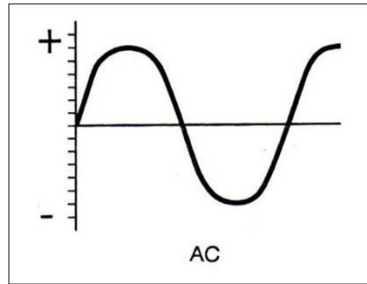
Direct Current

Direct current does not reverse like AC so that means that it has a positive and negative reading. When measuring voltage on the battery, the black lead is placed on the negative or ground and the red lead is placed on the positive terminal of the battery. This registers a positive 13.18 Volts on the meter. If the leads were reversed, the meter would read -13.18 Volts.



Alternating Current

When reading AC voltage, the meter is reading the RMS value of the voltage. When placing the meter leads on the terminals to measure AC voltage, it is important that one of the leads is placed on the terminal that is needing to be read and the other is placed on neutral and/or ground. If the meter is placed on the same wire but at different ends of the wire, the meter will register 0 Volts because there is no potential difference on that wire.



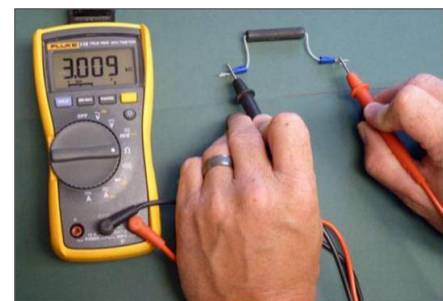
Amperage

The amp clamp shown here is measuring the current in one of the incoming power wires. The meter is displaying 9.2 amps AC. When using an amp clamp, it is crucial that the meter is placed around one wire at a time. It may be necessary to read multiple wires, but only one can be read at a time. An example of needing to read multiple wires would be on a 3 phase motor. If the motor is continually tripping an overload, the current in the motor may need to be tested. The procedure for this would be to place the meter around each wire one at a time and observe what the particular motor lead is registering. If all three are above the name plate of the motor, then it could be a bad motor or a wheel gear bound up causing the motor to pull too much amperage.



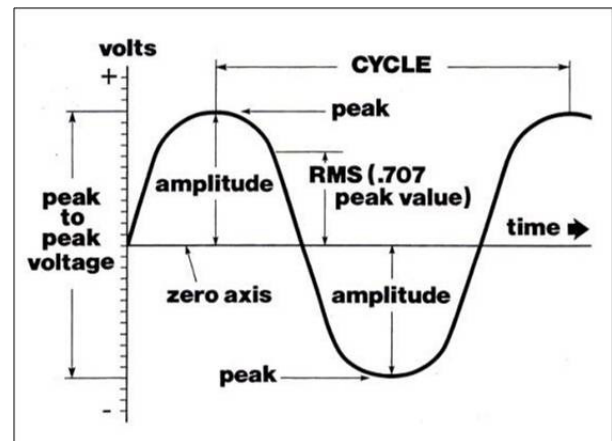
Resistance

When referring to resistance, there are two settings on the meter that can be used. The ohm sign (which the meter is set on in this slide), as you can see, is pointed at the omega sign on the meter or the setting one click down which is the continuity setting. The ohms setting will give you a more accurate reading; for example, when needing to know the actual resistance of a resistor, coil, or winding. The continuity setting is used when needing to know if there is a good electrical path between point A to point B. An example of a continuity test would be on a system power switch. The meter leads would be placed on the two terminals of the switch. When the switch is off, the meter should read O.L. for open line. When the switch is on, the meter should read approximately 0 or just above it. Also, with the switch on, the meter should have an audible beep.



Frequency

In the United States and most of North America, the Frequency is 60 hertz. In most of the rest of the world, the frequency is 50 hertz. To illustrate the difference, the typical 3 phase voltage in the U.S. is 480 volts A.C. The typical 3 phase voltage at 50 Hz is 380 volts A.C.



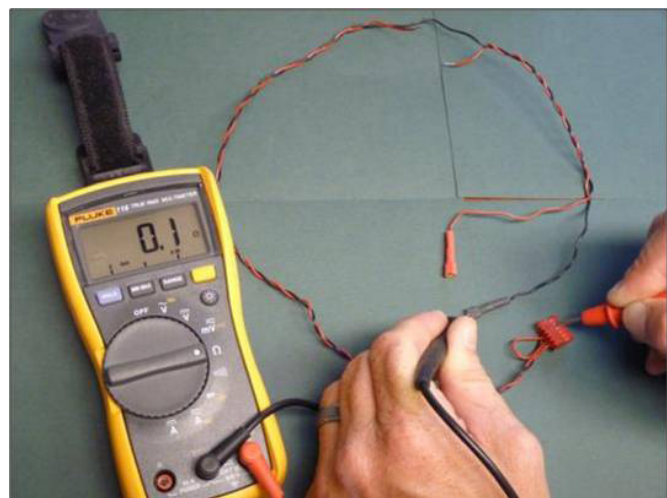
Ground

The ground is very important when wiring a system. The ground is for protection of the machine and anyone that is in contact with the system. If a shorted wire occurred, then the ground's job is to take the voltage to the earth and this would cause a fuse to blow or a breaker to trip. If the ground is not properly installed, then the same shorted wire could energize the entire machine and seriously electrocute someone. When walking up to a system, it's important to inspect the ground termination. Another good practice is to tap the main control panel with the back of the hand so that if the system is energized by the shorted wire the shock will not contract the muscles in the hand to the panel but away from the panel.



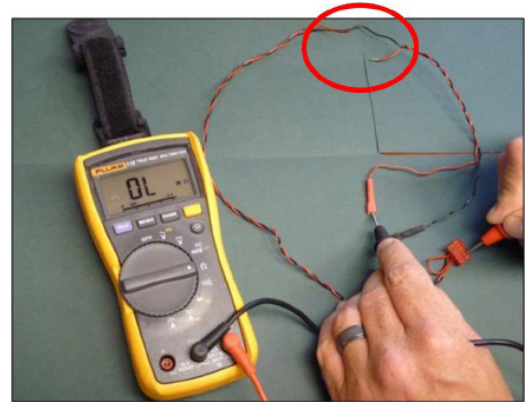
Continuity (Closed Circuit)

Testing the continuity is essentially checking the resistance of the wire. It can be done with the ohms setting which is shown in the picture or in the setting one click down. In the ohms setting, good continuity will show near zero but the meter does not have an audible beep. In the continuity setting, the meter will read the same but the meter will have an audible beep.



Discontinuity (Open Circuit)

Discontinuity means that there is no path for electricity to flow. As shown in the picture, the wire is cut. In this situation, the meter will read O.L. for open line.



Electrical Short

An electrical short means that two wires with different potentials have touched. When this happens, the wires essentially become a welder causing very high temperatures that will melt plastic and insulation as well as cause amperages to rise to very high levels which in turn will blow a fuse or breaker.



Electrical Short to Ground

A short to ground is similar to an electrical short, but the current carrying conductor is shorted to ground. The results are very similar with the heat and current going up. A short to ground can be very dangerous if the system is not properly grounded to earth causing the metal parts of the system to carry full system voltage.

The next page shows examples of abbreviations and symbols that may be seen on a Reinke schematic.

ABBREVIATION - SYMBOLS LEGEND

IMF = INTERLOCKING MAGNETIC FORWARD

IMR = INTERLOCKING MAGNETIC REVERSE

AR = AUTO REVERSE

M = VOLT METER

^WIMF = WINCH MOTOR-IMF

^WIMR = WINCH MOTOR-IMR

MS = MOTOR STARTER-WHEEL DRIVE

PP = PURGE PUMP STARTER

TD = TIMER DELAY RELAY

K2 = TIMER RELAY-AUTO RESTART

SS = SWING SPAN RELAY

DK = DIRECTION RELAY

DO DO TIMER = DELAY ON DROP OUT

DO PI TIMER = DELAY ON PULL IN

C = CLUTCH

M = MOTOR

SK = SENSOR RELAY

BK = BOOM TOWER RELAY

PK = POWER TOWER RELAY

FF = FLIP FLOP RELAY

EG = END GUN RELAY

BPK = BOOSTER PUMP RELAY OR P200 RELAY

VSK = 120 VOLT SOURCE RELAY

TB = TIMER BY-PASS RELAY



← MANUAL DISCONNECT CONTACTS



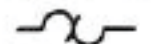
← FUSE



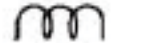
← CONTACTS, NORMALLY OPEN



← CONTACTS, NORMALLY CLOSED



← THERMAL OVERLOAD ELEMENT



← WINDING



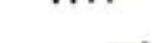
← COIL, STARTER



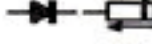
← GROUNDING SYMBOL



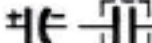
← RESISTOR



← +DC OUTPUT



← DIODE



← -DC OUTPUT



← CAPACITOR



← SPIKE SUPPRESSOR



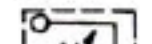
← ARC SUPPRESSOR



← WIRE CAP



← COIL, RELAY



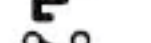
← SENSOR, PROXIMITY SWITCH



← LIGHT



← TEMPERATURE SWITCH



← FLOW SWITCH



← PRESSURE SWITCH



← TRANSFORMER


← ALTERNATE OR
OPTIONAL WIRING

SELECTOR SWITCH OPERATION TABLES

POSITION #1 IS COUNTER CLOCKWISE POSITION

X = CLOSED CONTACTS

O = OPEN CONTACTS

4 POSITION SELECTOR SWITCH-VM

	CONTACTS			
	1--2	3--4	5--6	7--8
OFF-POS. #1	O	O	O	O
1-2-POS. #2	X	O	O	X
2-3-POS. #3	O	O	X	X
3-1-POS. #4	O	X	X	O

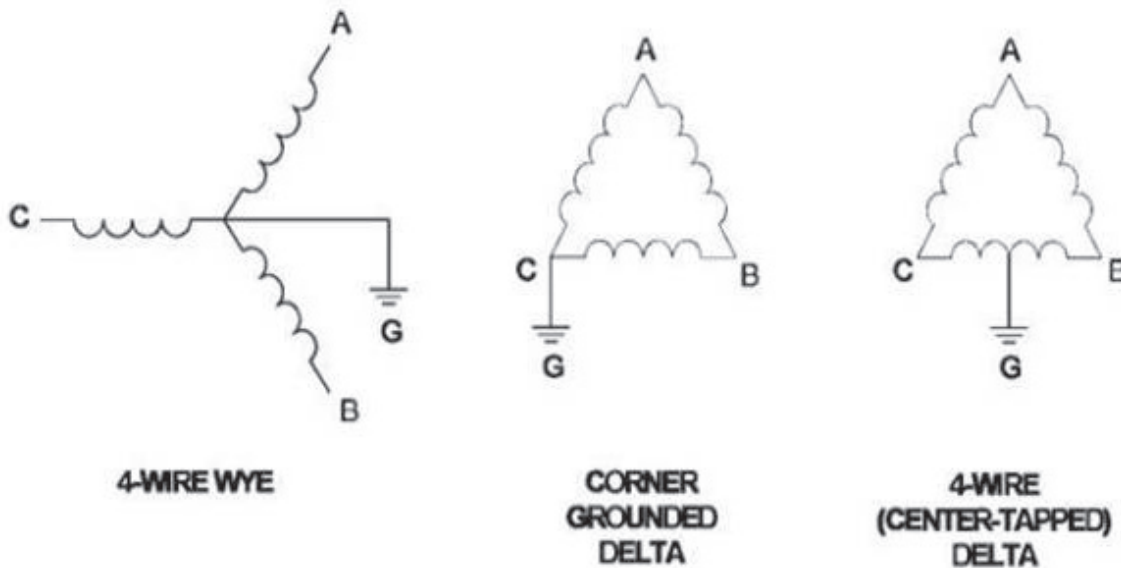
5 POSITION SELECTOR SWITCH-AR

	CONTACTS			
	1--2	3--4	5--6	7--8
REV.-POS. #1	X	O	O	O
OFF-POS. #2	O	O	O	O
FWD.-POS. #3	O	X	O	O
OFF-POS. #4	O	O	O	O
AR-POS. #5	X	X	X	X

5 POSITION SELECTOR SWITCH-AR-SACS

	CONTACTS					
	1--2	3--4	5--6	7--8	9--10	11--12
REV.-POS. #1	X	O	O	X	X	O
OFF-POS. #2	O	O	O	O	O	O
FWD.-POS. #3	O	X	O	X	O	X
OFF-POS. #4	O	O	O	O	O	O
AR-POS. #5	X	X	X	X	O	O

3-Phase Voltage Chart – 480 Volt



	A-G	B-G	C-G	A-B	B-C	C-A
4-Wire Wye	277	277	277	480	480	480
C G Delta	480	480	0	480	480	480
4-Wire Delta	415	240	240	480	480	480

On the left of the chart is a 4 wire wye, 3 phase system. The voltage on the legs to ground are all 277 volts. Voltage line to line is 480 volts.

The center 3 phase system is a corner-grounded delta. The voltage on the legs to ground on 2 of the phases is 480 volts and on the third “C” phase it is 0 volts. The reason for this is that there is no potential difference there. The corner-grounded delta is a dangerous system to work with. The grounded “C” phase will not measure voltage on it, but it still has current flowing on it. Make sure that if this system is used, the grounded phase goes on the center of the disconnect to ensure that the grounded phase is not a part of the control circuit. Voltage phase to phase is 480 volts.

The right hand system is a 4 wire delta also known as a center tap delta. The voltage on its phases to ground are 240 on 2 of them and approximately 415 on the third. The reason for the higher voltage is that the third leg is being measured through 1½ windings. All of the legs phase to phase is 480 volts.



110831
SWITCH-MAIN DISCO 194R-C30



110835
DISCONNECT CONTACT
BLOCK N.O.

Related P/N's:

P/N	ITEM DESCRIPTION
110831	SWITCH-MAIN DISCO 194R-C30
110832	DISCONNECT HANDLE-194R
110833	DISCONNECT SHAFT-7 3/4"
110834	DISCONNECT OPERATING GUIDE
110835	DISCONNECT CONTACT BLOCK N.O.
110836	DISCONNECT NON FUSE CARRIER

P/N 110836, DISCONNECT NON FUSE CARRIER

When installing a Reinke Main Control Panel on a Corner-Grounded Delta (Grounded B Phase) electrical service the use of 110836, DISCONNECT NON FUSE CARRIER, is required for compliance to the National Electrical Code. Please select this option when ordering through Reinke Order Entry. The NON FUSE CARRIER will fit only in the B phase position, replacing the fuse with a shorting bar. DO NOT use the NON FUSE CARRIER for a Center Grounded Delta (High or Wild-Leg Delta) or 480/277v electrical service.

When the voltage on any phase measures different than the voltage on the other two phases, the odd phase must be installed to the center screw terminal of the 3 phase disconnect switch. An example of this is the use of a phase converter to operate a system on single phase commercial power. The manufactured phase from the converter must be connected to the center terminal of the disconnect switch.

When working with a corner-grounded delta configuration, you must have a shorting bar installed on the grounded leg. This shorting bar (part #110836 DISCONNECT NON FUSE CARRIER) must be installed in the center position of the disconnect switch.

The main control panel reference chart (see next page) for an RPM panel (which is the blue panels) illustrates what the function of each color of wire is intended for. Note the asterisk below the chart for special considerations. Another point to note is that just because a wire is orange in color does not mean that it is a percent timer signal. It may mean that someone had to work on the panel and needs to replace a wire and only had orange with them so they ran an orange wire for another circuit. It is important that one knows the color code but also important to understand how a system works to be an effective troubleshooter.

Main Control Panel Color Code (RPM Panel) Wire Cross Reference Charts

COLOR	DESCRIPTION	VOLTAGE	WIRE SIZE
Red/Yellow*	Transformer Line	120	16 AWG
Yellow/Red*	Park Bypass	0	16 AWG
Light Blue*	Frost Control, Hot Safety	120	16 AWG
Blue/Yellow*	Pressure Switch	120	16 AWG
Black/Yellow*	Chem/Aux	120	16 AWG
Blue*	N.C. Well Kill	0-480	16 AWG
Blue/Red*	N.O. Well Kill	0-480	16 AWG
Yellow	End Gun	120	16 AWG
Orange	Percent Timer	120	16 AWG
Brown	Limit	0	16 AWG
Purple	Forward	120	16 AWG
Pink	Reverse	120	16 AWG
Blue	L1	480	14 AWG
Red	L2	480	14 AWG
Black	L3	480	14 AWG
Red**	Line	120	16 AWG
Gray**	Hot Limit	120	16 AWG
Green	Ground	0	14 AWG
White	Neutral	0	16 AWG

*Indicates internal wiring which can be found on the terminal strip.

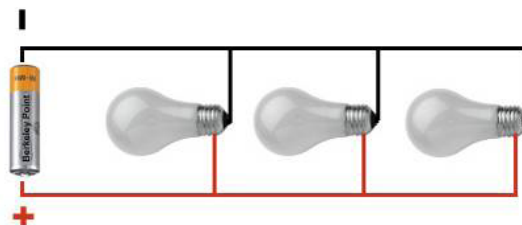
**Indicates wire that are not found on terminal strip, only internally.

CAUTION: Make all components resistance checks with the power OFF in the Main Control Panel.

CIRCUIT TYPES (SERIES & PARALLEL CIRCUITS)

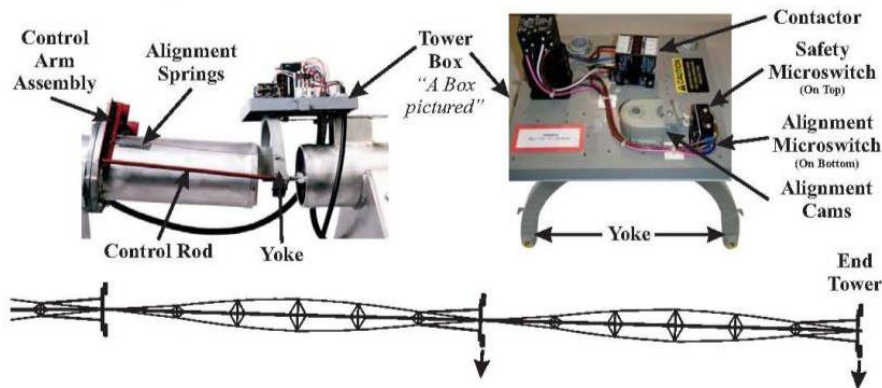
Parallel Circuit

A parallel circuit is interconnecting two or more components of a circuit with their neutral terminals joined to one conductor and the line terminals to another. This makes the potential difference applied to each component identical. The voltage from the source is branched out to each load in a parallel circuit giving each light bulb in this example the same amount of voltage.



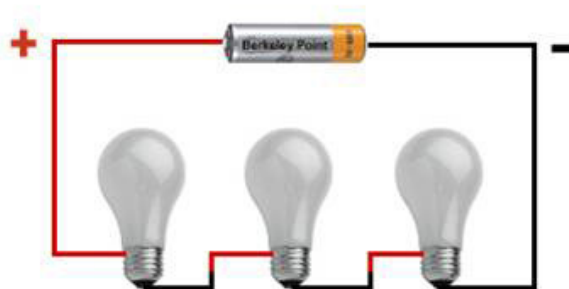
Alignment Basic Switch Circuit

An alignment basic switch circuit is a parallel circuit containing the purple (forward travel) and pink (reverse travel) wires that supplies power to the tower motor contactor coil.



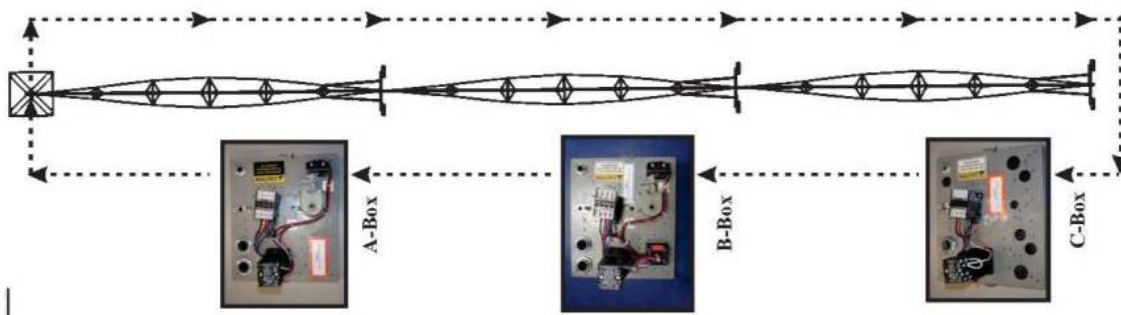
Series Circuit

A series circuit is joining components of a circuit end to end so the same current passes through each component. In a series circuit, the current will be the same on all components but the voltage will be different depending on the size of the individual load on the light bulbs in this scenario.



Safety Basic Switch Circuit

The safety basic switch circuit is a series circuit containing the white neutral wire and the brown limit wire. If the circuit is opened (caused by a switch being activated), the system shuts down.



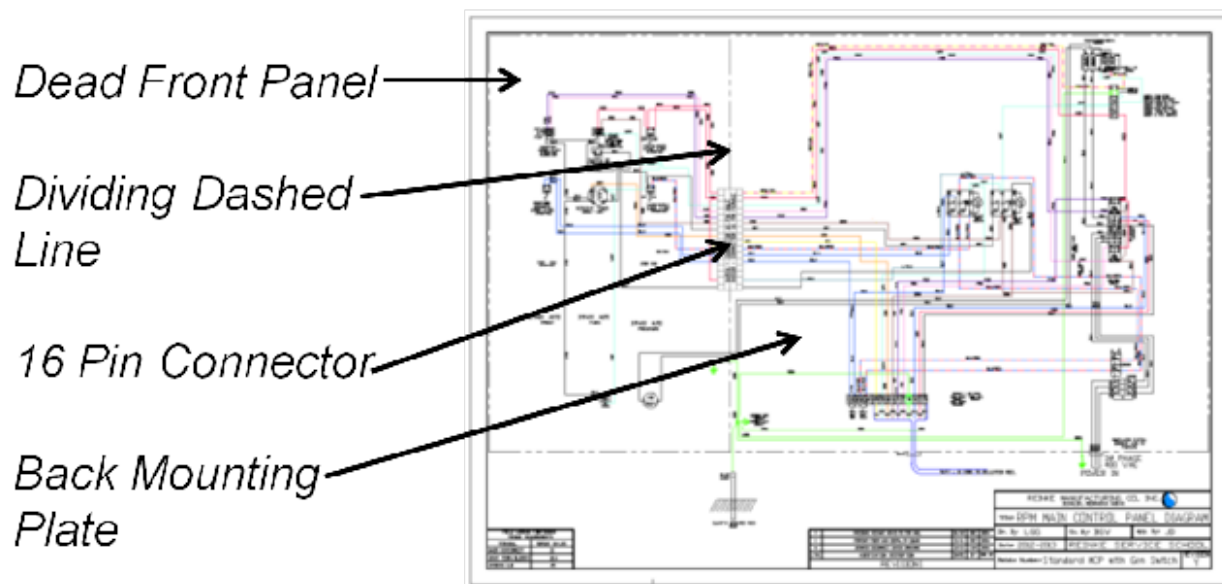
Primary Electrical Circuits on a Reinke System

CHAPTER 3

To understand the various electrical circuits and their functions, we will go through the primary electrical circuits one at a time. To follow along with these circuits, it is very important that you reference the proper electrical print in the reference section in this book. The progression for each circuit will start in a MCP (Main Control Panel) and then progress out through the various electrical boxes on the system.

REINKE PRINTS

All Reinke electrical prints are pictorial representations. This means that while not to scale, the print shows the components in the correct relative positions as they are located in the panel. The wiring is shown in the color of the wire used for that circuit.



Standard MCP W/ Gen Switch: Each MCP print is split into two parts separated by a dashed line. Everything shown to the right of the dashed line is physically located on the back mounting plate, and everything shown to the left of the dashed line is physically located on the dead front panel. The 16 pin connector is the electrical link between the two.

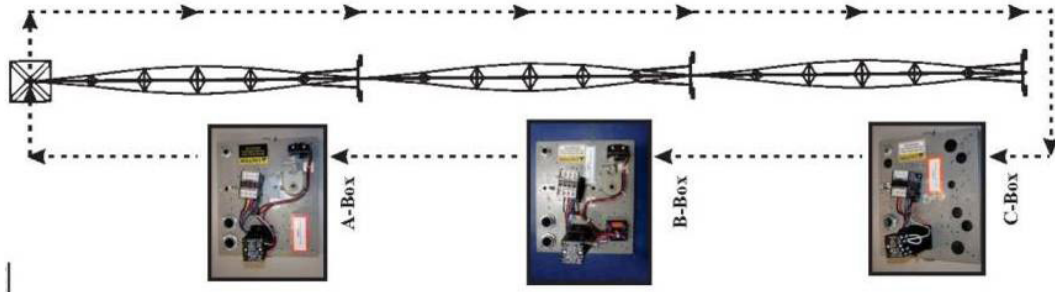
When looking at a tower box print (see print #[84610031](#)), you will notice “Span Cable In” and “Span Cable Out”. This refers to where the cable came from and where it is going, not necessarily how a particular circuit functions. We always assume that span wiring originates at the pivot center and moves out on the system; therefore, if this is a first A-Box, the “Span Cable In” comes from the collector reel, and the “Span Cable Out” goes to the next tower box.

PRIMARY ELECTRICAL CIRCUITS

The Safety Circuit

(brown and white wires)

In function, the safety circuit originates in the C-Box with a jumper from the white wire (which carries the neutral circuit from the MCP to the end of the system) and the brown wire (which is the actual safety circuit going back to the MCP) to complete a series circuit. On this path back, the brown wire goes through the limit switches in the B-Box and all of the A-Boxes. This is the one circuit flow that is contrary to the normal span cable progression from the MCP to the end of the system.



MAIN CONTROL PANEL

Follow along with the [MCP RPM](#) Print in the reference section in this book: In the circuit shown to the right, 120VAC is supplied to the system power switch from the transformer. If this switch is in the ON position, then power goes through it down to the 16-pin connector. There is a jumper from the red wire to the light blue wire, which is connected to terminal 3 on the safety relay. So anytime that the system power switch is in the ON position, then 120VAC is at terminal 3 on the safety relay.

The coil of the safety relay is connected to a brown wire on one side and a black wire on the other side.

The brown wire goes to the terminal strip, which is where it connects with the brown (safety) wire on the system. As long as the system is in line, this will supply the neutral for the coil of the safety relay.

The black wire will supply 120VAC when the start button is depressed, which will engage the safety relay closing the connection between terminals 1 & 3. This will allow the 120VAC from terminal 3 to go to terminal 1, which has the grey wire attached to it.

The grey wire goes through the 16-pin connector and is jumpered to the black wire that is connected to the coil of the safety relay. This is the latching circuit. As long as the safety on the machine is good, the safety relay will stay engaged allowing the machine to run.

The connection between terminal 4 & 6 on the safety relay will also be closed when the safety relay is engaged supplying a neutral for the coils of the reversing contactor.

Back to where the grey wire goes through the 16-pin connector there is also a grey jumper from that point to the water supply switch. This grey wire is connected to black wires that go to the start button and is jumpered over to the direction switch.

Depending on which position the switch is in, the 120VAC on the black wire will be connected to either the purple or pink wire. This voltage will then travel to the respective coil on the reversing contactor allowing the contactor to engage.

COLLECTOR REEL

Print #84610098: The brown wire comes in through the j-pipe from the MCP and goes to brush #4 to go out to the first A-Box. The white wire comes in through the j-pipe from the MCP and goes to brush #1 to go out to the first A-Box.

A-BOX

Print #84610031: The Brown wire “In” and “Out” wires are in series through the disconnect switch and the safety switch. The white wire “In” and “Out” wires join at the disconnect switch.

B-BOX

Print #84610399: The brown wire “In” and “Out” wires are in series through the disconnect switch, the off delay timer (overwatering timer), and the safety switch. The white wire “In” and “Out” wires join at the disconnect switch.

C-BOX

Print #84610029: This is where the brown and white wires are joined to complete the series circuit that goes from the MCP through all tower boxes and back to the MCP to provide neutral to the Contactor.

The Power Circuit

(red, blue, black wires)

The power circuit is powered by the incoming line service, switched at the main disconnect switch.

When the desired contactor of the reversing contactor engages, three phase 480VAC power exits the main control panel and collector reel on the black, red, and blue wires.

These wires are attached to the three main contacts of each contactor in the control boxes. When the alignment circuitry activates the contactor coil, 480V power is supplied to the tower drive motors.

MAIN CONTROL PANEL

Follow along with the [MCP RPM](#) Print in the reference section: The incoming 480VAC goes through the main disconnect and on up to the reversing contactor carried by black wires. Coming out of the reversing contactor, the wire colors are blue, red, and black. The phasing of the blue and black wires changes depending on which side of the reversing contactor is activated.

COLLECTOR REEL

Print #84610098: The red, blue, and black wires all come in through the j-pipe from the MCP and go out to the first A-Box. The red wire goes to brush #9. The blue wire goes to brush #8. The black wire goes to brush #10

A-BOX

Print #84610031: Red, Blue, and Black Wires

- 480 Volt 3-phase wires
- “In” and “Out” wires join at the disconnect switch
- Power also goes through the disconnect switch, the contactor (when activated), and fuse block to the motor. The fuse block is only needed if the system has a 5 H.P. booster pump ordered for the system per the NEC.

B-BOX

Print #84610399: Red, Blue, and Black wires

- 480 Volt 3-phase wires
- “In” and “Out” wires join at the disconnect switch
- Power also goes through the disconnect

C-BOX

Print #84610029: Red, Blue, and Black wires

- 480 Volt 3-phase wires
- Power goes through the disconnect switch, the contactor (when activated), and fuse block to the motor.

The Percent Timer Circuit

(orange wire)

The 120 V power to the percent timer circuit is supplied through an auxiliary contact on each contactor of the reversing controller. When the machine is “latched in”, power is sent to the percent timer by the black wire. When the percent timer engages (depending on the percent timer), power is sent to the contactor coil in the last tower control box. The contactor engages the tower drive motor for the amount of time the percent timer runs. The rest of the towers follow the last tower movement determined by each towers’ alignment basic switch cam setting.

MAIN CONTROL PANEL

Follow along with the [MCP RPM](#) Print in the reference section: The 120V power to the percent timer circuit is supplied through an auxiliary contact on each contactor of the reversing controller. When the machine is “latched in”, power is sent to the percent timer by the black wire. When the percent timer engages (depending on the percent timer), power is sent to the contactor coil in the last tower control box. The contactor engages the tower drive motor for the amount of time the percent timer runs. The rest of the towers follow the last tower movement determined by each towers’ alignment basic switch cam setting.

COLLECTOR REEL

Print #84610098: The percent timer circuit comes in through the j-pipe from the MCP and goes to brush #2 to go out to the first A-Box.

A-BOX

Print #84610031: The orange wire “In” and “Out” join at the disconnect switch. This circuit does not go anywhere else in this box.

B-BOX

Print #84610399: The orange wire “In” and “Out” join at the disconnect switch. This circuit does not go anywhere else in this box.

C-BOX

Print #84610029: The orange wire terminates at the disconnect with the incoming span cable from the B-box and then goes directly to the coil of the contactor. The percent timer setting dictates how long this tower will be energized thus how long it will move. The higher the percent timer setting, the faster the machine will move; for instance, if the percent timer was set at 50 percent, the last tower would be on for 30 seconds and off for 30 seconds. Every tower inside the last tower is a slave to the last tower.

The Alignment Circuit

(purple and pink wires)

MAIN CONTROL PANEL

Follow along with the [MCP RPM](#) Print in the reference section: The forward/reverse switch sends 120 VAC from the black wire to either the purple wire for forward travel or to the pink wire for reverse travel. 120 VAC from the forward/reverse switch on the purple or pink wire will supply power to the appropriate coil on the reversing contactor and will cause that side to pull in. 120 VAC that passes through the activated auxiliary contact on the reversing contactor will go out to the tower boxes on either the purple or pink wires.

COLLECTOR REEL

Print #84610098: The pink and purple wires come in through the j-pipe from the MCP and go out to the first A-Box. The pink wire goes to brush #7. The purple wire goes to brush #6.

A-BOX

Print #84610031: The purple (forward) and the pink (reverse) wires “In” and “Out” join at the disconnect switch. They also go to the alignment switch which will send power on the red wire to engage the contactor.

B-BOX

Print #84610399: The purple (forward) and the pink (reverse) wires “In” and “Out” join at the disconnect switch. They also go to the off delay timer and to the alignment switch which will send power on the red wire to engage the contactor. Power on the red wire resets the off delay timer.

C-BOX

Print #84610029: The purple (forward) and the pink (reverse) wires terminate at the disconnect switch as there is no alignment in the C-Box.

The End Gun Circuit

(yellow wire)

MAIN CONTROL PANEL

Follow along with the [MCP RPM](#) Print in the reference section: The 120VAC end gun circuit is powered by blue/red wire coming from the water supply switch. This circuit goes out on the system via the yellow wire.

COLLECTOR REEL

Print #84610098: The yellow wire comes in through the j-pipe from the MCP and goes to brush #3 to go out to the first A-Box

A-BOX

Print #84610031: The yellow wire “In” and “Out” join at the disconnect switch. This circuit does not go anywhere else in this box.

B-BOX

Print #84610399: The yellow wire “In” and “Out” join at the disconnect switch. This circuit does not go anywhere else in this box.

C-BOX

Print #84610029: The yellow wire terminates in the disconnect switch unless there is an end gun option on the system.

The Ground and Neutral Circuits

(green and white wires)

MAIN CONTROL PANEL

Follow along with the [MCP RPM](#) Print in the reference section: The ground circuit ties the MCP, tower boxes and motors back to earth ground. The ground circuit (green wire) is also joined with the neutral circuit (white wire) to the MCP.

The neutral circuit originates on the 120V side of the step down transformer. It goes to several components in the MCP including the start/safety override switch, the percent timer, the hour meter, and the safety OK light. It then goes out to the system.

COLLECTOR REEL

Print #[84610098](#): The green and white wires come in through the j-pipe from the MCP and go out to the first A-Box. The green wire goes to brush #5. The white wire goes to brush #1.

A-BOX

Print #[84610031](#): The white wire “In” and “Out” join at the disconnect switch. This circuit goes on to provide neutral to the contactor. The green wire provides ground and does not go through the disconnect switch.

B-BOX

Print #[84610399](#): The white wire “In” and “Out” join at the disconnect switch. This circuit goes on to provide neutral to the contactor and the off delay timer. The green wire provides ground and does not go through the disconnect switch.

C-BOX

Print #[84610029](#): The white wire goes through the disconnect switch to provide neutral to the contactor. It also joins the brown wire to complete the safety circuit. The green wire provides ground and does not go through the disconnect switch.

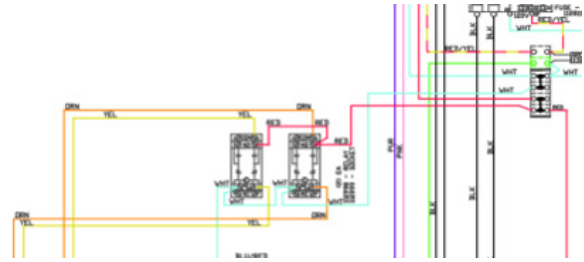
This Page Intentionally Left Blank

Common Options

CHAPTER 4

ISOLATION RELAYS

Isolation Relays for RAMS/Touch: Reinke is now putting isolation relays in all panels to protect the percent timer and end gun inside the panel. The percent timer and the end gun circuit now energize its own relay inside the panel and the transformer power is then sent out onto the system to turn on the functions of the percent timer and the end gun. The point of these relays is to protect the components inside the panel. If one of these circuits were to ground out, instead of damaging the controlling device they would just pop the fuse inside the MCP.



END GUN WITH BOOSTER PUMP

An end gun adds additional irrigation coverage area by spraying water beyond the end of the end boom (the exact distance is determined by water pressure). The end gun can be set to turn on and off as needed. This on/off function can be controlled by a switch, actuated by a stop, or by GPS location if the system is equipped with end of system GPS. A booster pump extends the water throw distance by raising the water pressure at the end gun.

MCP RPM W/ End Gun: The end gun override switch is used to disable the end gun when there is a physical switch that is actuated by a cam wheel or cam plate option. The reason for the switch is to turn the end gun off if it is running at undesirable times.

84610389: The end gun switch mounts on a cam wheel or cam plate at the pivot center. The yellow "out to system" wire is tied to the red wire and the blue wire is connected to terminal #3. This causes the yellow out wire to be grounded when the end gun switch is in the off position. This prevents any induced voltage to be able to operate the end gun.

84610446: The yellow wire goes through the pressure switch (which requires good water pressure for the end gun to work) to the coil of the Honeywell relay.

The direction wires (pink and purple) go through normally open contacts to the coils of the reversing contactor. The operation of this reversing contactor maintains the proper phasing for correct rotation of the booster pump when the system changes direction.



LOW PRESSURE SHUTDOWN

MCP RPM W/ Low Pressure Shutdown: The low pressure switch monitors the system pressure at the pivot center telling the main control panel if there is adequate pressure. If the system pressure drops below a preset level, the contacts of the pressure switch will open causing the main control panel to stop the machine because the “latch” circuit is de-energized, causing the reversing contactor to drop out, stopping the system.

There is also an override switch that will allow the machine to run dry when needed. When this switch is put into bypass, the pressure switch will be ignored; and when this switch is in auto, then this switch will be reading the pressure switch.

AUTO STOP

The picture to the right illustrates the auto stop / auto reverse mechanism that includes the physical barricade, auto stop / auto reverse box, and the arms that trip the switches in the boxes to control the machine.



MCP RPM W/ Cam Plate Stop Overdrive: The panel park switch is an option that can be ordered on a system. If this option is ordered, there will need to be an 11 ring collector ring. This switch will bypass a physical switch that would be located on a cam wheel or cam plate.

84610389: The auto stop switch is a physical switch on a cam wheel or cam plate. The brown (safety) wire is tied in series through the auto stop switch, causing the system to shut down if tripped.

84610550: The brown wire is run through the tower auto stop switch in series to interrupt the safety circuit if the auto stop mechanism is actuated.

84610432: The brown wire is run through the tower auto stop switch in series to interrupt the safety circuit if the auto stop mechanism is actuated. See print note #2: This jumper (white to brown) must be removed if using an auto stop or tower auto reverse on this tower.

AUTO REVERSE

RPM STANDARD MCP W/ Auto Reverse: The auto reverse box is used to allow a standard or basic panel auto reverse.



Standard Auto Reverse Controller Information

Input and Output operations:

- **Line:** The line terminal is the 120VAC power in to the controller box.
 - This power comes from the latched side of the black wire circuit. This means that this box will stay powered up as long as the system is told to run and the safety is good.
- **NEUT:** This is the neutral terminal to the controller box.
- **REV-I:** Reverse input
 - 120VAC power for this input comes from the direction switch when it is in either the reverse or the auto (reverse) positions.
- **FWD-I:** Forward input
 - 120VAC power for this input comes from the direction switch when it is in either the forward or the auto (reverse) positions.
- **SAF-I:** Safety input
 - Neutral is tied to this input when the safety is good and the safety relay is pulled in. This input monitors the brown safety wire, which is on the start button and the neutral side of the reversing contactor. This means that when the start button is pressed, it can override the safety, allowing the system to be overridden. This is the input that pulses to give the controller the indication that it is time to reverse directions. If this input is not good, then the WEL-O / constant output will not pull in and the system will not latch in. Also, since the safety input is monitoring contacts off of the safety relay instead of monitoring the safety on the terminal strip, it is a lot less susceptible to stray voltages that may be on the system.
- **WEL-O/CNST:** Constant output
 - This output comes on when there is power on either REV-I, FWD-I, or both (in auto reverse), and the SAF-I, safety input, is good. This output then energizes the relay that safely “latches in” the black wire circuit.
- **REV-O:** Reverse output
 - This output comes on when REV-I is energized, or when both FWD-I and REV-I inputs are energized (auto reverse mode) and the machine is running in the reverse direction.
- **FWD-O:** Forward output
 - This output comes on when the forward input is energized, or when both FWD-I and REV-I inputs are energized (auto reverse mode) and the machine is running in the forward direction.
- **ST-NO or A/R-NO (normally open) & ST-C or A/R (common):** Auto reverse contact terminals
 - This is the set of dry contacts that override the “latch in” safety relay contacts on the 120VAC side of the safety circuit, typically the gray and light blue wires. When running in auto reverse mode, both the FWD-I and REV-I inputs are on. This closes this set of contacts, overriding the “latch out” function of the safety relay so that when the limit is pulsed, the safety relay can drop out, then pull right back in, which in turn, through the neutral safety contacts on the safety relay, pulses the SAF-I input to the auto reverse controller. If this happens in less than approximately one second, the controller will drop the direction output currently on, dropping out the reversing contactor, and then power up the opposite direction output on the controller, pulling in the reversing contactor, starting the machine in the opposite direction. If the selector switch is in FWD or REV, then these contacts are open, allowing the safety circuit to “latch out” when the limit is tripped. This will drop the power to the latched side of the constant, or black wire, circuit, dropping the power to the auto reverse controller, which drops out the direction output and the reversing contactor, shutting down the system.
- **FWD & REV Lights:** These lights give an indication of which output is being turned on. If either of these lights is on, it is also an indication that there is power to the controller. This is especially handy when troubleshooting and in the auto reverse mode since the controller will run in the same direction as it was last run. The lights are on their own circuit, so if the light is on, it may not necessarily mean that the output is on, only that it should be on.

84610434: The Honeywell relays (PN 112969) are actuated by a switch (PN 113025) mounted on a cam wheel or cam plate. When the machine is moving forward, it trips the switch opening the contacts from terminal #5 to terminal #6 causing the safety circuit to open for a brief second. When the main control panel senses this blip on the safety circuit, it takes the voltage off the direction wire (purple) and the safety is restored by the normally closed contacts at terminals #4 and #5 on the right hand Honeywell relay. When going in reverse, the same events take place but on the contacts to terminal #3 and #4 on the 113025 switch. This time the safety is restored through the left hand Honeywell relay with the pink wire to the coil.

84610148: This is an A-box with a tower auto reverse mechanism and box installed on it. The tower box works the same way as a standard A-box with the addition of a 7 conductor span cable wired in. The pink and purple wires of the 7 conductor cable wires into the same place as the pink and purple of the regular span cables. Yellow of the 7 conductor cable is wire nudded to the brown out wire. The brown wire from the 7 conductor is wired in place of the “brown out”, which is wire nudded to the yellow. The orange wire is not used. The white wire is tied to the rest of the neutrals and the green wire is tied to ground. The reason for the brown and yellow wiring in this application being different from conventional wiring is so that we can keep the safety circuit as a series circuit. Since the safety circuit originates at the C-box, it will enter on the “brown out” which is tied to the yellow wire on the 7 conductor. It then goes into the auto reverse tower box and returns on the brown 7 conductor wire. From there it goes through the safety limit switch and back to “brown in” and continues to the previous A-box or the collector reel depending on which A-box tower this is installed on.

84610263: The difference between this C-box and a standard C-box is the addition of the 7 conductor cable that goes to the tower auto reverse switch box. Note #1 indicates that the white to brown jumper must be removed if the tower auto reverse is mounted on this tower.

84610279: The auto reverse switch box is the tower box installed on the auto reverse mechanism (the arms over the wheels on the tower). The pink and purple wires supply 120 VAC to the coils of the Honeywell relays depending upon the direction of travel. The brown wire is the safety return wire back to the tower box above it. It runs through a series of contacts that open or close depending on the status of the “arms” in relation to the barricade. The yellow wire is the safety supply and is terminated on the bottom safety limit switch. The white wire supplies neutral to the coils of the Honeywell relays.

When the system is moving forward, the purple wire is energized which actuates the left hand Honeywell relay. The safety is coming in on the yellow wire. It is tied to the brown/yellow wire through the bottom switch. It then goes to the right hand Honeywell relay where it travels through the middle switch and goes from brown/yellow to brown/white. Next it goes to the left hand Honeywell relay where it can't go through the normally closed contacts because the relay is energized and the contacts are still open. The brown/white wire goes to the top switch and ties to the brown wire, returns to the disconnect switch, and back to the tower box. This completes the series circuit. When the arm hits the barricade, it will trip the top switch. It trips in a clockwise rotation at 6°. This will cause the safety to open. The main control panel will sense this and de-energize the forward direction. The safety will return via the normally closed contacts on the left hand Honeywell relay (#4 and #5). The main control panel will see the safety come back and will then energize the reverse circuit if the panel is set to auto reverse.

The system will travel in the reverse direction until it hits the reverse barricade, and it will trip the middle switch. It trips in a counter-clockwise direction at 6°. The contacts in the right hand Honeywell relay (#4 and #5) will allow the safety to return. The bottom switches act as a limit. They will open if the arm trips more than 8° either clockwise or counter-clockwise. This will immediately open the safety circuit and shut the system down.

MAIN CONTROL PANEL OPTIONS

Use the [MCP RPM W/ Standard Options](#) print located in the reference section to view the following common options.

Frost Control:

The frost control override switch allows the user to bypass the frost control if needed. The frost control device is what shuts down the system when the temperature goes below a preset level allowing the system to not apply water if there is a chance of the water freezing. The thermistor is basically a thermostat that tells the controller what the outside temperature is.

Low Voltage Relay:

The low volt relay is a three-phase voltage monitor that will keep the machine from running if the incoming three-phase is not adequate. The main control (120 volt) power is ran through a set of contacts that will open if the low volt relay does not see appropriate voltage.

Auto Restart:

The auto restart relay monitors the incoming control voltage. When the voltage is not present, the relay knows that the power to the machine is off. When it sees the 120 volts return, it will then activate the relay and start the machine. There is also a set of contacts on the relay that will bypass the pressure switch for a set amount of time to allow the pump to build pressure so that there is not a pressure shut down right after the machine restarts.

Pressure Restart:

The pressure restart relay works like the auto restart relay; but it is not reading the power, it is reading the pressure switch. When pressure becomes great enough to close the pressure switch, it will send 120 volts to the pressure restart relay and the relay will then start the machine. This allows whoever is operating the machine to start the well and not have to go to the main control panel and press the start button. In order for this function to work, the system power switch must be in the on position, and the pressure bypass/auto switch must be in the auto position.

See following print:

For Power Auto Restart, the timer relay on the left, the relay is turned on when system power is restored to the system via the red wire on the coil at the top of the relay. Once this happens, the “Start” and “Pressure” contacts close since we are using the open and common contacts on the relay for both of them. The start contacts have the light blue and grey on them, paralleling the contacts on the safety relay, for 3 seconds and then they reopen. This sends power to the latch circuit which starts the system if the safety is good. The pressure contacts bypass the pressure switch with the black and grey wires. This side of the relay is closed for 10 minutes after the relay is energized.

The Pressure Restart relay, the one on the top right works the same as the power restart but the trigger to energize the relay is when the pressure switch closes and sends power to the timer. The start relay is closed for 3 seconds and that is what sends power to the latch circuit and starts the system. The pressure switch will be bypassed for 10 minutes as well to make sure there is no fluctuation in the pressure on start up.



Lightning Arrestor:

The lightning arrestor is installed on the load side of the main disconnect. Its function is to take high voltage and current that would be caused by a lightning strike and absorb it so that damage will be kept to a minimum to the electronic components on the system.

Large Service Wire Adapter:

Larger service wire adapters are used when the incoming power wired into a main control panel is larger than #6 AWG. The largest size of wire that the main disconnect can accept is #6. The large service adapter is capable of #14 to 2-0.

GFI / RCD:

The GFI, which means ground fault interrupter, or RCD, which means residual current device, is installed in a main control panel to monitor the three-phase power to make sure that the load from the system is balanced. If one of the devices senses that the load is not balanced properly, then the device will open up the contacts not allowing the power to get through thus shutting down the system.

Remote Panel:

This is a pivot junction box. It is used when the main control panel is located at a remote location (not at the pivot center). It is a junction box with a disconnect that will allow the span cable from the collector ring to be disconnected from the span cable coming from the main control panel.



WRAP SPAN

84610339: This is a print of the 15° pivot tower box that would go in place of a regular A-box on a 15° wrap span.

142878: Operation of the 15° wrap span (#84610339): The Honeywell relay (PN 112969) is an isolation relay to provide power to the second switch from the top. That switch controls a valve that will turn sprinklers off when in wrap mode. The top switch is the tower run switch. The pink and purple wires on the switch do not come from the disconnect like normal tower boxes. They run from the disconnect switch on orange for the forward wire and yellow for the reverse wire to the terminal block. From there, they go to the 15° wrap switch. When the switch is in its normal state, orange ties to purple via terminals #3 and #4, and yellow ties to pink via terminals #5 and #6. From there they go back to the terminal block and on to the run micro switch. They then tie to the red wire that pulls in the contactor to run the tower. By running the direction wires through the 15° wrap span, the contacts will open when the tower hits the wrap barricade. This will prevent power on the direction wires from getting to the micro switch to engage the contactor to make the tower move. The third switch is the wrap safety switch. It is tied into the series safety circuit and will open the contacts and shut down the system if the towers wrap too far. The bottom switch is the same safety switch you would see in a regular tower box. This is used when the system is in normal operation (not wrapping). When the machine gets into a wrap situation, it will close either #7 to #8 or #1 to #2 depending on the direction on the 15° wrap switch. This bypasses the safety on the bottom switch and allows the tower to walk out of normal alignment. In summary, when the system walks into the wrap barricade, the switch on the mechanism will bypass the direction wires on that tower and any tower back toward the pivot center. This will prevent them from moving. It also bypasses the regular safety switch from shutting down the machine.

84610449: This is a print of the 90° pivot tower box that would go in place of a regular A-box on a 90° wrap span. The Honeywell relay is an isolation relay to provide 120 VAC to the second cam from the top to control a valve to shut off sprinklers on the spans that are not wrapping. The top switch is the tower run switch and is responsible for pulling in the contactor with the red wire while the machine is not in a wrap situation. The third switch from the top is the wrap safety switch. This switch will open the safety circuit if the machine wraps too much in either direction. It is set at the factory to 45° both directions. The bottom switch is the regular safety switch and is set to trip at 3° in both directions.

When the system engages the wrap barricade, the bottom safety switch is bypassed in the 90° wrap box located on the AR / AS mechanism. The voltage on the direction wires is also interrupted in the 90° wrap box resulting in that tower not moving.

84610450: The 8 pin relays (PN 113244) are time delay relays for the safety to prevent the system from dropping out as it engages the barricade.

The top switch is the forward bypass switch. When the system engages the barricade, the top switch is open and doesn't allow the tower to move.

The second switch is the reverse bypass switch.

Switches #3 and #4 are safety switches, that will shut down the system if that tower doesn't stop when it engages the barricade.

The bottom switch is the safety override and is the first cam to trip in both directions. When it is tripped, it will bypass the regular safety in the tower box, allowing the system to wrap.

SINGLE PHASE

MCP RPM Single Phase: This is a single phase main control panel print. The only difference between this and a three phase main control panel is that there is no transformer. The reason for no transformer is because with a single phase system there is already 120 volts to ground on the incoming legs so the need for a transformer is gone. There is a fuse coming off of one of the incoming lines to protect the control components of the system.

84610552: On a single phase system, changing directions is not as simple as swapping the phasing of the power wires. To change the direction of rotation on a single phase motor, you have to change the way the windings of the motor are connected together. Reinke has installed a Honeywell relay in all single phase tower boxes that will change the motor winding connections depending on the desired direction of rotation.

84610306: The B-box is the same as a regular B-box with the exception of the relay and terminal block that make it capable of changing directions.

84610307: The C-box is just like a regular C-box but has the relay and terminal strip in it to allow for travel in both directions for the single phase system.

This page intentionally left blank.

Advanced / PAC III

CHAPTER 5

RPM ADVANCED

In the reference section of this book, you will find three schematics that show the internal wiring of an advanced panel. These are ADV MCP WITH NO OPTIONS; ADV MCP WITH PS, GPS; and ADV MCP WITH MOST OPTIONS.

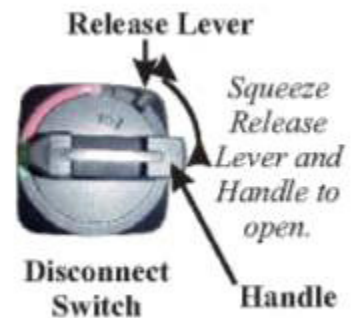
MAIN CONTROL PANEL LAYOUT

The Reinke Precision Management Advanced Main Control Panel consists of a Reinke Precision Application Controller III, also referred to as a PAC III, as the main control module. All the components pictured below are located inside of the outer door.



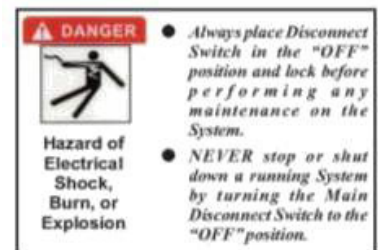
Main Disconnect Switch

The main disconnect switch controls the 480 or 240 VAC power supply to the system. The switch is mounted on the back plate of the main control panel and is interlocked with a disconnect handle on the main control panel inner door. The switch must be in the “On” position for the system to operate. To open the inner panel door and view the inside of the control panel, place the disconnect switch in the “Off” position. If an operator is doing any maintenance, or it is the end of the season, a padlock in the “Off” position should be used. This will ensure that the system does not operate unintentionally. Do not turn the main disconnect switch to the “On” position with the inner door open.



System Power Switch

The system power switch must be in the “On” position for the system to operate in either direction. When the switch is in the “Off” position, the system will not operate. Place the switch in the “Off” position to shut down the system.



Hour Meter

The hour meter displays the number of hours that the system has operated (excluding the pump) since it was new.



Voltmeter

The voltmeter is mounted on the main control panel inner door and monitors the voltage between two incoming legs of the power supply. This meter should read 456-504 VAC at 60 HZ, (380-420 VAC, 50 HZ) on 480 VAC 3 phase systems or 230-250 VAC at 60 HZ on 240 VAC single phase systems. In no event should the operational voltage read outside these parameters.



Generator Switch (optional)

If your system is equipped with an engine driven generator or pump, the generator switch is used to bypass the engine shutdown during start up. The “Start” position is used when starting the engine. Then the switch is turned to “Run” after the system is running. If this switch is left in the “Start” position, the engine shutdown is bypassed and the engine will not kill in the event the system stops.



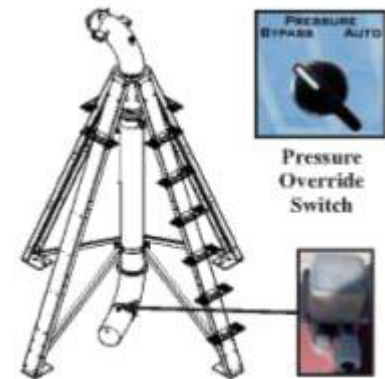
Water Supply Switch

The water supply switch breaks the well kill circuit and allows the system to operate without the “pump on” for REA systems. When the switch is placed in the “Off” position, the well kill circuit is open and the pump will not start when the system is started. When the switch is placed in the “On” position, the well kill circuit is closed and the pump will start when the system is started. For generator systems (engine driven well), always leave the water supply switch “On”.



Pressure Override Switch (optional)

If your system is equipped with an optional low pressure shutdown, a two-position switch labeled “Pressure Bypass/Auto” will be included on the main control panel. In the “Bypass” position, the pressure switch is bypassed. Use this position when you start the system or want to move the system without supplying water to it. In the “Auto” position, the pressure switch is activated and the system will shut down if the pressure falls below the preset level. Be sure the switch is in the “Auto” position after pressure has built in the system. This assures proper function of the switch during water application and normal application.



Direction Switch

The direction switch is used to select the system’s direction of operation. To run the system clockwise, the switch must be placed in the “Forward” position before pressing the start button. To run the system counterclockwise, the switch must be placed in the “Reverse” position before pressing the start button. In the “Auto” position, the system will reverse direction when the tower auto-reverse switch is tripped by the physical barricade or the system reaches the programmed barrier from the setup menu. If the direction switch is placed in the “Forward” or “Reverse” position, the system will stop when the tower auto-reverse switch is tripped by the physical barricade or the system reaches the programmed barrier from the setup menu; thus acting as an automatic system stop. To change the system’s direction between the reversing trip positions, move the direction switch to the “Forward” or “Reverse” position and press the start button. The direction switch may be moved to the “Auto” position after “Forward” or “Reverse” so the system reverses when it contacts the barricade. If the system is started with the direction switch in the “Auto” position, the system will move in the direction it was last moving. Note: Auto-reverse option must be enabled in the PAC III setup menu for the system to start with the direction switch placed in the “Auto” position.



Safety OK Indicator Light

The safety OK indicator light will illuminate when the safety circuit is complete and the system is running. If the safety circuit is open, the problem must be corrected before the system can safely start. One reason for a “no go” safety circuit could be a stuck tower that has caused misalignment. Call your dealer or authorized service person for assistance.



Start Button

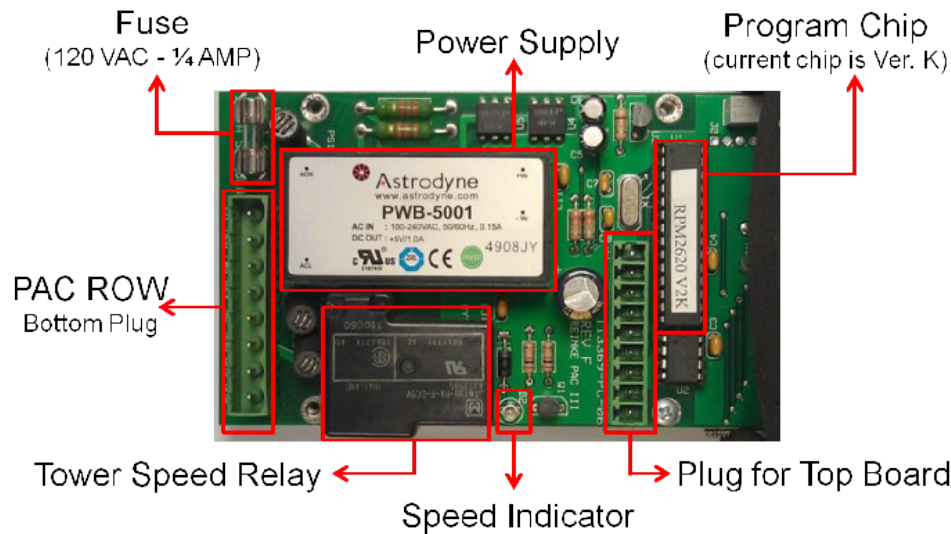
After the direction switch has been placed in the “Forward”, “Reverse”, or “Auto” positions, movement of the system is initiated by engaging the start button. This activates the appropriate magnetic contactor in the main control panel. The contactor “latches” if all safety circuits indicate “closed” or “go” condition. You can usually hear the contacts engage when the button is engaged. However, if after you release the button, another “thump” is heard, the contactor did not stay engaged (latched) and the system will not continue to run. This indicates the safety circuits have an “open” or “no go” condition that needs correcting. Do not hold your finger on the button, forcing the system to override the safety circuit. This can and will cause damages to your system in as little as two or three seconds. One reason for a “no go” safety circuit could be a stuck tower that has caused misalignment. Call your dealer or authorized service person for assistance.



The precision application controller (PAC III) is a micro-processor controlled module that controls the following functions:

- Digital Percent/Depth of Application Control
- Displays Position (degrees) utilizing Resolver or Navigator GPS Option
- Displays Status Updates
- Auto-Restart
- Pressure Shutdown (w/optional Pressure Switch)
- Frost Shutdown (w/optional Temperature Probe)
- Frost Restart (w/optional Temperature Probe)
- Auto-Reverse (by degrees or w/optional Tower Auto-Reverse)
- Stop Delay
- Remote Communication (optional)
- 2 End Gun or Auxiliary Programs
- 2 Sector Programs (up to 10 sectors per program)
- Park Command (by degrees)

PAC III BOARD



This is the PAC III main circuit board. On this board, there are the following components:

1/4 Amp Fuse: This is used to protect the electronic components of the board.

Power Supply: This is used to convert the incoming AC voltage to DC voltage, controlling the electronic components of the PAC timer.

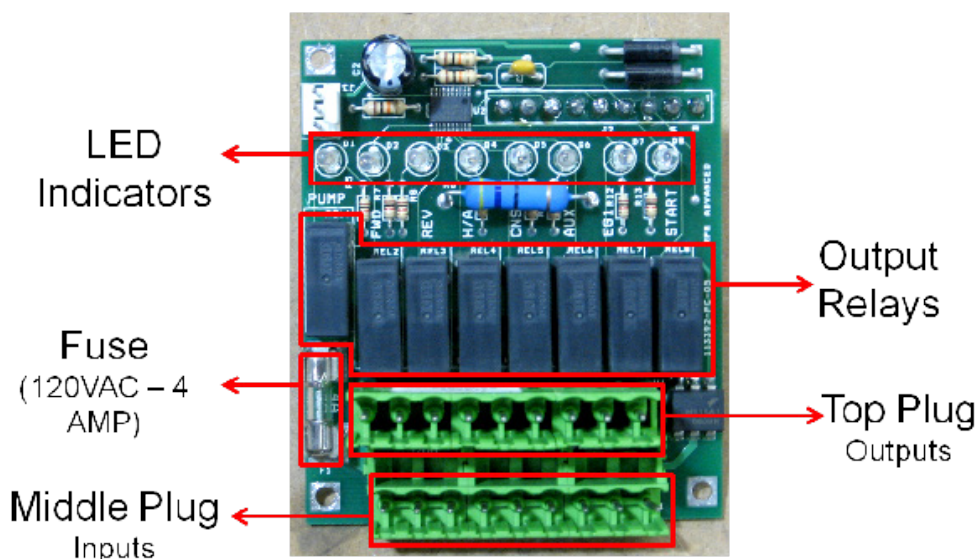
Program Chip: This chip contains the firmware used to control the PAC timer. Version K is the current version installed in all new PAC timers.

Top Board Plug: This is where a top board is connected to the PAC board to allow for communication between boards. There are two types of top boards available: the PAC II end gun board and the advanced board.

Speed Indicator: This is an LED light that displays when the percent timer output is active.

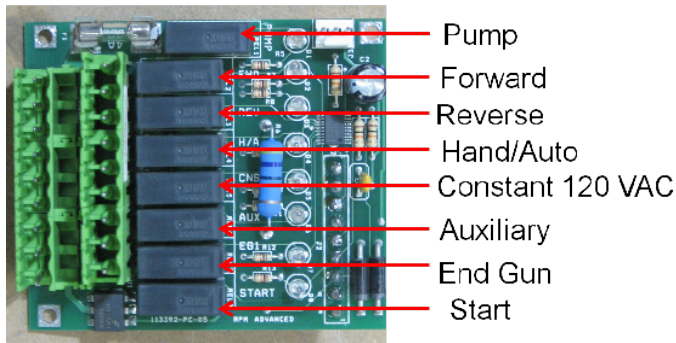
Tower Speed Relay: This relay closes contacts for a set amount of time depending on the depth selected on the PAC timer, allowing the percent timer signal to be sent to the last tower.

PAC Bottom Row Plug: This contains the inputs and the percent timer output.



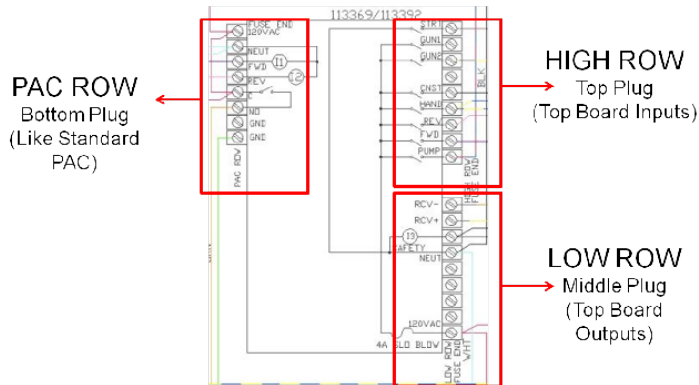
Above, this is the PAC III advanced relay board. There are eight different output relays controlling different functions of the machine. The LED indicator lights illuminate when the respective output relays are active. The 4 amp fuse, located beside the top plug, is used to fuse all the outputs of the advanced board. The top plug and middle plug are used for output and input connections.

Output Relays



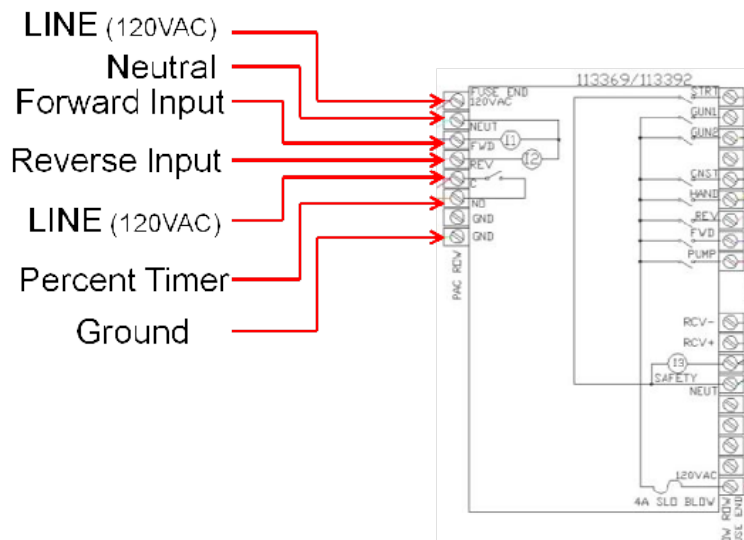
The 8 output relays on an advanced board include the pump relay (which controls the well kill relay), and the forward and reverse output relays (which control the coils of the reversing contactor, allowing the machine to operate in desired direction). The hand/ auto relay is used for the pressure bypass function. The constant 120V output relay supplies 120V to the latching black wire circuit. The auxiliary output relay is used to supply voltage to a second end, chem pump, or any other auxiliary device. The end gun output relay is used to supply voltage to the yellow end gun wire. The start output relay is used to supply neutral to the reversing contactor coils during the start pulse.

PAC III Plug Layout



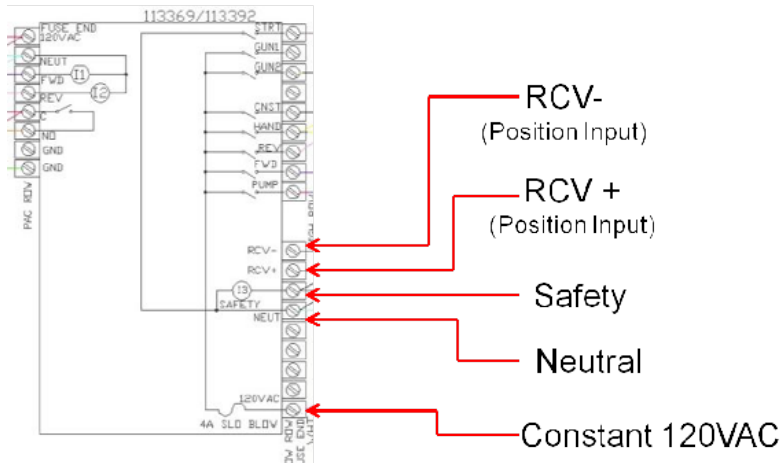
This image depicts the PAC timer as it is shown on an electrical schematic. The PAC ROW plug is the bottom plug on the main PAC timer circuit board. The HIGH ROW and LOW ROW plugs are located on the advanced relay board.

PAC Row (bottom plug)



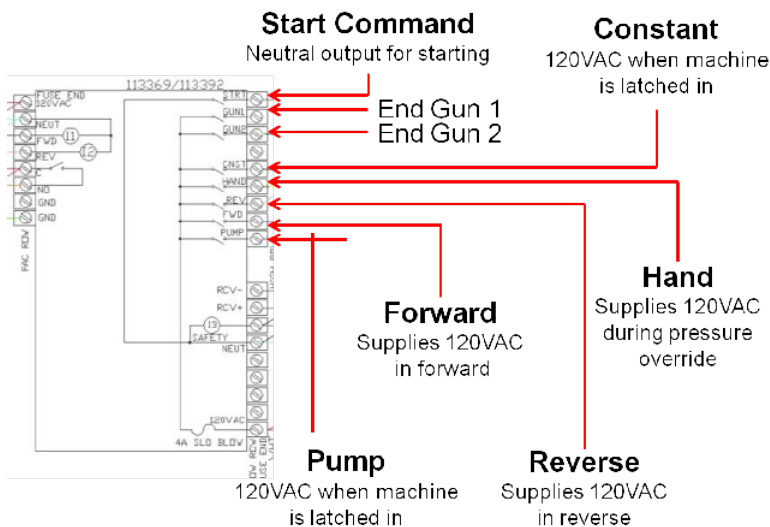
On the bottom plug is the input for the incoming line voltage, which is used to power the PAC timer and is also jumpered over to the other line terminal to input 120V for the percent timer output. Below the top line terminal is the neutral. There are also forward and reverse inputs that come from the direction switch to inform the PAC board which direction the machine is set to run. The percent timer terminal is an output which controls the end tower. The last terminal on the bottom plug is the ground terminal.

Low Row (middle plug)



The low row, or middle plug, is located on the advanced relay board. On this plug, there are terminals RCV- and RCV+, which are used to receive data from positioning devices such as a resolver or end of system GPS. Also located on this plug is a safety input, which comes from the gray wire off of the safety relay. The last two terminals are constant 120V and neutral, which are used to power the advanced board.

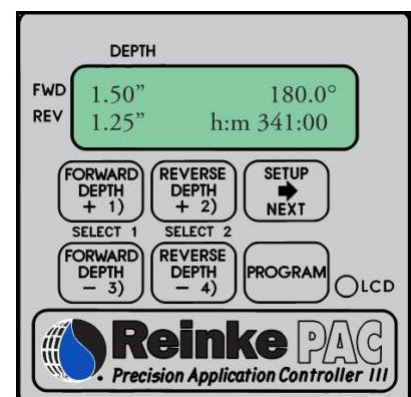
High Row (top plug)



The top plug is also called the high row. All of the terminals on this plug are outputs, which were previously covered on the output relay.

PAC III NORMAL OPERATION

The PAC display will be on unless the main disconnect switch on the main control panel is turned to the “Off” position. The normal operation screen shown here will be the default screen. From the normal operation screen, the forward (clockwise) or reverse (counterclockwise) depth may be modified at any time while the system is running or stopped. The PAC unit uses four input keys to set the forward and reverse application depths. To adjust the depth in the forward and/or reverse direction(s), press the desired key as indicated by the image shown. The depth will change by 0.01” increments. The hours per revolution or feet per hour will adjust as the depth changes on the screen. The speed will be automatically calculated. Note: Increasing the depth applied decreases the end tower run time, which increases the hours per revolution for pivot systems and reduces the feet per hour rate on lateral move systems. Also, be aware that other factors such as (but not limited to) wheel slippage, improper tire pressure, worn nozzles, or adverse field conditions can affect the amount of water that is applied.



Press the “Setup/Next” key and hold for 3 seconds to enter the setup mode.

Press the “Program” key and hold 3 seconds to enter the program mode.

Press the “Forward Depth + 1” key to increase depth in the forward (clockwise) direction.

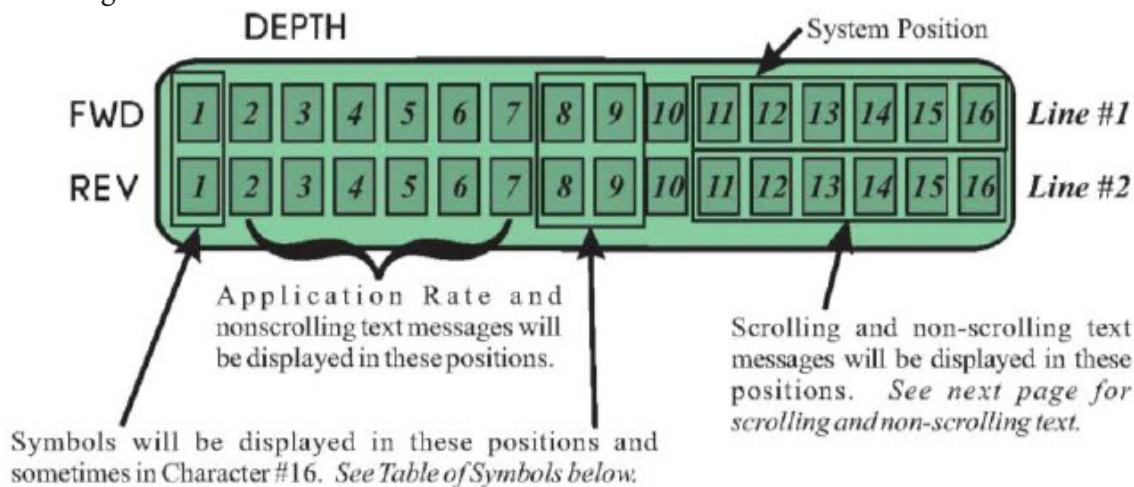
Press the “Reverse Depth +2” key to increase depth in the reverse (counter-clockwise) direction.

Press the “Forward Depth – 3” key to decrease depth in the forward (clockwise) direction.

Press the “Reverse Depth – 4” key to decrease depth in the reverse (counter-clockwise) direction.

The screen (LCD) can be adjusted to correct any viewing problems that may occur. The screw adjuster labeled LCD can be adjusted with a small screwdriver any time the PAC Timer is powered. Simply adjust the screw adjuster until the screen is most visible for your application. The height of the main control panel and the angle it is sitting on the pivot center will affect the view of the screen.

Hint: For rapid adjustments, press and hold the depth key. The display will rapidly scroll through the adjustment range.



During normal operation, the screen will show various types of information. This section will describe in detail what is shown on the display in both pivot and linear modes. This display screen consists of two lines with each line capable of displaying 16 characters. Characters are numbered in this image to indicate position in the Table of Symbols, which is shown below. These symbols are used on the PAC timer display screen to inform the user the current status of the machine.

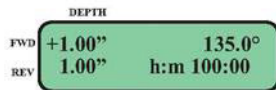
Symbol	Definition	Position
→	Indicates the System is running in the selected direction.	1 - Either Line
+	The End Tower is moving in the selected direction.	1 - Either Line
>	Indicates direction System will move when started.	1 - Either Line
E ₁	Indicates End Gun #1 is "ON".	8 - Top Line
A ₁	Indicates Auxiliary #1 is "ON".	9 - Top Line
100%	Indicates the Minimum Depth has been set and the System is moving at 100%.	1 - Either Line
OFF	Indicates the Maximum Depth has been set and the System will not move.	1 - Either Line
⌚	Indicates the System is waiting (1 to 15 minutes) after an automatic direction change has occurred. The number will count down as time expires and the hourglass will flash. "dly ??/??" will be displayed in Characters #12-#16. The first numbers indicate the time elapsed while the second set of numbers indicates the time to elapse before the System will restart.	1 - Either Line 12 - 16 Bottom Line
a	Indicates System is within an "active" Programmed Sector that is turned "ON".	1 - Either Line
flashing °	Indicates PAC III Module is sending and receiving data from a Communication Device.	16 - Top Line
X	Indicates PAC III Module has lost Resolver Signal.	16 - Top Line
0	Indicates improper PAC setup.	1 - Either Line

Scrolling and Nonscrolling Text Messages

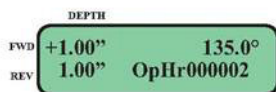
Scrolling and nonscrolling text messages will appear in the lower right section of the display screen.

Nonscrolling text messages will also appear in the lower left section of the display screen in some situations.

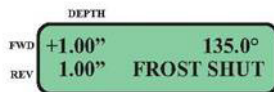
The screens below illustrate the possible scrolling and nonscrolling text messages. The displayed message will vary depending on the options running at that time.



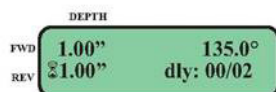
The scrolling "h:m 100:00" is the amount of time in hours and minutes it will take the system to make a complete revolution at the depth specified. *(Center Pivot Systems only)*



The scrolling "OpHr000002" is the operating hours or the amount of time in hours the system has been moving.



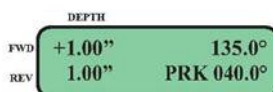
The scrolling "FROST SHUT" means the system has shut down because the temperature is below the current "FROST STOP" setting from the SETUP menu. *(Systems with Temperature Probe only)*



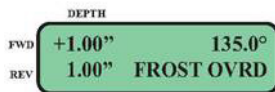
The nonscrolling "dly 00/02" indicates the system is waiting 2 minutes after a direction change. The "0" indicates 0 minutes have elapsed of the 2 minutes. The flashing hourglass symbol indicates which direction the system will move next.



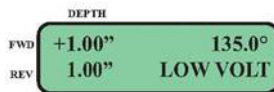
The scrolling "015%" is the percent of time the last tower is moving.



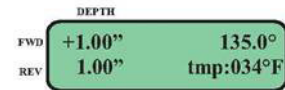
The scrolling "PRK 040.0°" indicates the park program has been turned "ON" and set to shut down at 40.0°.



The scrolling "FROST OVRD" indicates the system was shut down because the temperature was below the current "FROST STOP" setting from the SETUP menu. Pressing the start button will allow the system to move "dry" without the pump "ON". The pump will start once the temperature is above the current "FROST STOP" setting from the SETUP menu. *(Systems with Temperature Probe only)*



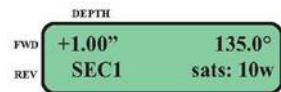
The nonscrolling "LOW VOLT" indicates low voltage has caused the voltage percent indicator to fall below the voltage percent safety indicator entered in the setup menu.



The scrolling "tmp:034°F" is the current temperature. *(Systems with Temperature Probe only)*



The scrolling "sats:10w" indicates the number of GPS satellites the PAC III is receiving signal from. The "w" indicates the PAC III is receiving the WAAS signal for differential correction. *(Systems with GPS only)*



The nonscrolling "SEC1" indicates sector program #1 is turned "ON".

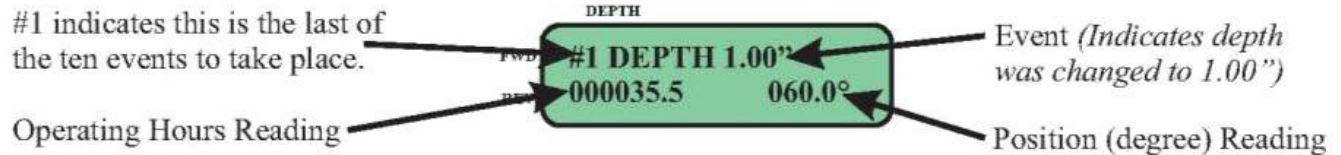


The nonscrolling "Start:029s" indicates low voltage has caused the voltage percent indicator to fall below the voltage percent safety indicator entered in the setup menu and the system shut down. The voltage has since came up and the system is counting down before it restarts.

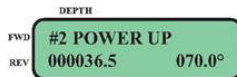
Logs

The PAC III timer has a unique feature that logs the last 10 events. Press the "Setup/Next" key and hold for 3 seconds from the normal operation screen to advance to the "SETUPBUILD2K" screen shown to the right.

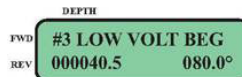
Each log screen has the operating hour and position (degree) readings where and when the event occurred. Press the "Reverse Depth +2" key to view the first of ten log screens.



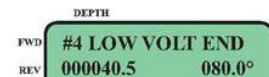
The other possible log screens that may be displayed are illustrated below and on the next page in random order for explanatory purposes. Press the "Setup/Next" key to view the next log screen.



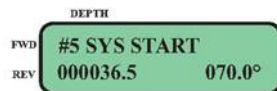
The "POWER UP" log screen indicates the main disconnect switch was turned "ON" at the displayed operating hour and position (degree) readings.



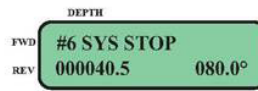
The "LOW VOLT BEG" log screen displays the operating hour and position (degree) reading where low voltage has caused the voltage percent indicator to fall below the voltage percent safety indicator entered in the setup menu.



The "LOW VOLT END" log screen displays the operating hour and position (degree) reading where the voltage has caused the voltage percent indicator to rise above the voltage percent safety indicator entered in the setup menu.



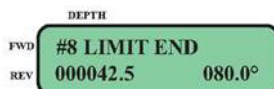
The "SYS START" log screen indicates the system was started at the displayed operating hour and position (degree) readings.



The "SYS STOP" log screen indicates the system was stopped at the displayed operating hour and position (degree) readings.



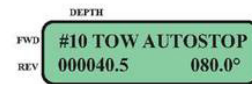
The "LIMIT BEGIN" log screen indicates the system safety circuit was opened at the displayed operating hour and position (degree) readings.



The "LIMIT END" log screen indicates the system safety circuit was closed at the displayed operating hour and position (degree) readings.



The "PARK STOP" log screen indicates the system was programmed to stop and it stopped at the displayed operating hour and position (degree) readings.



The "TOW AUTOSTOP" log screen indicates the tower auto-stop mechanism has contacted the tower barricade at the displayed operating hour and position (degree) readings.



The "TOW AUTOREV" log screen indicates the tower auto-reverse mechanism has contacted the tower barricade at the displayed operating hour and position (degree) readings. *Remember, the auto-reverse option must be enabled for the system to start with the direction switch placed in the "AUTO" position.*



The "PIV AUTOSTOP" log screen indicates the system reached the barrier setting from the setup menu. The system didn't auto-reverse because the direction switch was not in the "AUTO" position. Instead, it stopped at the displayed operating hour and position (degree) readings.

DEPTH	
FWD	#3 PIV AUTOREV
REV	000040.5 080.0°

The "PIV AUTOREV" log screen indicates the system reached the barrier setting from the setup menu. The system auto-reversed at the displayed operating hour and position (degree) readings because the direction switch was in the "AUTO" position. *Remember, the auto-reverse option must be enabled for the system to start with the direction switch placed in the "AUTO" position.*

DEPTH	
FWD	#6 END GUN 1 ON
REV	000036.5 070.0°

The "END GUN 1 ON" log screen displays the operating hour and position (degree) readings where the system reached a position (degree) setting in the end gun #1 program where end gun #1 was programmed to turn "ON".

DEPTH	
FWD	#9 END GUN 2 OFF
REV	000036.5 070.0°

The "END GUN 2 OFF" log screen displays the operating hour and position (degree) readings where the system reached a position (degree) setting in the end gun #2 program where end gun #2 was programmed to turn "OFF".

DEPTH	
FWD	#2 PRESSURE SHUT
REV	000036.5 070.0°

The "PRESSURE SHUT" log screen displays the operating hour and position (degree) readings where the system shut down due to low pressure.

DEPTH	
FWD	#4 FORWARD DIR
REV	000042.5 080.0°

The "FORWARD DIR" log screen indicates the system direction was switched to forward at the displayed operating hour and position (degree) readings.

DEPTH	
FWD	#7 END GUN 1 OFF
REV	000036.5 070.0°

The "END GUN 1 OFF" log screen displays the operating hour and position (degree) readings where the system reached a position (degree) setting in the end gun #1 program where end gun #1 was programmed to turn "OFF".

DEPTH	
FWD	#10 REM START
REV	000036.5 070.0°

The "REMSTART" log screen displays the operating hour and position (degree) readings where the system was started by a remove device such as a Reinke Phone Link or remote control and monitoring software.

DEPTH	
FWD	#5 REVERSE DIR
REV	000045.5 080.0°

The "REVERSE DIR" log screen indicates the system direction was switched to reverse at the displayed operating hour and position (degree) readings.

DEPTH	
FWD	#8 END GUN 2 ON
REV	000036.5 070.0°

The "END GUN 2 ON" log screen displays the operating hour and position (degree) readings where the system reached a position (degree) setting in the end gun #2 program where end gun #2 was programmed to turn "ON".

DEPTH	
FWD	#1 REM STOP
REV	000036.5 070.0°

The "REM STOP" log screen displays the operating hour and position (degree) readings where the system was stopped by a remote device such as a Reinke Phone Link or remote control and monitoring software.

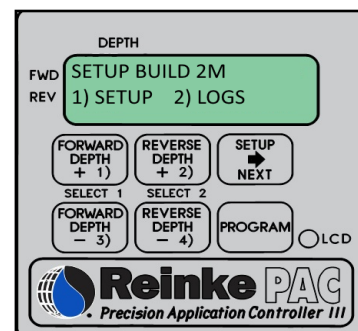
STARTING THE SYSTEM

The following procedures are for typical installations. If your installation is non-typical, consult your dealer or service person. See appropriate system manual for instructions on optional equipment. **If the system has just been installed, the generator or its direction of rotation has been changed, or the power service has been worked on or changed; have your service person check the system for proper phasing.**

1. Observe field and system conditions to be sure the system and system's path are clear of people and any other obstacles.
2. If you want to run water with the system, ensure that the electric pump control is enabled and safe to start as per the pump manufacturer's recommendations. Then turn the pump disconnect switch to the "On" position.
3. If your system utilizes an optional generator switch to control a combustion engine driven well and/or generator, turn the generator switch to the "Start" position. Start the engine following all engine manufacturer's safety and operating instructions.
4. Turn the main disconnect switch to the "On" position.
5. Observe the voltage on the main control panel voltmeter. This meter should read 456-504 VAC, 60 HZ (380-420 VAC, 50HZ) on 480 VAC systems or 230-250 VAC, 60 HZ on 240 VAC systems. In no event should the operational voltage read outside these parameters.
6. Turn the system power switch to the "On" position.
7. Ensure that the "Safety OK" indicator light is on. If it is not on, the safety circuit is open.
8. Turn the water supply switch to the "On" position if you want to run water with the system. Turn the water supply switch to the "Off" position if you want to run the system dry.
9. If your system is equipped with an optional low pressure shut down, turn the pressure override switch to the "Bypass" position.
10. Set the PAC III timer to desired speed.
11. Turn the direction switch to the desired direction. "Forward" = clockwise, "Auto" = auto-reverse, and "Reverse" = counterclockwise. Note that the auto-reverse option must be enabled in the PAC III setup menu for the system to start with the direction switch placed in the "Auto" position.
12. Push the start button. You should hear the magnetic contactor engage (snap) and the system should start. If not, recheck this procedure to make sure you have followed it correctly. Release the button. If another snap is heard, a malfunction is indicated. Do not hold the start button in for more than 3 seconds as this will override the safety circuits.
13. If your system is equipped with an optional low pressure shut down, once the system has built up pressure, turn the pressure override switch to the "Auto" position.
14. If your system utilizes an optional generator switch to control a combustion engine driven well or generator, turn the generator switch to the "Run" position.

PAC III SETUP

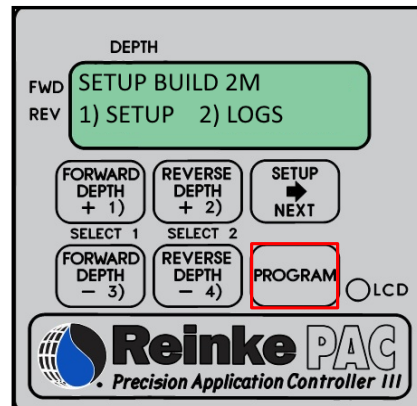
The PAC timer must be properly setup to ensure accurate depth of application. Press and hold the "Setup/Next" key for 3 seconds to enter the setup menu. The screen shown here will be displayed. If any of the setup screens are left idle for more than two minutes, the PAC will return to the normal operation screen without saving any changes. **Do not change setup options while the system is moving or undesirable operation will occur.** Press and hold the "Setup/Next" key for 3 seconds to enter the setup menu and to move through the setup screens. The "Forward Depth" key toggles through the setup menu options. The "Program" key can be used as a back button when in the setup menu. Use the number keys to change numeric values.



To enter the "SETUP" menu press and hold the  key for 3 seconds, Then press  to select 1) SETUP to view and alter setup.

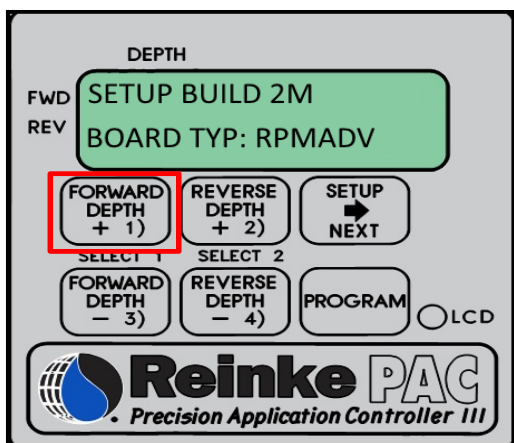
Board Type

The first setting that needs to be set is the board type. To access the board type setting, press and hold the "setup/next" key for three seconds. Once the screen shown below is displayed, press the "program" key to access the board type setting.



To change the Board Type press and hold the  key for 3 seconds, once the screen displays the options shown above press the  key to access the Board type setting.

There are three board types available for selection. The type that needs to be selected will be based on what type of top board the PAC timer is utilizing. For more information on the different board types, see the reference packet. For an advanced panel, the board type will need to be set to RPMADV. If the PAC is being used in a standard panel, PAC II or NONE will be selected.



SETUP BUILD 2M
BOARD TYP: RPMADV

RPMADV

Used when the PAC is being utilized in an Advanced Panel

SETUP BUILD 2M
BOARD TYP: PACII

PACII

Used when the PAC is being utilized in a Standard panel with a PACII End Gun Board

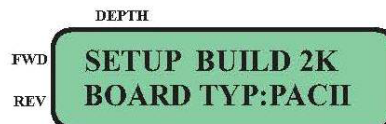
SETUP BUILD 2M
BOARD TYP: NONE

NONE

Used when the PAC is being utilized in a Standard panel with no top board

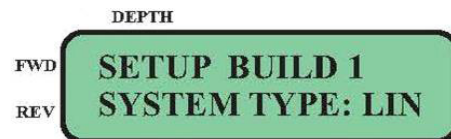
At the Board Type selection screen use the  key to change between

RPMADV , PACII, and NONE



System Type

Press the "Setup/Next" key to advance to the system type screen shown here. Press the "Forward Depth + 1" key to choose between the "PIV" for pivots and "LIN" for linear move machines.



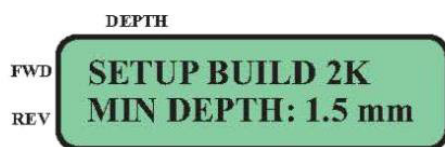
Units

Now press the "Setup/Next" key to advance to the units screen shown below. Press the "Forward Depth + 1" key to choose between "ENGLISH" and "METRIC" units.



Minimum Depth

Now press the "Setup/Next" key to advance to the minimum depth screen as shown here. Press either the "Forward Depth + 1" key to increase or the "Forward Depth - 1" key to decrease the minimum depth by 0.01" increments for "ENGLISH" units or 0.1 mm increments for "METRIC" units. This value is provided on the speed chart located in the main control panel. Use the depth @100% for center pivot systems and depth @90% for linear move systems. If using "METRIC" units, round to the nearest tenth (0.1) of a millimeter.



Metric Units



English Units

Cycle Time

Now press the "Setup/Next" key to advance to the cycle time screen. Press the "Forward Depth + 1" key to choose the desired cycle time among "030S" for 30 seconds or "060S" for 60 seconds. This is the percent timer cycle time that determines how long the end tower will move and stop. See the end tower movement section later in this chapter for calculations on how the base cycle time is utilized. Most center pivots and linear move systems will utilize a 60 second cycle.



Hours Per Revolution

Now press the “Setup/Next” key to advance to the hours per revolution screen for center pivot systems, feet per hour screen for linear move systems with English units, or the meters per hour screen for linear move systems with metric units. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease the hours per revolution setting by one hour increments for pivots or to increase or decrease the feet per hour setting by one foot or one meter increments for linear move systems. Press either the “Reverse Depth + 2” key to increase or the “Reverse Depth – 4” key to decrease the minute setting by one minute increments for center pivot systems or to increase or decrease the feet per hour setting by 0.01 foot or meter increments for linear move systems. This value is provided on the speed chart inside the main control panel.



Position Input

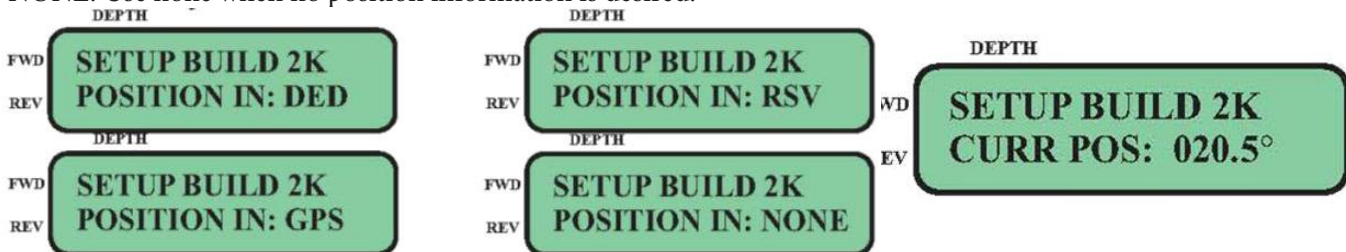
Now press the “Setup/Next” key to advance to the position input screen. Press the “Forward Depth + 1” key to choose from “DED” for dead reckoning, “RSV” for the collector reel resolver option, or “GPS” for the Reinke navigator GPS end tower position option. Linear move systems must be set to “GPS” only.

DED (Dead Reckoning): The location of the system is estimated using the known max speed and percent timer settings. This relies on the hours per revolution value of the system. If “DED” is selected, press the “Setup/Next” key to advance to the current position screen and input current position data. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease the current position setting by 10 degree increments. Press either the “Reverse Depth + 2” key to increase or the “Reverse Depth – 4” key to decrease the current position setting by 0.1 degree increments.

RSV (Collector Ring Resolver Option): The location of the system is derived from the resolver (a position reading device) that is connected to the collector ring. During installation of the system, the dealer or set-up crew will have to ensure that the resolver is set correctly in relation to the rest of the system.

GPS (Reinke Navigator End Tower GPS Position Option): A navigational system using satellite signals to fix the location of a receiver on or above the earth’s surface. The location of the system is determined by the GPS (Global Positioning System) receiver on the end tower. Due North is always zero degrees.

NONE: Use none when no position information is desired.



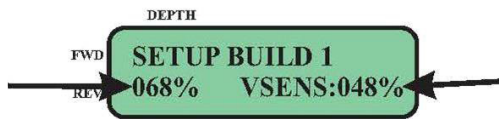
Auto Restart

Now press the “Setup/Next” key to advance to the auto restart screen. Press the “Forward Depth + 1” key to choose between “On” or “Off” as shown. If “On” is selected, press the “Setup/Next” key to advance to the power delay screen. The power delay is the amount of time the system will delay once power is restored to the system. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease the power delay time setting by one second increments. The power delay can be set from 5 seconds to 600 seconds.



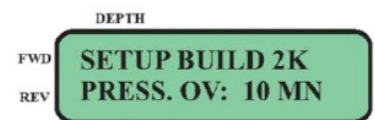
Voltage Sensitivity

Now press the “Setup/Next” key to advance to the voltage sensitivity screen shown here. The number on the left is a voltage percent indicator that is unique to every system. The number on the right is the voltage percent safety indicator or the value where the system will shut down. If the system voltage drops, the voltage percent indicator will drop. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease the voltage percent safety indicator. This value should be 20 points below the voltage percent indicator for REA systems and 30 points below the voltage percent indicator for generator systems. When vsens is set to 100%, the voltage percent indicator is disabled.



Pressure Switch

Now press the “Setup/Next” key to advance to the pressure switch screen. Press the “Forward Depth + 1” key to choose between “YES” if your system is equipped with a pressure switch or “NO” if your system is not equipped with a pressure switch. If “YES” is selected, press the “Setup/Next” key to advance to the pressure override screen shown here on the bottom. Press either the “Forward Depth + 1” key to increase or “Forward Depth – 3” key to decrease the pressure override time setting by one minute increments. This is the amount of time needed to override the pressure switch as the system is building pressure. The pressure override time setting can be set from 1 minute to 30 minutes.



English Units



Metric Units

Frost Sensor

Now press the “Setup/Next” key to advance to the frost sensor screen. Press the “Forward Depth + 1” key to choose between “YES” if your system is equipped with a frost sensor (temperature probe) or “NO” if your system is not equipped with a frost sensor (temperature probe). If “YES” is selected, press the “Setup/Next” key to advance to the frost stop screen shown. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease the frost stop setting by one degree increments. This is the temperature which the system will shut down. The frost stop setting can be set from 32 degrees Fahrenheit to 50 degrees Fahrenheit (English units) and from 0 degrees Celsius to 12 degrees Celsius (metric units). **Warning: Setting the frost stop below 43 degrees Fahrenheit may result in damage to the system!!**



English Units



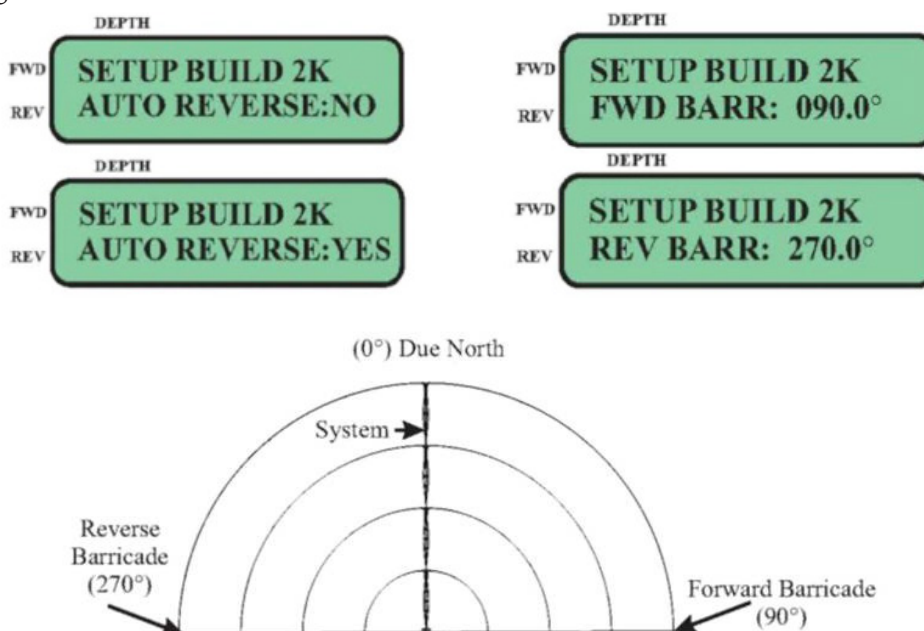
Metric Units

Frost Restart

Now press the “Setup/Next” key to advance to the frost restart screen. Press the “Forward/Next” key to choose between “YES” if you would like the frost restart option enabled or “NO” if you would like the frost restart option disabled. If “YES” is selected, press the “Setup/Next” key to advance to the frost restart @ screen. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease the frost restart @ setting by one degree increments. This is the temperature which the PAC will allow the system to restart. The frost restart @ setting can be set from 44 degrees Fahrenheit to 50 degrees Fahrenheit (English units) and from 7 degrees Celsius to 12 degrees Celsius (metric units).

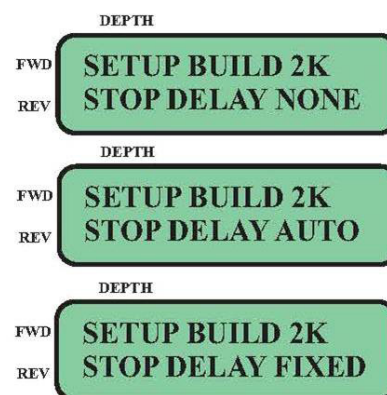
Auto Reverse

Now press the “Setup/Next” key to advance to the auto reverse screen. Press the “Forward Depth + 1” key to choose between “YES” if you would like the auto-reverse option enabled or “NO” if you would like the auto-reverse option disabled. The auto-reverse option must be enabled for the system to start with the direction switch placed in the “Auto” position. If “YES” is selected, press the “Setup/Next” key to advance to the auto reverse forward barricade screen. This is the location relative to 0 degrees where the system must stop and auto-reverse when running in the clockwise direction. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease the auto-reverse forward barricade setting by 10 degree increments. Press the “Reverse Depth + 2” key to increase or the “Reverse Depth – 4” key to decrease the auto-reverse forward barricade setting by 0.1 degree increments. Now press the “Setup/Next” key to advance to the auto reverse reverse barricade. This is the location relative to 0 degrees where the system must stop and auto-reverse when running in the counterclockwise direction. Press either the “Forward Depth + 1” to increase or the “Forward Depth – 3” key to decrease the auto-reverse reverse barricade setting by 10 degree increments. Press either the “Reverse Depth + 2” key to increase or the “Reverse Depth – 4” key to decrease the auto-reverse reverse barricade setting by 0.1 degree increments.



Stop Delay

Now press the “Setup/Next” key to advance to the stop delay screen. Press the “Forward Depth + 1” key to choose from “NONE”, “AUTO”, and “FIXED”. The stop delay function is used on part-circle systems. When “FIXED” stop delay is selected, the pivot will delay (not move) a set amount of time up to sixty minutes. This time is selected for both forward and reverse directions. When the “AUTO” stop delay is selected, the system will delay (not move) a set amount of minutes before moving in the opposite direction. This soaks the ends of the field on part-circle systems. The number of minutes the PAC delays is dependent upon the amount of water the time was applying before the switch was tripped. At a depth of 0.25”, the system will delay approximately four minutes. At a depth of 0.50”, the system will delay approximately eight minutes. At a depth of 0.75”, the system will delay approximately twelve minutes. Any depth above 1.00” will delay fifteen minutes maximum.



****DISCLAIMER****

Reinke disclaims any and all liability (including any liability created pursuant to the Irrigation Systems Warranty) with regard to damage to the irrigation system, or to other property, or personal injury or death, caused by improper installation or maintenance of Reinke supplied auto-reverse or auto-stop switches or tower barricades, or by use of customer supplied barricades.

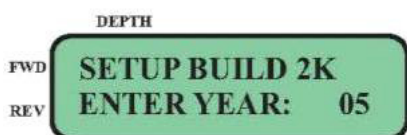
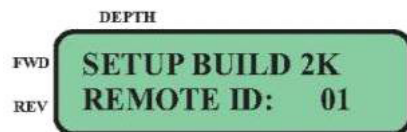
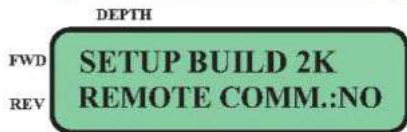
Auto Stop Delay

If "AUTO" is selected, press the "Setup/Next" key to advance to the delay at screen. Press the "Forward Depth + 1" key to choose from "BOTH", "FWD", or "REV". When the "DELAY AT BOTH" is selected, the system will delay in both the forward and reverse directions when switching to the opposite direction. When the "DELAY AT FWD" is selected, the system will only delay in the forward direction when switching to the reverse direction. When the "DELAY AT REV" is selected, the system will only delay in the reverse direction when switching to the forward direction.



Fixed Stop Delay

If "FIXED" is selected, press the "Select/Next" key to advance to the forward delay screen. Press either the "Forward Depth + 1" key or the "Forward Depth - 3" key to select the forward delay time in one minute increments up to sixty minutes. Now press the "Select/Next" key to advance to the reverse delay screen. Press either the "Forward Depth + 1" or the "Forward Depth - 3" key to select the reverse delay time in one minute increments up to sixty minutes.

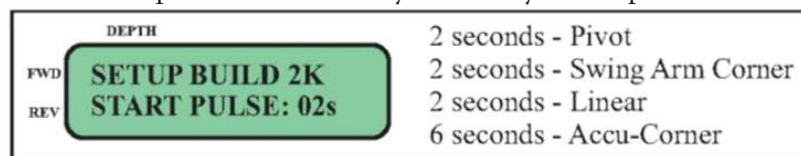


Remote Communication

Now press the "Setup/Next" key to advance to the remote communication screen. Press the "Forward Depth + 1" key to choose between "YES" if the system is equipped with an optional Reinke phone link or controlled by Reinke remote control monitoring base station software or "NO" if the system is not equipped with an optional Reinke phone link or controlled by Reinke remote control monitoring base station software. Now press the "Setup/Next" key to advance to the remote ID screen. Press either the "Forward Depth + 1" or the "Forward Depth - 3" key to set the remote ID to "01" if the system is equipped with an optional Reinke phone link. If the system is controlled by Reinke remote control monitoring base station software, the remote ID must coincide with the remote ID assigned to that particular system in the remote control monitoring base station software. Now press the "Setup/Next" key to advance to the enter year screen. Press either the "Forward Depth + 1" or the "Forward Depth - 3" key to set the last two digits of the current year. This gives the data a year so it is compatible with the Reinke remote control monitoring base station software.

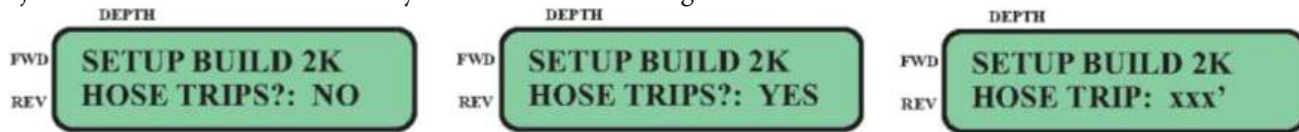
Start Pulse

Now press the "Setup/Next" key to advance to the start pulse screen shown here. This is the amount of time that a start command is supplied to the main control panel to power up the system during an auto-reverse or auto-restart. Press either the "Forward Depth + 1" key to increase or the "Forward Depth - 3" key to decrease the start pulse by one second increments. The start pulse can be set from two seconds to ten seconds. For center pivot systems, swing arm corner systems, or linear systems, set the start pulse to "02" seconds. For accu-corner systems, set start pulse settings to "06" seconds. If you're unsure of your system's start pulse setting, contact your local Reinke dealer for assistance. Use caution when setting this parameter as the start pulse overrides the system safety for the pulse duration.



Hose Trips for Lateral Move Systems with Canal Feed Only

Now press the "Forward Depth + 1" key to change from "NO" to "YES" if hose trips are desired. Hose trips are designed to shut down the linear system every time it reaches the limit set. If yes, then press the "Setup/Next" key. This will allow you to enter the distance (using the key numbers) the system should travel before it shuts down for a hose re-adjustment. Check what units the system is set to either English or Metric.



Save Setup Changes

Now press the "Setup/Next" key to advance to the writing all data to memory screen as shown. This means the PAC III setup menu is complete and the data entered is being written to the PAC III memory chip. You can abandon setup changes without saving. To do this at any point in the setup menu, wait two minutes and the timer will leave the setup menu without saving the changes.

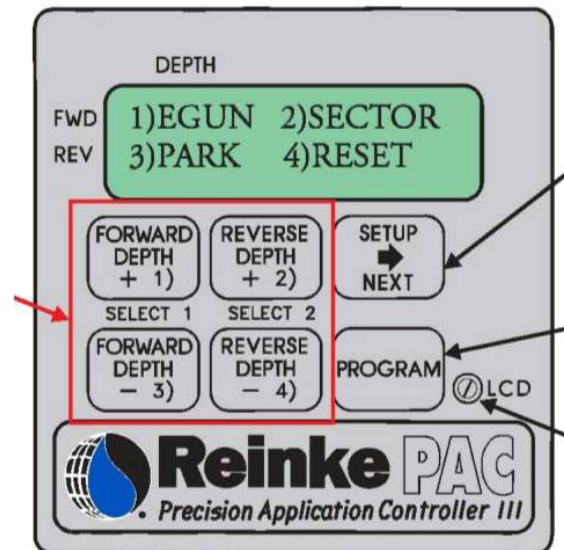


PAC III PROGRAMMING

This section will guide the operator how to program the PAC III. The PAC III can be programmed to control the following:

- 1 End Gun and 1 Auxiliary Program
- 2 Sector Programs with 10 Sectors per Program
- 1 Park Position
- Reset Command

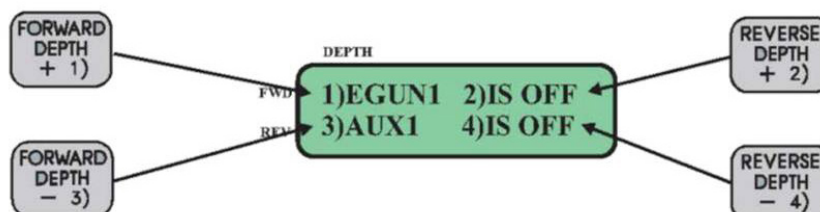
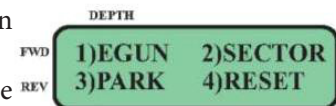
Press and hold the "Select/Next" key for 3 seconds to enter and navigate the setup menu. Press and hold the "Program" key for 3 seconds to enter program mode. Press the number keys to select the program menu item.

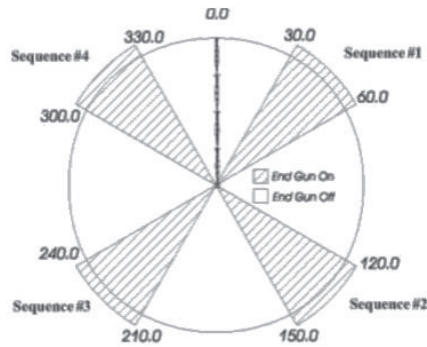


End Gun / Auxiliary Programming

Press the "Program" key and hold for 3 seconds to advance to the main program screen shown here.

Now press the "Forward Depth + 1" key to advance to the end gun status screen. At the end gun status screen, the operator can either configure or override the end gun/auxiliary programs. Press the "Forward Depth + 1" key to configure end gun #1 program or press the "Forward Depth - 3" key to configure auxiliary #1 program. Once all the end gun/auxiliary programming steps are completed, the program will run. To manually control end gun #1, press the "Reverse Depth + 2" key to toggle the end gun #1 on/off. To manually control AUX1, press the "Reverse Depth - 4" key to toggle auxiliary #1 on/off. The altered end gun/auxiliary program will remain in the override status until the system reaches a position in the program where it changes status. Remember, it is against the law in most states to spray water on federal, state, and county roadways.





DEPTH
FWD **EGUN #1 PROGRAM**
REV **NUMBER OF SEQ: 4**

In the example shown above, assume end gun #1 needs to be programmed as shown. Always configure end gun/auxiliary programs with a clockwise rotation from 0 degrees. Sequences should also be numbered in consecutive order in a clockwise rotation from 0 degrees. Press the “Forward Depth + 1” key to configure end gun #1 program and advance to the end gun #1 program – number of sequences screen. This is the number of areas the end gun will be “on”. Press either the “Forward Depth + 1” or the “Forward Depth – 3” key to set the number of sequences. For the example shown here, the end gun would be on four different sequences.

DEPTH
FWD **EGUN #1 PROGRAM**
REV **SEQ01 BEG: 030.0°**

Now press the “Setup/Next” key to advance to the end gun #1 program – sequence #1 – beginning degree screen. This is the position or degree setting of the first sequence where the end gun needs to turn “on”. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease this setting by 10 degree increments.

Press either the “Reverse Depth + 2” to increase or the “Reverse Depth – 4” key to decrease this setting by 0.1 degree increments. For the example shown above, the end gun #1 program – sequence #1 – beginning degree setting would be 30 degrees.

DEPTH
FWD **EGUN #1 PROGRAM**
REV **SEQ01 END: 060.0°**

Now press the “Setup/Next” key to advance to the end gun #1 program – sequence #1 – end degree screen. This is the position or degree setting of the first sequence where the end gun needs to turn “off”. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease this setting by 10 degree increments. Press either the “Reverse Depth + 2” to increase or the “Reverse Depth – 4” key to decrease this setting by 0.1 degree increments.

For the example shown above, the end gun #1 program – sequence #1 – end degree setting would be 60 degrees.

DEPTH
FWD **EGUN #1 PROGRAM**
REV **SEQ02 BEG: 120.0°**

Now press the “Setup/Next” key to advance to the end gun #1 program – sequence #2 – beginning degree screen. This is the position or degree setting of the second sequence where the end gun needs to turn “on”. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease this setting by 10 degree increments.

Press either the “Reverse Depth + 2” to increase or the “Reverse Depth – 4” key to decrease this setting by 0.1 degree increments. For the example shown above, the end gun #1 program – sequence #2 – beginning degree setting would be 120 degrees.

DEPTH
FWD **EGUN #1 PROGRAM**
REV **SEQ02 END: 150.0°**

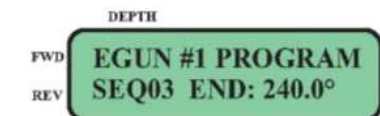
Now press the “Setup/Next” key to advance to the end gun #1 program – sequence #2 – end degree screen. This is the position or degree setting of the second sequence where the end gun needs to turn “off”. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease this setting by 10 degree increments.

Press either the “Reverse Depth + 2” to increase or the “Reverse Depth – 4” key to decrease this setting by 0.1 degree increments. For the example shown above, the end gun #1 program – sequence #2 – end degree setting would be 150 degrees.

DEPTH
FWD **EGUN #1 PROGRAM**
REV **SEQ03 BEG: 210.0°**

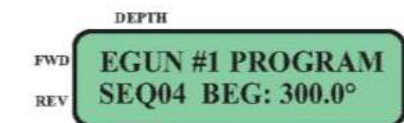
Now press the “Setup/Next” key to advance to the end gun #1 program – sequence #3 – beginning degree screen. This is the position or degree setting of the third sequence where the end gun needs to turn “on”. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease this setting by 10 degree increments.

Press either the “Reverse Depth + 2” to increase or the “Reverse Depth – 4” key to decrease this setting by 0.1 degree increments. For the example shown above, the end gun #1 program – sequence #3 – beginning degree setting would be 210 degrees.



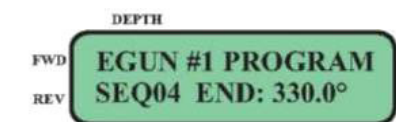
Now press the "Setup/Next" key to advance to the end gun #1 program – sequence #3 – end degree screen. This is the position or degree setting of the third sequence where the end gun needs to turn "off". Press either the "Forward Depth + 1" key to increase or the "Forward Depth – 3" key to decrease this setting by 10 degree increments.

Press either the "Reverse Depth + 2" to increase or the "Reverse Depth – 4" key to decrease this setting by 0.1 degree increments. For the example on the previous page, the end gun #1 program – sequence #3 – end degree setting would be 240 degrees.



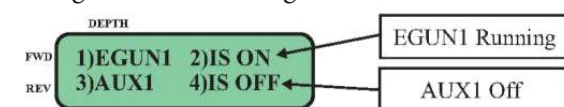
Now press the "Setup/Next" key to advance to the end gun #1 program – sequence #4 – beginning degree screen. This is the position or degree setting of the fourth sequence where the end gun needs to turn "on". Press either the "Forward Depth + 1" key to increase or the "Forward Depth – 3" key to decrease this setting by 10 degree increments. Press either the "Reverse Depth + 2" to increase or the "Reverse

Depth – 4" key to decrease this setting by 0.1 degree increments. For the example on the previous page, the end gun #1 program – sequence #4 – beginning degree setting would be 300 degrees.



Now press the "Setup/Next" key to advance to the end gun #1 program – sequence #4 – end degree screen. This is the position or degree setting of the fourth sequence where the end gun needs to turn "off". Press either the "Forward Depth + 1" key to increase or the "Forward Depth – 3" key to decrease this setting by 10 degree

increments. Press either the "Reverse Depth + 2" to increase or the "Reverse Depth – 4" key to decrease this setting by 0.1 degree increments. For the example on the previous page, the end gun #1 program – sequence #4 – end degree setting would be 330 degrees.

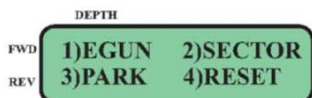
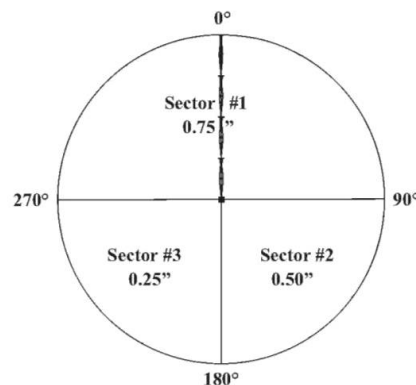


Now press the "Setup/Next" key to advance to the end gun status screen. End gun #1 is now programmed and will operate as per the program unless the status is changed. Remember, it is against the law in most states to spray water on state and county roadways. Auxiliary

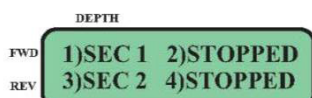
1 can be programmed and activated the same way. If you want to program AUX1, press the "Forward Depth – 3" key and follow the same steps as described for programming end gun #1 above. To return to the normal operation screen, press the "Setup/Next" key.

Sector Programming

Sector programming is a useful tool when you want to water certain sections (sectors) of the field with different application rates. There are two different sector programs with the capability to water up to ten different sectors per sector program. You must pick the beginning and ending degrees of each sector and the forward and reverse application depths of each sector. Only one sector program can run at any given time. In the example shown here, sector program #1 needs to be programmed as per the following conditions. The system needs to apply 0.75" between 270° and 90° (sector #1). The system needs to apply 0.50" between 90° and 180° (sector #2). The system needs to apply 0.25" between 180° and 270° (sector #3). Always configure the sector programs with a clockwise rotation from 0°. Sequences should also be numbered in consecutive order in a clockwise rotation from 0°. The application depth on the normal operation screen will apply to areas not covered by an active program.



Press and hold the "Program" key for 3 seconds to advance to the main program screen.



Now press the "Reverse Depth + 2" key to advance to the sector program status screen. At the sector program status screen, the operator can either configure the settings for both sector programs or toggle each sector program on/off. Press the "Forward Depth + 1" key to configure sector program #1 or press the "Forward Depth – 3" key to configure

sector program #2. Press the "Reverse Depth + 2" key to change the status of sector program #1 to either "STOPPED" or "RUNNING". Press the "Reverse Depth – 4" key to change the status of sector program #1 to either "STOPPED" or "RUNNING". Do not change the status of sector programs to "RUNNING" until the sector programs are properly configured as per the following instructions. Remember, only one sector program can run at any given time.



Press the “Setup/Next” key to configure sector program #1 and advance to the sector program #1 – number of sectors screen. This is the number of areas the system needs to water with different application rates and positions. Press either the “Forward Depth + 1” or the “Forward Depth – 3” key to set the number of sectors. For the example shown on the previous page, the number of sectors would be 3.



Now press the “Setup/Next” key to advance to the sector program #1 – sector #1 – beginning degree screen. This is the position or degree setting of the first sector where the system needs to start applying the application depths programmed for sector program #1 – sector #1. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease this

setting by 10° increments. Press either the “Reverse Depth + 2” key to increase or the “Reverse Depth - 4” key to decrease this setting by 0.1° increments. For the example on the previous page, the sector program #1 – sector #1 – beginning degree setting would be 270°.



Now press the “Setup/Next” key to advance to the sector program #1 – sector #1 – end degree screen. This is the position or degree setting of the first sector where the system needs to stop applying the application depths programmed for sector program #1 – sector #1. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease this setting by 10° increments. Press

either the “Reverse Depth + 2” key to increase or the “Reverse Depth - 4” key to decrease this setting by 0.1° increments. For the example on the previous page, the sector program #1 – sector #1 – end degree setting would be 90°.

Now press the “Setup/Next” key to advance to the sector program #1 – sector #1 – forward application depth screens. This is the application depths the system will apply running clockwise between the two degree values previously



English Units



Metric Units

programmed for sector program #1 – sector #1. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease the depth by 0.01” increments for “ENGLISH” units or 1 mm increments for “METRIC”

units. This depth must be greater than the minimum application depth entered during the setup process and less than 5.00 inches (127 mm). For the example shown earlier, the sector program #1 – sector #1 – forward application depth would be 0.75” (19.05 mm).



English Units



Metric Units

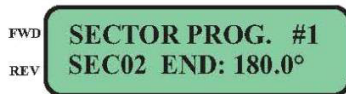
Now press the “Setup/Next” key to advance to the sector program #1 – sector #1 – reverse application depth screens. This is the application depths the system will apply running counter clockwise between the two degree values previously programmed for sector program #1 – sector

#1. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease the depth by 0.01” increments for “ENGLISH” units or 1 mm increments for “METRIC” units. This depth must be greater than the minimum application depth entered during the setup process and less than 5.00 inches (127 mm). For the example shown earlier, the sector program #1 – sector #1 – reverse application depth would be 0.75” (19.05 mm).



Now press the “Setup/Next” key to advance to the sector program #1 – sector #2 – beginning degree screen. This is the position or degree setting of the second sector where the system needs to start applying the application depths programmed for sector program #1 – sector #2. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease this setting by

10° increments. Press either the “Reverse Depth + 2” to increase or the “Reverse Depth – 4” key to decrease this setting by 0.1° increments. For the example shown earlier, the sector program #1 – sector #2 – beginning degree setting would be 90°.



Now press the “Setup/Next” key to advance to the sector program #1 – sector #2 – end degree screen. This is the position or degree setting of the second sector where the system needs to stop applying the application depths programmed for sector program #1 – sector #2. Press either the “Forward Depth + 1” key to

increase or the “Forward Depth – 3” key to decrease this setting by 10° increments. Press either the “Reverse Depth + 2” to increase or the “Reverse Depth – 4” key to decrease this setting by 0.1° increments. For the example shown earlier, the sector program #1 – sector #2 – end degree setting would be 180°.

DEPTH
FWD
REV
SECTOR PROG. #1
SEC02 fDPH: 0.50
English Units

DEPTH
FWD
REV
SECTOR PROG. #1
SEC02 fDPH: 12.7
Metric Units

Now press the “Setup/Next” key to advance to the sector program #1 – sector #2 – forward application depth screens. This is the application depths the system will apply running clockwise between the two degree values previously programmed for sector program #1 – sector #2. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease the depth by 0.01” increments for “ENGLISH” units or 1 mm for “METRIC” units. This depth must be greater than the minimum application depth entered during the setup process and less than 5.00 inches (127 mm). For the example shown earlier, the sector program #1 – sector #1 – forward application depth would be 0.75” (19.05 mm).

DEPTH
FWD
REV
SECTOR PROG. #1
SEC02 rDPH: 0.50
English Units

DEPTH
FWD
REV
SECTOR PROG. #1
SEC02 rDPH: 12.7
Metric Units

Now press the “Setup/Next” key to advance to the sector program #1 – sector #2 – reverse application depth screens. This is the application depths the system will apply running counter clockwise between the two degree values previously programmed for sector program #1 – sector #2. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease the depth by 0.01” increments for “ENGLISH” units or 1 mm for “METRIC” units. This depth must be greater than the minimum application depth entered during the setup process and less than 5.00 inches (127 mm). For the example shown earlier, the sector program #1 – sector #1 – reverse application depth would be 0.75” (19.05 mm).

DEPTH
FWD
REV
SECTOR PROG. #1
SEC03 BEG: 180.0°

Now press the “Setup/Next” key to advance to the sector program #1 – sector #3 – beginning degree screen. This is the position or degree setting of the third sector where the system needs to start applying the application depths programmed for sector program #1 – sector #3. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease this setting by 10° increments. Press either the “Reverse Depth + 2” to increase or the “Reverse Depth – 4” key to decrease this setting by 0.1° increments. For the example shown earlier, the sector program #1 – sector #3 – beginning degree setting would be 180°.

DEPTH
FWD
REV
SECTOR PROG. #1
SEC03 END: 270.0°

Now press the “Setup/Next” key to advance to the sector program #1 – sector #3 – end degree screen. This is the position or degree setting of the third sector where the system needs to stop applying the application depths programmed for sector program #1 – sector #3. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease this setting by 10° increments. Press either the “Reverse Depth + 2” to increase or the “Reverse Depth – 4” key to decrease this setting by 0.1° increments. For the example shown earlier, the sector program #1 – sector #3 – end degree setting would be 270°.

DEPTH
FWD
REV
SECTOR PROG. #1
SEC03 fDPH: 0.25
English Units

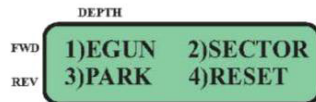
DEPTH
FWD
REV
SECTOR PROG. #1
SEC03 fDPH: 6.35
Metric Units

Now press the “Setup/Next” key to advance to the sector program #1 – sector #3 – forward application depth screens. This is the application depths the system will apply running clockwise between the two degree values previously programmed for sector program #1 – sector #3. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease the depth by 0.01” increments for “ENGLISH” units or 1 mm increments for “METRIC” units. This depth must be greater than the minimum application depth entered during the setup process and less than 5.00 inches (127 mm). For the example shown earlier, the sector program #1 – sector #3 – forward application depth would be 0.25” (6.35 mm).

DEPTH
FWD
REV
SECTOR PROG. #1
SEC03 rDPH: 0.25
English Units

DEPTH
FWD
REV
SECTOR PROG. #1
SEC03 rDPH: 6.35
Metric Units

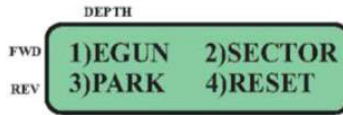
Now press the “Setup/Next” key to advance to the sector program #1 – sector #3 – reverse application depth screens. This is the application depths the system will apply running counter clockwise between the two degree values previously programmed for sector program #1 – sector #3. Press either the “Forward Depth + 1” key to increase or the “Forward Depth – 3” key to decrease the depth by 0.01” increments for “ENGLISH” units or 1 mm increments for “METRIC” units. This depth must be greater than the minimum application depth entered during the setup process and less than 5.00 inches (127 mm). For the example shown earlier, the sector program #1 – sector #3 – reverse application depth would be 0.25” (6.35 mm).



Now press the “Setup/Next” key to advance to the sector program status screen. Sector program #1 is now programmed. To run sector program #1, you must be sure the status is set to “RUNNING”. If the status of sector program #1 is set to “STOPPED”, then press the “Reverse Depth + 2” key to change the status of sector program #1 to “RUNNING”. Do not change the status of sector programs to “RUNNING” until

sector programs are properly configured as per instructions in this section. Remember, only one sector program can run at any given time. Now press the “Setup/Next” key to advance to the normal operation screen.

Park Programming



The park program is used to stop the system at a particular degree setting. To configure the park program, press the “Program” key and hold for 3 seconds to advance to the main program screen shown here.



Now press the “Forward Depth -3” key to advance to the park program status screen. At the park program status screen, the operator can either configure the settings for the park program or toggle the park program on and off. Press the “Forward Depth + 1” key to configure the park program. Press the “Forward Depth - 3” key to change

the status of the park program to either “PARK ON” or “PARK OFF”. Press the “Reverse Depth - 4” key to exit back to the normal operation screen. Do not change the status of the park program to “PARK ON” until the park program is properly configured as per the following instructions. Remember, once the system reaches the park destination, the park program status will revert back to “PARK OFF”. The operator must change status of park program to “PARK ON” each time it is desired to have the system park at the programmed position.

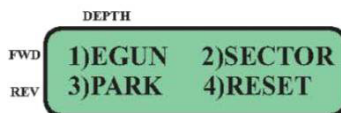


Press the “Forward Depth + 1” key to advance to the set park position screen. Press either the “Forward Depth + 1” key to increase or the “Reverse Depth + 2” key to decrease this setting by 10° increments. Press either the “Forward Depth - 3” key to increase or the “Reverse Depth - 4” key to decrease this setting by 0.1° increments.



Now press the “Setup/Next” key to advance back to the park program status screen. Press the “Forward Depth - 3” key to change the status of the park program to “PARK ON” to have the system park at the programmed position. Now press the “Reverse Depth - 4” key to exit back to the normal operation screen.

Reset Program



To reset end gun and auxiliary #1, sector programs #1 and #2, setup options, or revert back to factory defaults, press the “Program” key and hold for 3 seconds to advance to the main program screen. Now press the “Reverse Depth - 4” key to advance to the first of three reset screens.



1) RST EG1



RST EG1: Press the “Forward Depth + 1” key to advance to the screen to reset the end gun #1 program. Press the “Reverse Depth - 4” key to revert back to the main program screen. Press the “Forward Depth - 3” key to reset the end gun #1 program. The screen shown will quickly appear to indicate the end gun program #1 has been reset.

3) RST AUX1

DEPTH	
FWD	RESET AUX1?
REV	3)YES 4)NO
DEPTH	
FWD	*****RESET*****
REV	*****END AUX1*****

RST AUX1: Press the “Forward Depth - 3” key to advance to the screen to reset auxiliary #1 program. Press the “Reverse Depth - 4” key to revert back to the main program screen. Press the “Forward Depth - 3” key to reset auxiliary #1 program. The screen shown will quickly appear to indicate the auxiliary #1 program has been reset.

4) MOR

DEPTH	
FWD	1)RST SEC1
REV	3)RST SEC2 4)MOR
DEPTH	
FWD	1)RST FACT DEFLT
REV	3)RST SET 4)EXIT

MOR: Press the “Reverse Depth - 4” key to advance to the second of three reset screens. Press the “Forward Depth + 1” key to reset sector program #1 or the “Forward Depth - 3” key to reset sector program #2. Sector programs will reset the same way as the end gun programs illustrated. Press the “Reverse Depth - 4” key to advance to the third of three reset screens. Press the “Forward Depth + 1” key to reset all data back to factory defaults. This includes all setup menus, end gun programs, sector programs, etc. Press the “Forward Depth - 3” key to reset the setup menu only. These commands will reset the same way as the end gun programs illustrated previously. Press the “Reverse Depth - 4” key to exit back to the normal operation screen.

Appendix A – Calculating End Tower Movement

$$\frac{\text{Minimum Depth Setting}}{\text{Application Depth Setting}} \times 100 = \% \text{ of End Tower Movement}$$

EXAMPLE: $\frac{\text{Minimum Depth Setting}}{\text{Application Depth Setting}} = \frac{0.10''}{0.20''} \times 100 = 50\%$

$$\frac{\text{Minimum Depth Setting}}{\text{Application Depth Setting}} \times \text{Cycle Time Setting} = \# \text{ of Seconds End Tower will Move}$$

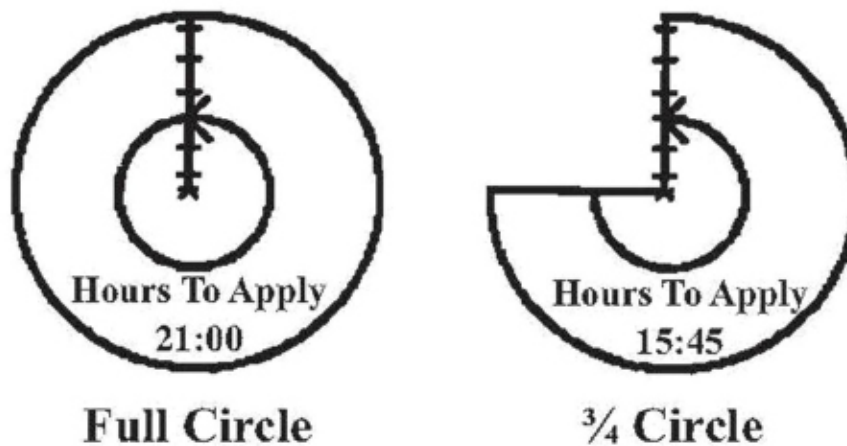
EXAMPLE:

$$\frac{\text{Minimum Depth Setting}}{\text{Application Depth Setting}} = \frac{0.10''}{0.20''} \times 60 = 30 \text{ Seconds that End Tower will Move}$$

The actual end tower movement of the system is based on percentage timing calculated by the PAC module. The standard cycle period for the timer is sixty seconds, but thirty seconds is an optional setting. The percentage cycle the end tower will move is calculated by dividing the minimum depth setting by the application depth setting as shown above. The number of seconds the end tower will move is calculated by dividing the minimum depth setting by the application depth setting and multiplying the cycle time setting as shown.

If the system is on a part circle, the hours to apply can be setup to the number of hours it will take to move the partial revolution. For example, if the speed chart says the system will take 21 hours to complete one circle at 100%, but the pivot only moves $\frac{3}{4}$ of a revolution, the number of hours can be recalculated by multiplying 21 times $\frac{3}{4}$, which equals 15 hours and 45 minutes. See the diagram shown below.

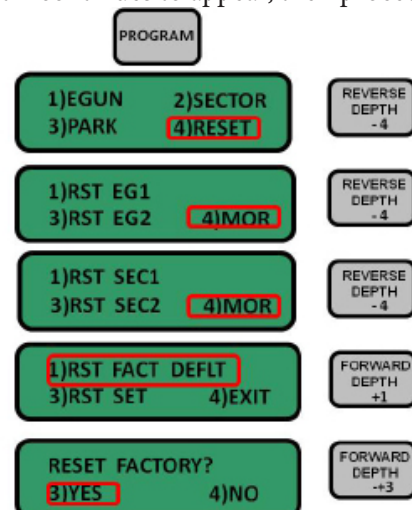
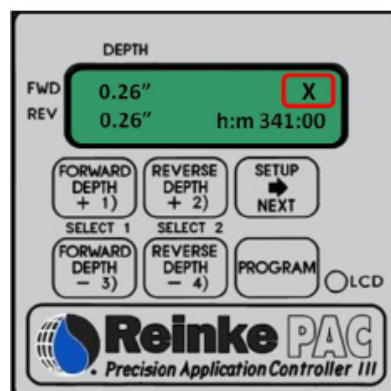
Appendix B – Calculating Part Circle Systems

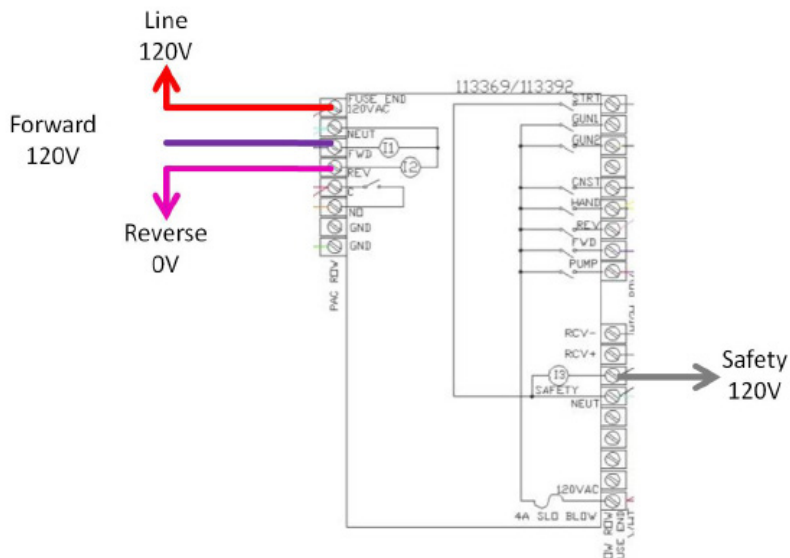


The recalculated number of hours to apply may be entered as the “HOURS PER REVOLUTION” setting in the setup menu. The PAC timer can then make a better approximation of when the pivot will traverse the part circle. The ideal setup would be to actually time the system as it moves around the circle at 100%, and enter that exact time as the “HOURS PER REVOLUTION” setting in the setup menu.

PAC III TROUBLESHOOTING

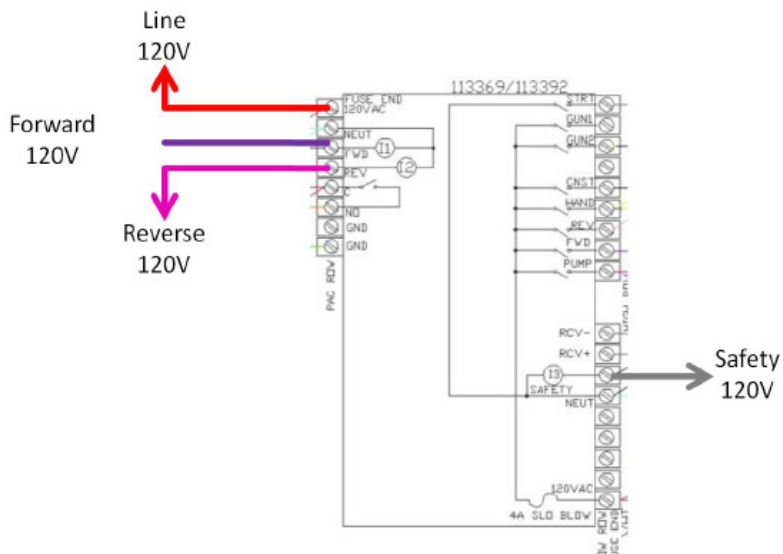
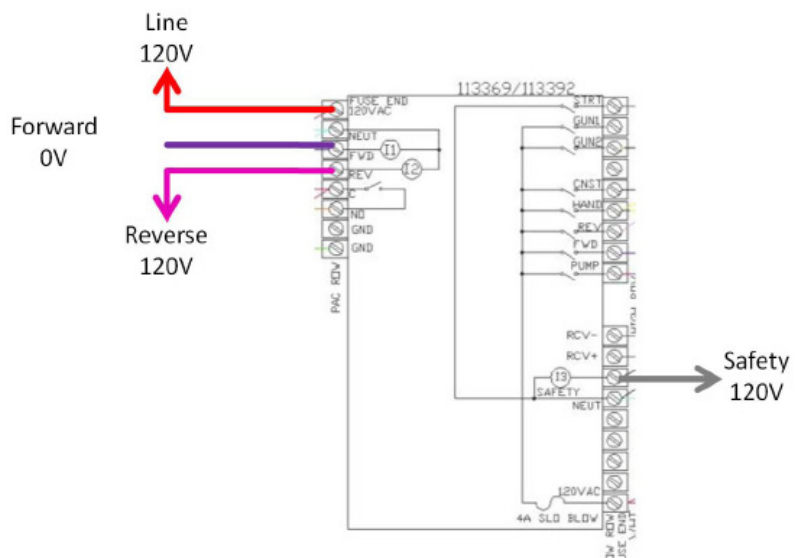
If “X” appears in the system position display, check GPS or resolver. If positioning device appears to be functioning properly and the “X” is still present and system is utilizing ontrac plus units or radio telemetry, record all PAC setup and program settings and reset PAC to factory defaults. If remote communication is not being utilized, make sure that “REMOTE COMM” in the setup menu is set to “NO”. If the “X” still continues to appear, then proceed with resetting to factory defaults as mentioned above.



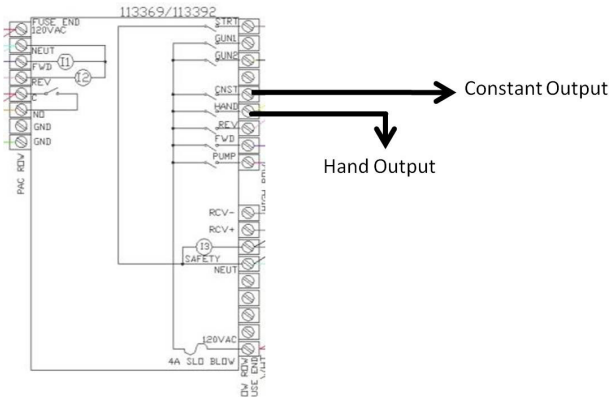


There should be 120V on the purple wire attached to the forward input when the machine is latched in and set to run in the forward direction. If voltage is not present on the purple wire, check the direction switch.

There should be 120V on the pink wire attached to the reverse input when the machine is latched in and set to run in the reverse direction. If voltage is not present on the pink wire in this scenario, check the direction switch.

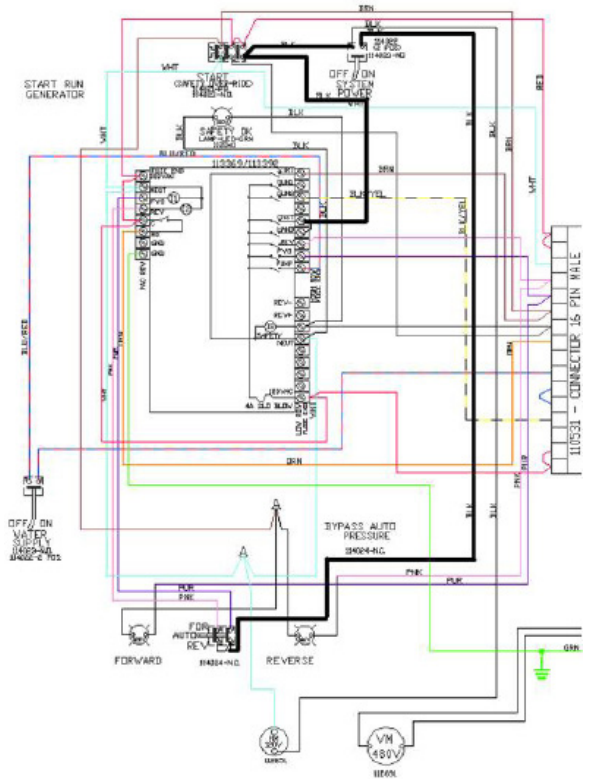


If the machine is equipped with auto reverse and the direction switch is set to the "AUTO" position, there should be 120V on both forward and reverse inputs. If voltage is not present on these terminals, check the direction switch.

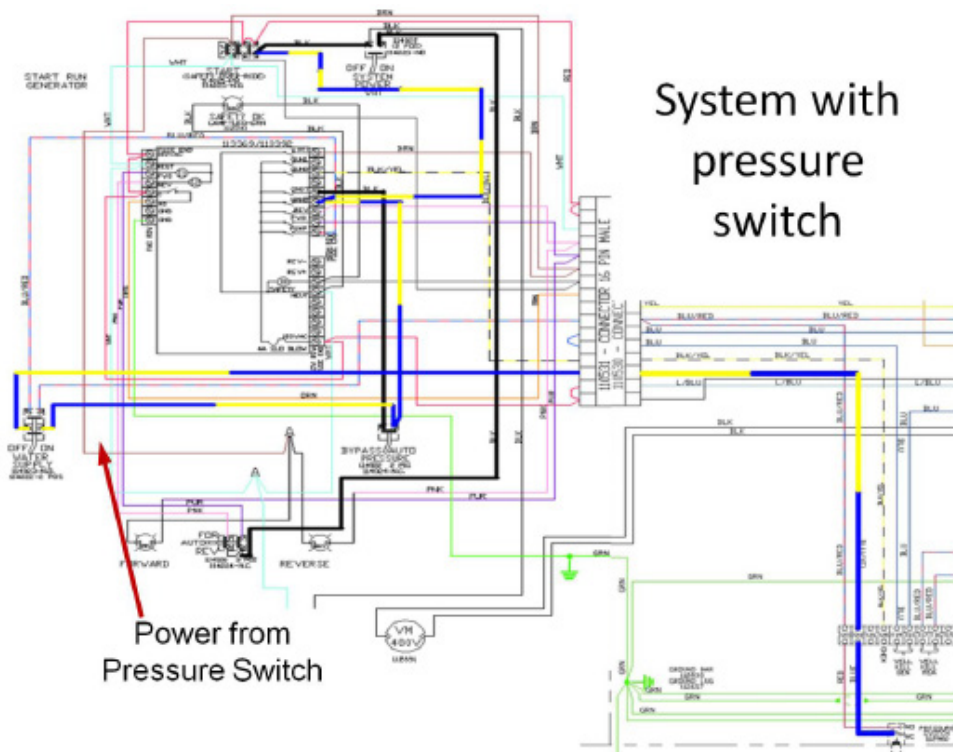


There are two outputs that are used to keep the PAC timer panel latched in. They are "Constant" and "Hand" outputs. The one used depends on what kind of options are in the panel.

In a system without a pressure switch, the "Constant" outputs feeds power to the system power switch, which then feeds to the direction switch during run time. If constant output is not outputting 120V, the machine will not latch in.



System with pressure switch



In a system with a pressure switch, the "Constant" feeds power to the system power switch, which feeds the bypass switch so either the pressure switch or bypass switch can feed the direction switch. If 120V is not present on this output, the machine will not latch in or run.

This page intentionally left blank.

Troubleshooting

CHAPTER 6

TROUBLESHOOTING THEORY

Making necessary repairs on an irrigation system to get it running is usually relatively easy when you know what is wrong. The toughest job for any service technician is diagnosing the root cause of the problem. Often, the problem will be obvious; and with your experience and training, the proper diagnosis will occur quickly. However, sometimes this is not the case. Here are some tips for when things get a little tougher:

1. Know what the system is, or is not, doing before you start.

Ask questions of the system owner or operator. Getting good and complete information of what the system was doing before, during, and after the problem can give you valuable clues and save you a great deal of diagnostic time. Make sure and inquire about anything the operator has done to the system prior to calling you for help. Sometimes a customer trying to fix things himself can mask the real situation and cause you some confusion.

2. Approach the problem logically and safely.

Use your experience and knowledge. Consult your service manual or service school book. Remember, your safety needs to be your top priority.

3. Look for the root cause of the problem.

Until the root cause of the problem is defined and eliminated, you have not really fixed anything. Don't be a parts changer. If you just change a bunch of parts that "could" have caused the problem, you may have made the symptoms go away; however, you not only failed to find the root cause, but you have probably changed out mostly good components with no understanding of what really happened. This is not the sign of a good technician.

4. Use the process of elimination to narrow down the potential area of the problem.

Not all troubleshooting needs to start at the main control panel. You need to determine the following:

- What symptoms are you seeing?
- What components could cause these specific symptoms?
- Where are these components located?
- Do they need to be checked with or without power?

Then check the most obvious or common of these components first. If that isn't the problem, proceed to the next most obvious or common component.

5. Don't get caught up in the "black box syndrome".

Often, due to a lack of understanding of an advanced component, a technician will become convinced that any problem on that system has to be that expensive, high-tech electronic device. He replaces it, only to find out that the problem was really something simple (and less expensive) like a relay that he should have checked first. Don't forget to use your basic troubleshooting skills.

6. Don't miss the obvious.

A system that limits out because it is out-of-line could be an electrical problem or it could just be a tower that got stuck due to a flat tire. A system that won't latch in after you have replaced alignment micro switches could be an electrical problem or you may have forgotten to turn the tower box disconnect back on in one of the tower boxes.

Heat, long hours, and frustration can cause any person to lose focus on their troubleshooting skills. Sometimes, you need to step back: take some time to regroup, refresh, and refocus. Remember: your safety is at risk when you push yourself beyond reasonable physical limits.

Consult with your co-workers. A fresh set of eyes or another opinion might be just what you need. This problem might be the same as one that somebody else in your dealership has experienced previously. You can also call the Reinke Technical Support Department. The service technicians will be happy to help walk you through the problem over the phone. Please be prepared with detailed information and the history of the problem.

SERVICE BULLETINS AND ALERTS

The service alert is sometimes used as a quick notification and is followed by a service bulletin with more details and solutions when available. Dealership action is usually not required. Some examples are product changes or enhancements, discovery of a possible product issue that has not been totally verified or does not have a solution yet, and other service related issues.

Some examples of a service bulletin are product recalls or required service issues, notification of a discovered and verified problem (normally includes a diagnostic method and/or solution), and distribution of new assembly prints or electrical drawings that should be used to update the dealership service manuals.

DATE: 18/11/2013

BY: A3-Robin Dierker

SUBJECT: Teachware Framework - Problem & Solution

The Teachware framework version 1.2.20 will add a problem, where running both the VRM and console components will fail with the error message: "The Teachware framework is broken and displays an error message".

The following Teachware patch will create version 1.2.20. Therefore:

- Any Teachware installed before version 1.2.20 will be replaced by RRD5 between November 1, 2012 and November 1, 2013.
- Patched or future patched Teachware from version 1.2.20 will be replaced by RRD5 between November 1, 2012 and November 1, 2013.

As Teachware version 1.2.20 will be replaced by RRD5, Teachware will be updated to version 1.2.3.

Symptoms:

The Teachware framework, which both VRM and console communications operate simultaneously. The patch will display either a "configuration exception" or a "shell and socket" error message and program termination will occur. The following is the error message:

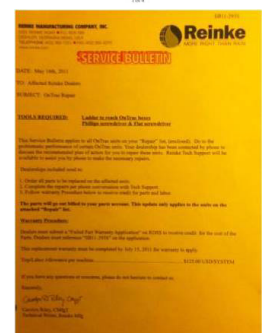
Resolution:

The Teachware framework version 1.2.20 (3.11.3) will be replaced by version 1.2.20.2 and then be downloaded to local computers or flash drives.

The Teachware Framework version can be downloaded from RRD5. We will find the download link at the RRD5 website: www.rrd5.com. Refer to the following table on the following pages.

VRM Element:

In most cases, VRM operation fails. Teachware framework version 1.1.22 and 1.14.04 binary version will be replaced by version 1.2.20.2 for the window operation.



REFERENCE MATERIAL

The resistance charts shown below and on the next page show what resistance (ohms) particular components should read. It also has the voltage that should be used to operate each component. Take note of the Honeywell relays. The first is a 480 volt coil and its resistance is 4000 ohms. The “BPK”, “BK”, and “PK” relays indented below that are still honeywell relays but are the 120 volt coil. They have a resistance of 240 ohms. In the real world, if you were troubleshooting the safety relay in a main control panel, then you would read the resistance with the power off on the coil and if it reads 240 ohms then the relay is not the problem.

Resistance Chart

<u>ITEM</u>	<u>VOLTAGE</u>	<u>RESISTANCE</u>
Contactor Coils:		
Benedict & Jaeger	120	205
G.E. Def. Pur.-Black	120	80
G.E. MCP-White	120	190
G.E. TWR-White	120	210
Square D MCP-50 AMP	120	120
Square D TWR-25 AMP	120	200
Square D BST-25 AMP	120	120
MCP-Telemecanique	120	120
End Gun Valves:		
Skinner	120	180
Asco	120	220
Shutdown Valve:		
6" Shutdown Solenoid	120	180
Transformers:		
No. 9155Y52	480 Input	3.4
	120 Output	0.2
Micron .750 KVA	480 Input	3.4
	120 Output	0.2
Micron .500	480 Input	7.0
	120 Output	0.4
Buck Boost-15 KVA (step-up) all three .500 KVA	480 Input + 10% Out	6
Square D .500 KVA Single Phases (H1-H2, H3-H4)	480 Input	2.2
Square D .500 KVA Single Phase (X1-X2, X3-X4)	480 + 10% Output	0.8
Relays:		
Honeywell	480	4000
“BPK”, “BK”, “PK” Relays	120	240
Clear Plastic -11 Pin, Potter & Brumfield, “SK”	120	1590
Clear Plastic - 8 Pin, Potter & Brumfield	120	2023
Memory Latch - 11 Pin, Potter & Brumfield	120	1025
Schrack	120	1470
Omron (RAMS M300, GPS RX Board)	12 VDC	265
ATC Time Delay (Series 417)	120	Open Line
Signaline Time Delay (On/Off Delay)	120	Open Line
SAC Antenna:		
Pickup Coil (Forward, Reverse, Limit)	0	180

Resistance Chart

<u>ITEM</u>	<u>VOLTAGE</u>	<u>RESISTANCE</u>
Timers (mechanical): ATC Timer - DO DO - "B" Box		
Clutch Coil (Terminals 1 & 2)	120	360
Motor (Terminals 2 & 6)	120	900
ATC Percent Timer		
60 Second (Terminals 1 & 4)	120	420
30 Second (Terminals 1 & 4)	120	90
Tower Motors:		
Doerr 1 HP	480	15
Emerson ½ HP (SAC Steering)	480	14.8
Franklin 1 HP (Old Style)	480	12
Franklin 1 HP (Stainless Steel)	480	11
Franklin 1 ½ HP (Stainless Steel)	480	8
G.E. 1 HP	480	15
Leeson 1 HP	480	13
Leeson 1 HP	480	8
Century-Gould 1 HP	480	11
US Syncrogear 1	480	18
US Syncrogear ¾ HP	480	26
Reinke Helical Drives		
Red	480	23.1
Blue	480	15.9
Black	480	6.9
Green (No Thermal Protection)	480	15.8
Brown (No Thermal Protection)	480	6.8
UMC Power Saver 3.5	480	9.3
Booster Pump Motors:		
Franklin 2 HP + 1.5 Service Factor (3 HP)	480	4.6
Century 2 HP + 2.4 Service Factor (5 HP)	480	3.2
Pentek	480	4.6
Air Compressor:		
Gast	480	25.1
Hourmeters:		
ENM	120	Open Line
Forth Dimension	120	820
Hobbs	120	1250

*Voltage listed is VAC unless otherwise noted.

*Solid State components have Open Line resistance.

The KW ratings for the type of motor used is to get an idea of the size of generator that it would take to run a particular machine. If there was a pivot that was a seven tower machine with a 2HP booster pump and the motors on the towers were standard speed UMP powersavers, the total KW rating would be $.85 \text{ KW} \times 7 = 5.95 \text{ KW} + 3.42 = 9.37 \text{ KW}$. This system would need, at the minimum, a 10 KW generator to run the system.

KW Ratings for Generator Sizing

Ratings for the US Helical Motors

0.65 KW for Low Speed Motors on Pivot Towers
0.97 KW for Standard Speed Motors on Pivot Towers
1.94 KW for High Speed Motors on Pivot Towers
0.81 KW for Low Speed Motors on Standard Linear Towers
1.22 KW for Standard Speed Motors on Standard Linear Towers
2.43 KW for High Speed Motors on Standard Linear Towers
1.62 KW for Standard Speed Motors on S-Towers or Linear Power Towers
3.24 KW for High Speed Motors on S-Towers or Linear Power Towers
0.65 KW for the Steering Motor on an S-Tower (EII)

Ratings for the UMC Powersaver 3.5 Helical Motors

0.85 KW for Low and Standard Speed Motors on Pivot Towers
1.70 KW for High Speed Motors on Pivot Towers
1.06 KW for Low and Standard Speed Motors on Standard Linear Towers
2.13 KW for High Speed Motors on Standard Linear Towers
1.42 KW for Low and Standard Speed Motors on S-Towers or Linear Power Towers
2.84 KW for High Speed Motors on S-Towers or Linear Power Towers
0.85 KW for the Steering Motor on an S-Tower (EII)

5.70 KW for a 5 Hp Booster Pump
3.42 KW for a 2 Hp Booster Pump
0.46 KW for the Emerson Steering Motor on an S-Tower (E60)
0.68 KW for an Air Compressor on the S-Tower

The approximate motor horsepower to drive the generator is calculated by multiplying the total generator KW by 1.6

$$\text{Motor HP} = 1.6 \times \text{Gen KW}$$

You will find the prints listed below in the reference section of this book.

[85152107](#): This print shows how to connect motor leads to the motor wiring.

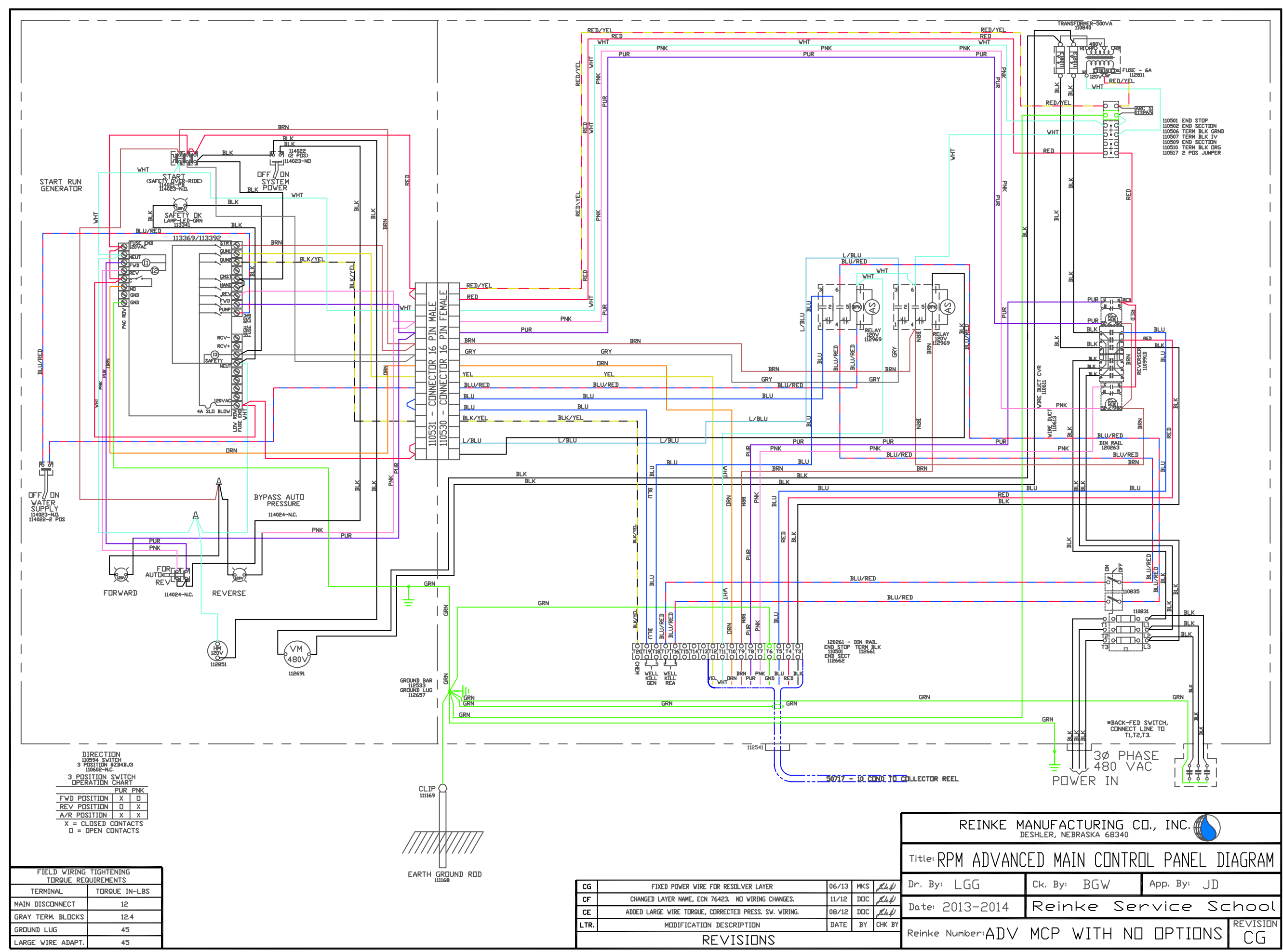
[142514](#): This print shows how to connect a Leeson single-phase motor.

RST-02 Pivot Electrical & Troubleshooting

Reference Packet

The prints found in this reference packet are for this course only. As product changes and enhancements occasionally occur, these may not be the most current revisions. Please consult your Electrical Service Manual for current prints to use for actual field troubleshooting reference.





DIRECTION
110594 SWITCH
3 POSITION #ZB4B3J3
110602-N.C.

	PUR	PNK
FWD POSITION	X	0
REV POSITION	0	X
A/R POSITION	X	X

X = CLOSED CONTACTS
0 = OPEN CONTACTS

FIELD WIRING TIGHTENING TORQUE REQUIREMENTS	
TERMINAL	TORQUE IN-LBS
MAIN DISCONNECT	12
GRAY TERM. BLOCKS	12.4
GROUND LUG	45
LARGE WIRE ADAPT.	45

CG	FIXED POWER WIRE FOR RESOLVER LAYER	06/13	MKS	
CF	CHANGED LAYER NAME, ECH 76423. NO WIRING CHANGES.	11/12	DOC	
CE	ADDED LARGE WIRE TORQUE, CORRECTED PRESS. SW. WIRING.	08/12	DOC	
LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY

REINKE MANUFACTURING CO., INC.
DESHLER, NEBRASKA 68340

Title:RPM ADVANCED MAIN CONTROL PANEL DIAGRAM

Dr. By: LGG

ck. By: BGW

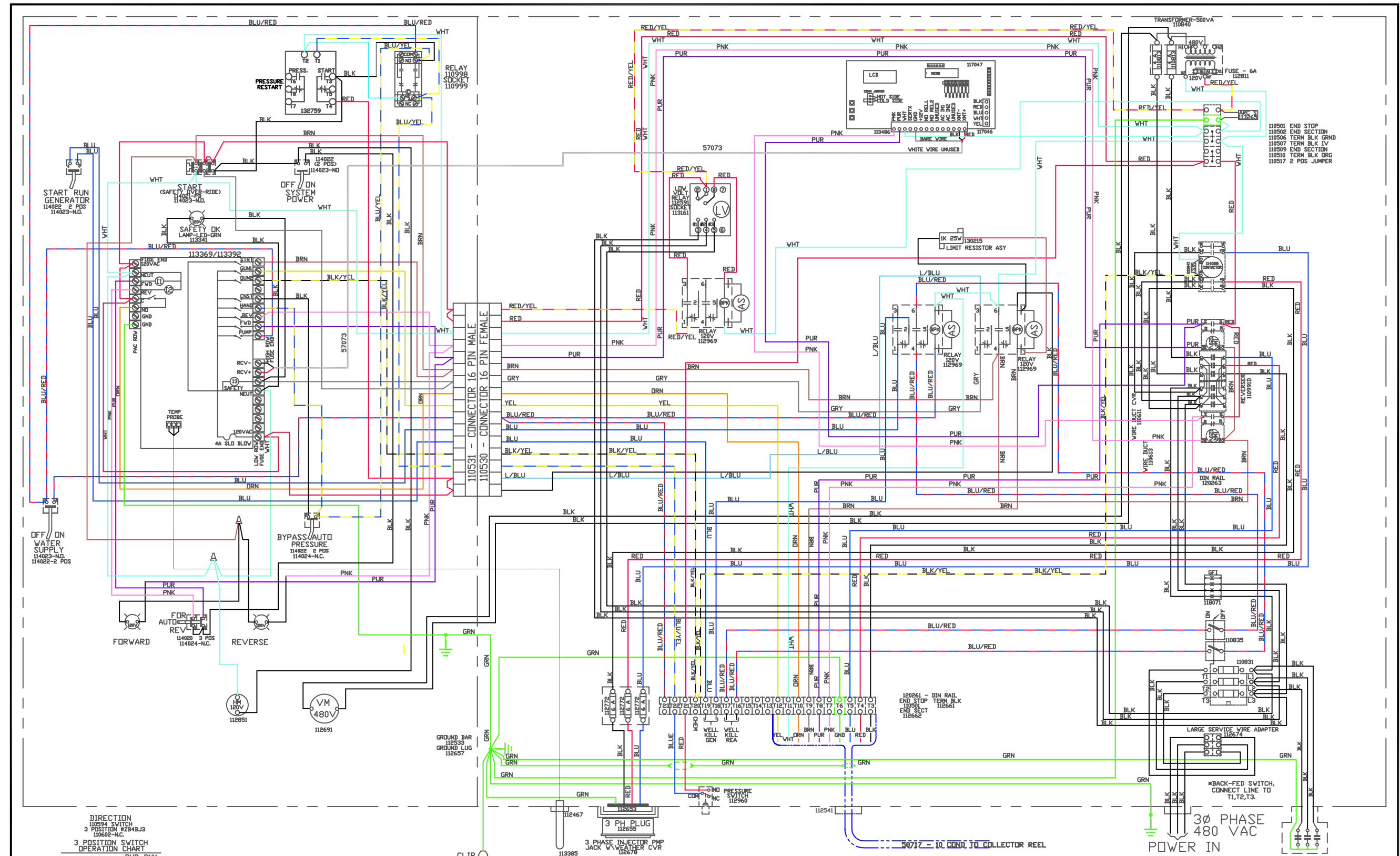
App. By: JD

Date: 2013-2014

Reinke Service School

Reinke Number:ADV MCP WITH NO OPTIONS

REVISION
CG



DIRECTION
110594 SWITCH
3 POSITION #ZB4B3J3
110602-N.C.

3 POSITION SWITCH
OPERATION CHART

	PUR	PNK
FWD POSITION	X	0
REV POSITION	0	X
A/R POSITION	X	X

X = CLOSED CONTACTS
0 = OPEN CONTACTS

FIELD WIRING TIGHTENING TORQUE REQUIREMENTS	
TERMINAL	TORQUE IN-LBS
MAIN DISCONNECT	12
GRAY TERM. BLOCKS	12.4
GROUND LUG	45
LARGE WIRE ADAPT.	45

CG	FIXED POWER WIRE FOR RESOLVER LAYER	06/13	MKS	
CF	CHANGED LAYER NAME, ECH 76423. NO WIRING CHANGES.	11/12	DOC	
CE	ADDED LARGE WIRE TORQUE, CORRECTED PRESS. SW. WIRING.	08/12	DOC	
LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY

REVISIONS

REINKE MANUFACTURING CO., INC.
DESHLER, NEBRASKA 68340

Title: RPM ADVANCED MAIN CONTROL PANEL DIAGRAM

Dr. By: LGG

ck. By: BGW

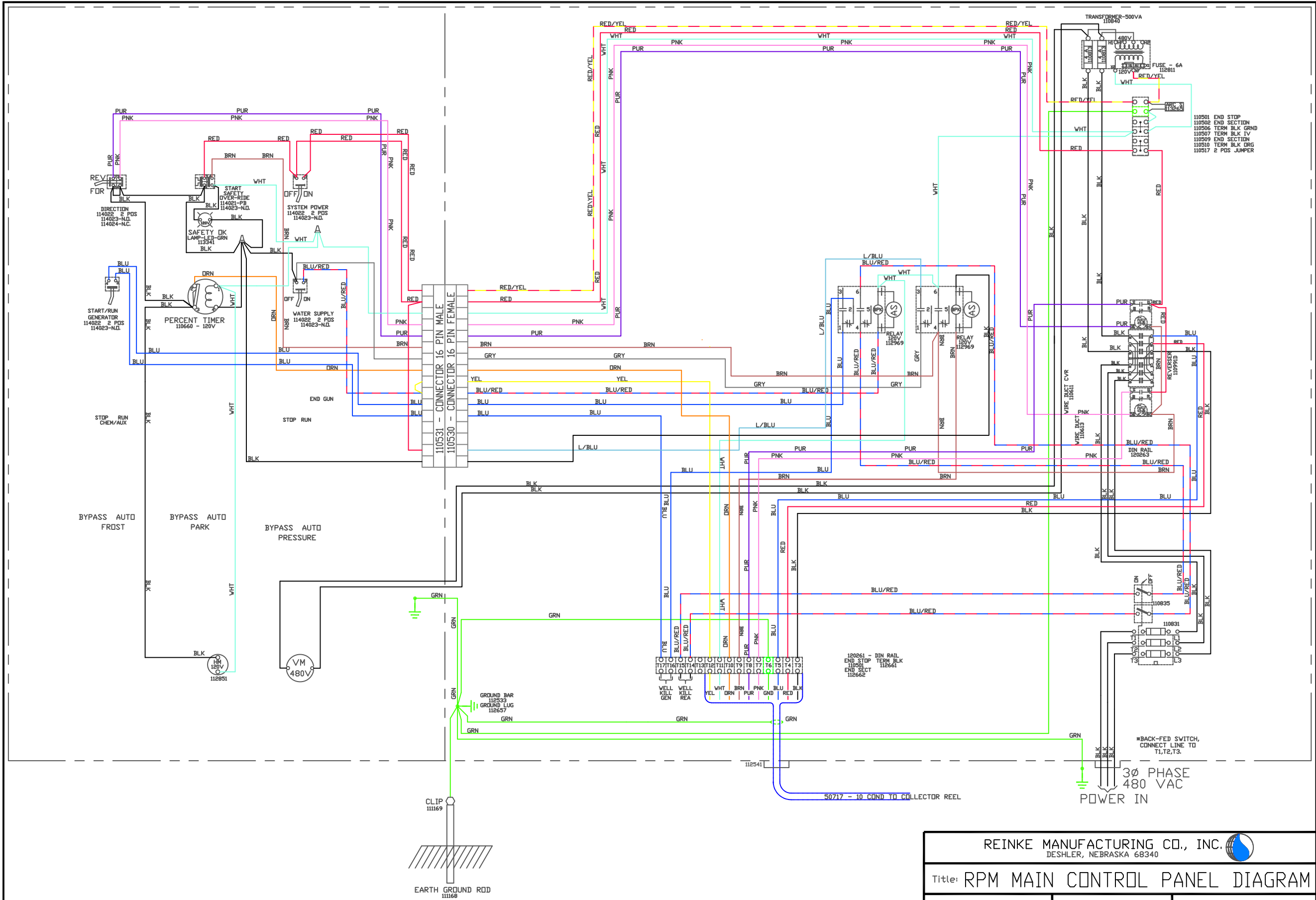
App. By: JD

Date: 2013-2014

Reinke Service School

Reinke Number: ADV MCP WITH MOST OPTIONS

REVISION
CG



FIELD WIRING TIGHTENING TORQUE REQUIREMENTS	
TERMINAL	TORQUE IN-LBS
MAIN DISCONNECT	12
GRAY TERM. BLOCKS	12.4
GROUND LUG	45

Y	DESCRIPTION	DATE	BY	CHK BY
X	PRESSURE RESTART MOVED TO TOP RAIL	01/12	DDC	BGV
X	PERCENT TIMER WAS: 112784, IS: 110660	11/11	DDC	BGV
W	UPDATED DISCONNECT SWITCH DRAWING	10/11	DJM	BGV
LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY

REVISIONS

REINKE MANUFACTURING CO., INC.
DESHLER, NEBRASKA 68340

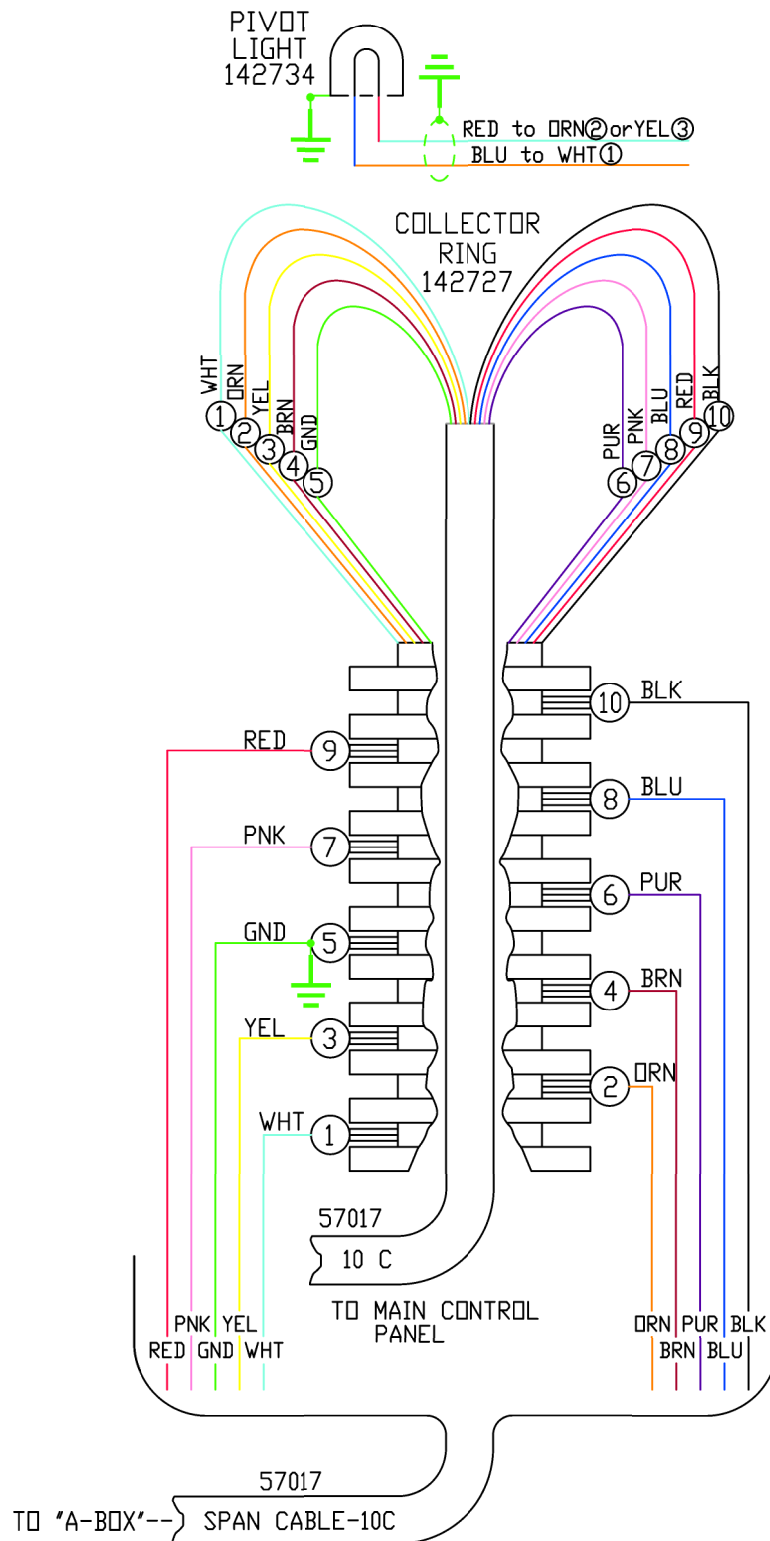
Title: RPM MAIN CONTROL PANEL DIAGRAM

Dr. By: LGG Ck. By: BGV App. By: JD

Date: 2012-2013 REINKE SERVICE SCHOOL

Reinke Number: Standard MCP with Gen Switch REVISION Y

Relinke Electrical Document Number: 84610031 LABEL #112696



Copyright, 1994 , Reinke Manufacturing Co., Inc

REINKE MANUFACTURING CO., INC.
Deshler, Nebraska 68340



Title: COLLECTOR REEL ASSEMBLY

Dr. By: JGL/DGK

Ck. By: *[Signature]*

App. By: *[Signature]*

Date: 1/13/94

Reference Bill of Materials: 142727

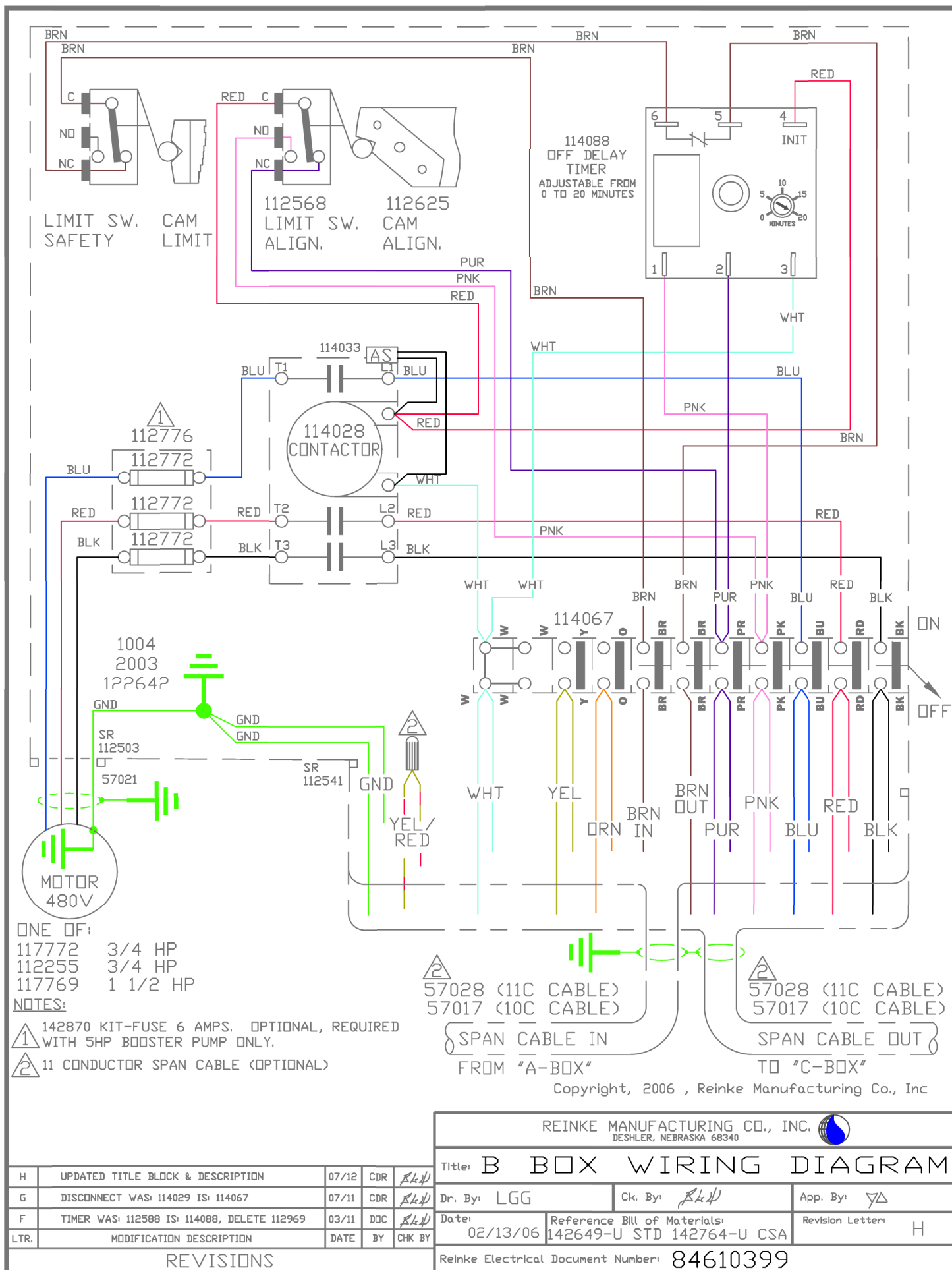
Revision Letter: A

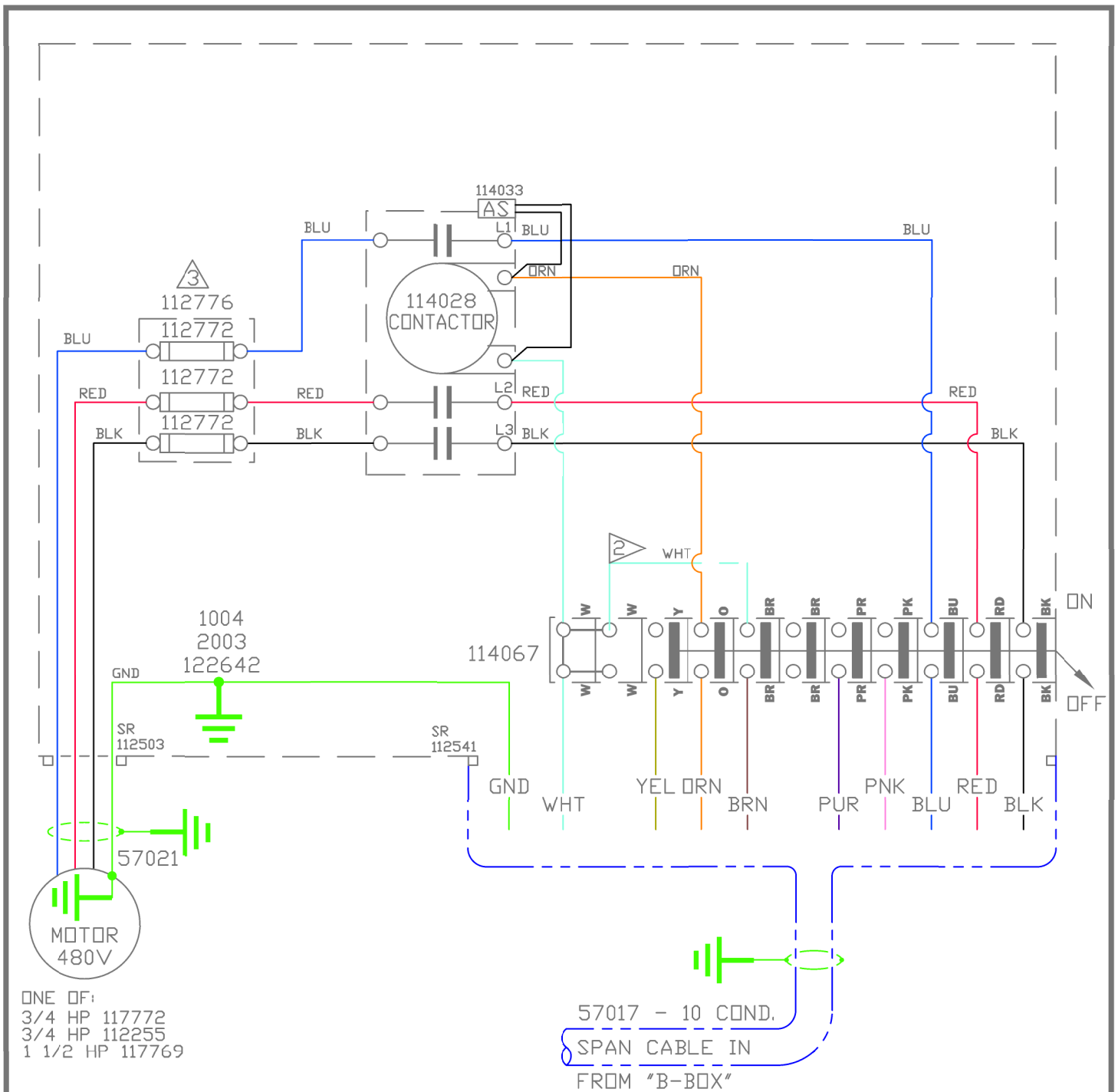
Reinke Electrical Document Number: 84610098

REVISION A

LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY
A	UPDATED TITLE BLOCK	2/12/94	CDR	<i>[Signature]</i>

REVISIONS





NOTES:

1.) SEE 84610097 DIAGRAM FOR C BOX WIRING FOR LINEAR SYSTEMS

2.) THIS JUMPER WIRE MUST BE REMOVED IF USING AN AUTO STOP ON THIS TOWER.

3.) 142870 KIT-FUSE 6 AMPS REQD W/5 HP BSTR PM

REINKE MANUFACTURING CO., INC.
 DESHLER, NEBRASKA 68340



Title: C BOX LAST TOWER

G	UPDATE COMPONENTS TO CURRENT PER ECN# 74823	05/12	CSR	RLH
F	UPDATE DIAGRAM TO BE AS BUILT	10/08	RLH	YD
E	UPDATE DIAGRAM TO CALL FOR CORRECT CTR DRV	10/07	RLH	YD
LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY

Dr. By: DGK	Ck. By: RLH	App. By: YD
Date: 06JUN91	Reference Bill of Materials: 142650-U STD 142766-U CSA	Revision Letter: G

REVISIONS

Reinke Electrical Document Number: 84610029

PIVOT
LIGHT
142734

RED to WHT ①
BLU to ORN ②

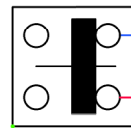
BLU
RED

COLLECTOR
RING
142727

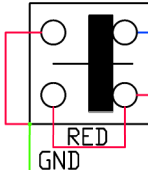
WHT ①
ORN ②
YEL ③
BRN ④
GND ⑤

PUR ⑥
PNK ⑦
BLU ⑧
RED ⑨
BLK ⑩

AUTO
STOP SWITCH
LSA3K-2C-112750



END
GUN SWITCH
C2-JK528-112746



IF NO END GUN YELLOW
WIRE TO GO TO BRUSH 3

*NOTE: THE BRN WIRE FROM
THE SPAN CABLE IS
CONNECTED TO TERM
#4 ON THE COL RING
IF THE AUTO STOP
SWITCH IS NOT USED

57017
10 C

TO MAIN CONTROL
PANEL

57017

TO "A-BOX"-- SPAN CABLE-10C

Copyright, 1980 , Reinke Manufacturing Co., Inc

REINKE MANUFACTURING CO., INC.
Deshler, Nebraska 68340

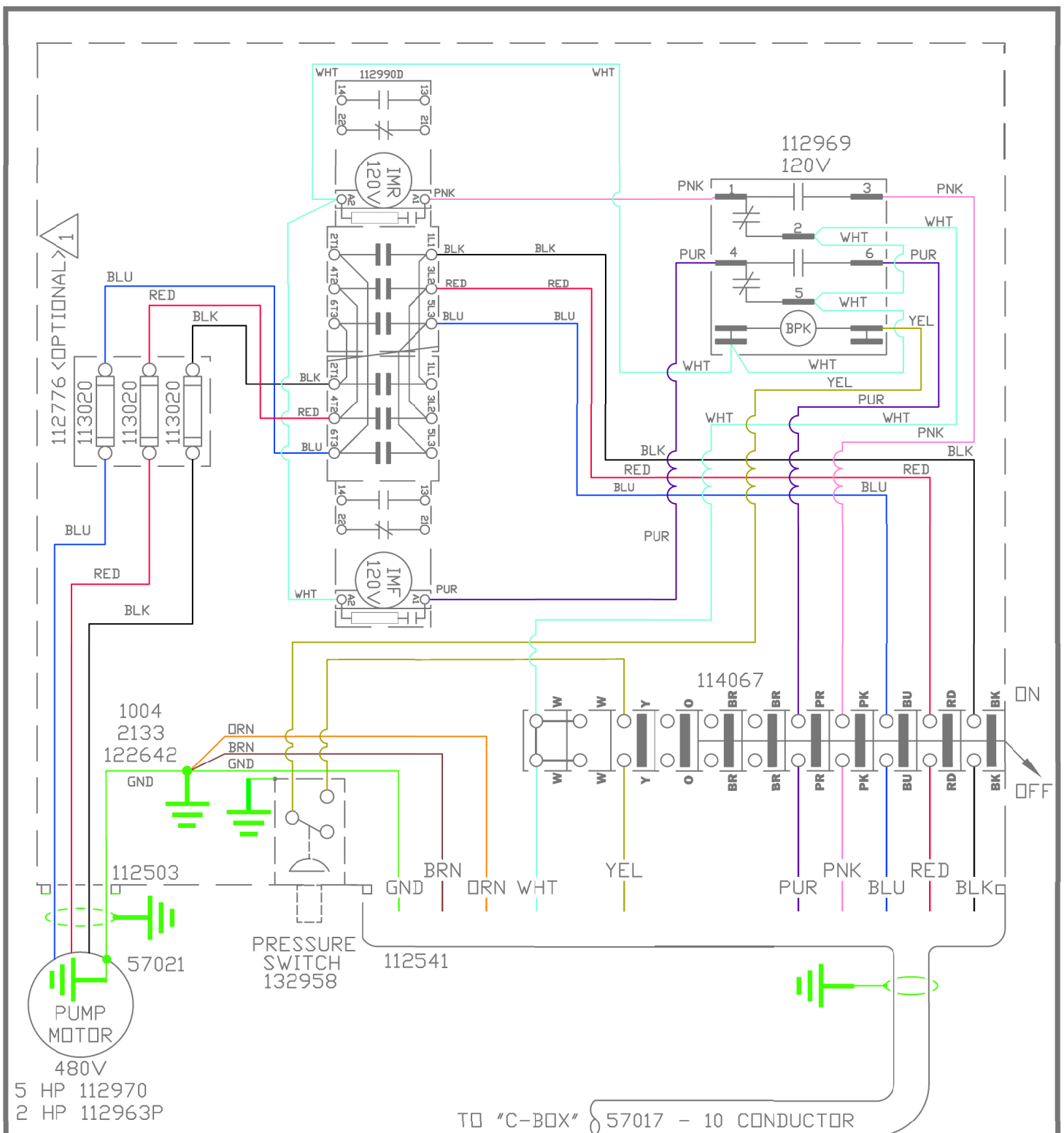
Title: COLLECTOR REEL E/G A/S

Dr. By: AVS Ck. By: *[Signature]* App. By: *[Signature]*

Date: 11/25/80 Reference Bill of Materials: 142727 Revision Letter: A

Reinke Electrical Document Number: 84610389 REVISION A

LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY
A	ADDED REFERENCE BOM	2/12	CDR	<i>[Signature]</i>
REVISIONS				



1. FUSE BLOCK REQUIRED FOR 5 HP PUMP

REF BOM: 145995/145996/145997/145998

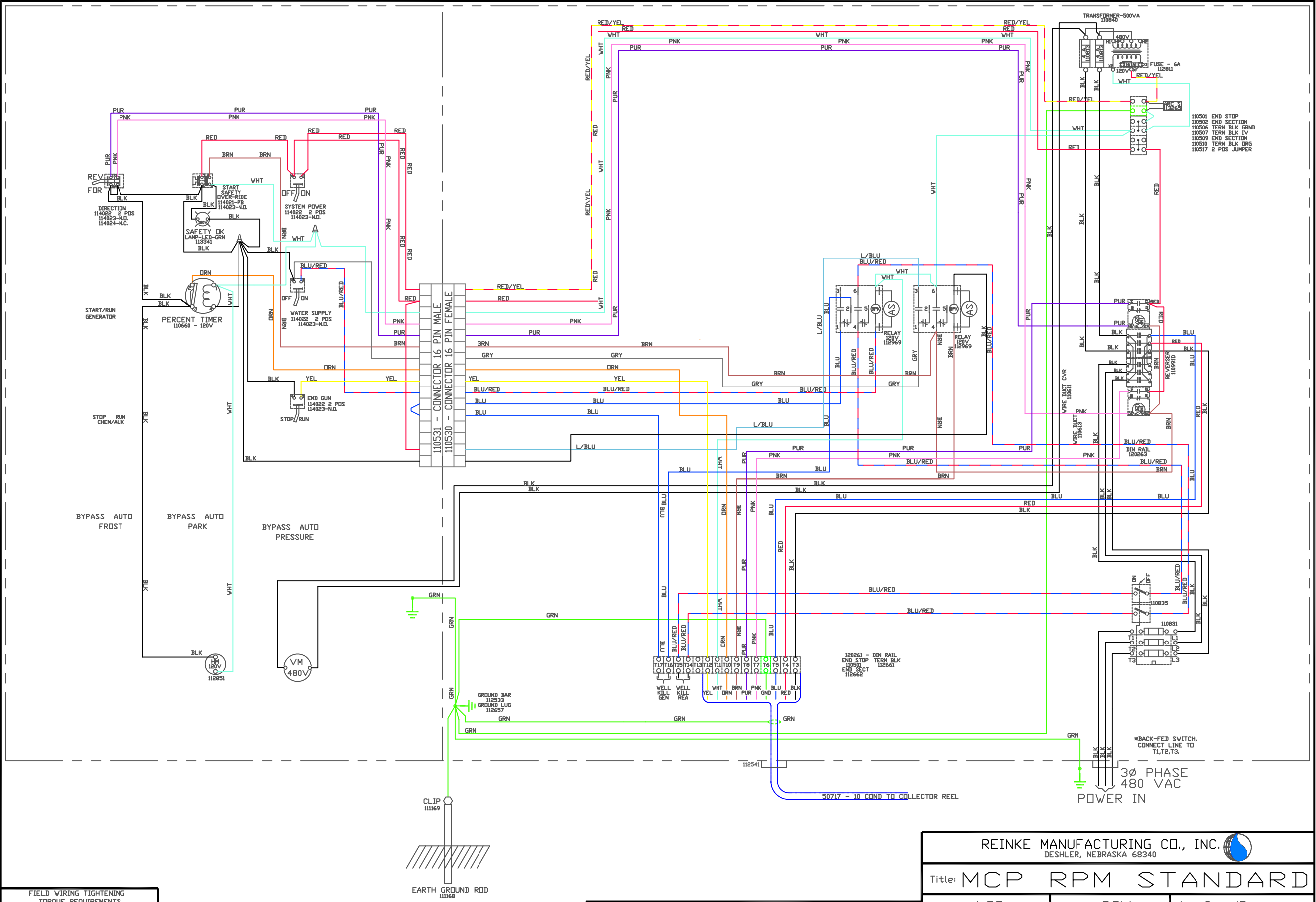
REINKE MANUFACTURING CO., INC.
DESHLER, NEBRASKA 68340

D	PRESSURE SW WAS 112958 & 2HP MTR WAS 112953	07/11	CDR	JKL
C	DISCONNECT P/N WAS: 114029 NOW: 114067	07/11	CDR	JKL
B	MOVED WHITE WIRE IS ON WHITE, WAS ON BROWN TERM.	5/11	DOC	JKL
A	DISCONNECT SV. WAS: 112615 IS: 114029	12/10	DOC	JKL
LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY

REVISIONS

Title: BOOSTER PUMP BOX 2HP & 5HP		
Dr. By: RJD	Ck. By: DC	App. By: YAD
Date: 06/29/09	Reference Bill of Materials: SEE ABOVE	Revision Letter: D

Reinke Electrical Document Number: 84610446

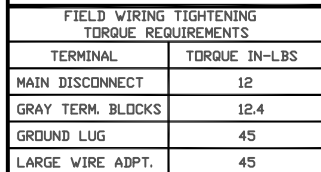


FIELD WIRING TIGHTENING TORQUE REQUIREMENTS	
TERMINAL	TORQUE IN-LBS
MAIN DISCONNECT	12
GRAY TERM. BLOCKS	12.4
GROUND LUG	45
LARGE WIRE ADPT.	45

AA	ADD LARGE SERVICE WIRE TORQUE	08/12	DOC	
Y	PRESSURE RESTART MOVED TO TOP RAIL	01/12	DOC	BGW
X	PERCENT TIMER WAS: 112784, IS: 110660	11/11	DOC	BGW
LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY

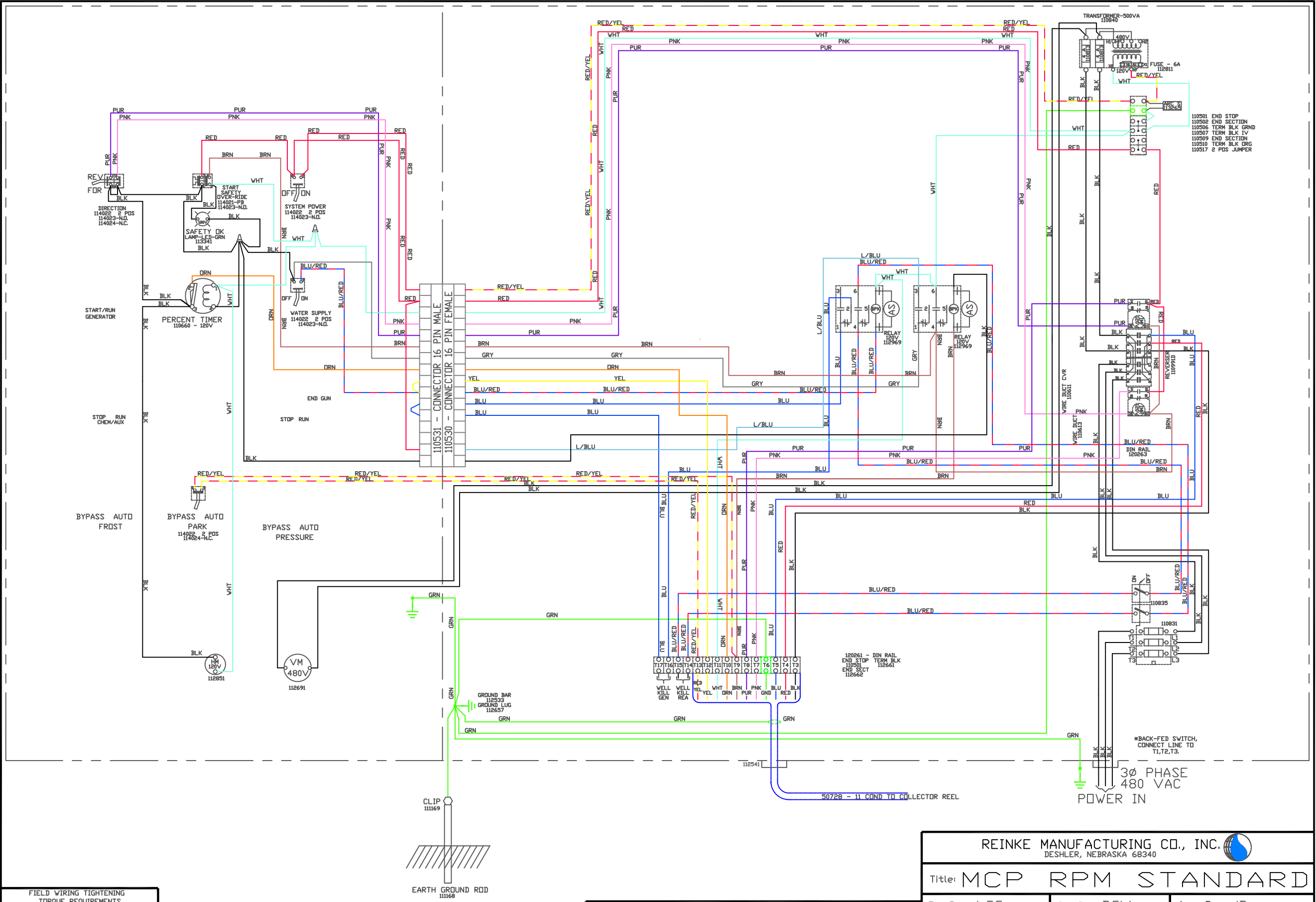
REVISIONS

REINKE MANUFACTURING CO., INC.			
DESHLER, NEBRASKA 68340			
Title: MCP RPM STANDARD			
Dr. By: LGG		Ck. By: BGW	App. By: JD
Date: 2013-2014		REINKE SERVICE SCHOOL	
Reinke Number: RPM STANDARD W/ END GUN SW			REVISION AA



AA	ADD LARGE SERVICE WIRE TORQUE	08/12	DOC	
Y	PRESSURE RESTART MOVED TO TOP RAIL	01/12	DOC	BGW
X	PERCENT TIMER WAS: 112784, IS: 110660	11/11	DOC	BGW
L.T.R.	MODIFICATION DESCRIPTION	DATE	BY	CHK B
REVISIONS				

Reinke Number:	RPM STANDARD W/ LOW PRESS SHUTDOWN	REVISION AA
----------------	------------------------------------	----------------

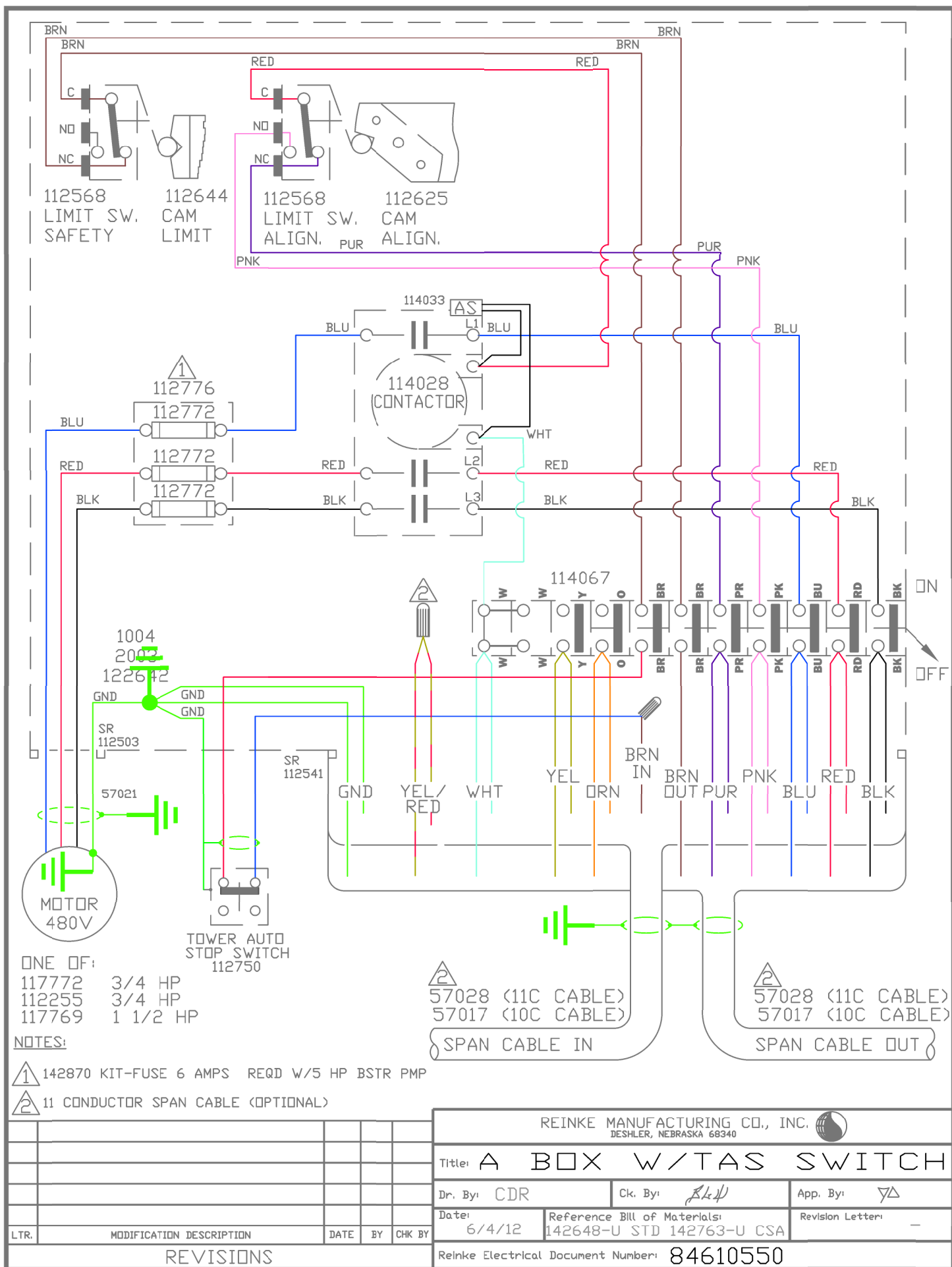


FIELD WIRING TIGHTENING TORQUE REQUIREMENTS	
TERMINAL	TORQUE IN-LBS
MAIN DISCONNECT	12
GRAY TERM. BLOCKS	12.4
GROUND LUG	45
LARGE WIRE ADPT.	45

AA	ADD LARGE SERVICE WIRE TORQUE	08/12	DOC	
Y	PRESSURE RESTART MOVED TO TOP RAIL	01/12	DOC	BGW
X	PERCENT TIMER WAS: 112784, IS: 110660	11/11	DOC	BGW
LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY
REVISIONS				

REINKE MANUFACTURING CO., INC. Deshler, Nebraska 68340		
Title: MCP RPM STANDARD		
Dr. By: LGG	ck. By: BGW	App. By: JD
Date: 2013-2014	REINKE SERVICE SCHOOL	
Reinke Number: RPM STANDARD W/ CAM PLATE STOP OVERRIDE		REVISION AA

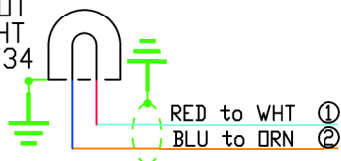




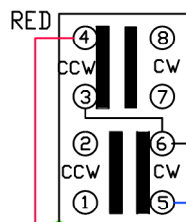


REINKE MANUFACTURING CO., INC. DESHLER, NEBRASKA 68340			
Title: C BOX LAST TOWER W/OPTIONS			
Dr. By: LGG		Ck. By: <i>ELV</i>	
		App. By: <i>YD</i>	
Date: 05/23/07	Reference Bill of Materials: 142650-U; 142766-U		Revision Letter: F
Reinke Electrical Document Number: 84610432			

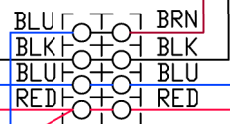
PIVOT
LIGHT
142734



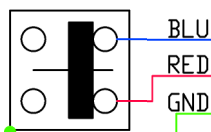
AUTO REV SW.
LSM 7N -113025



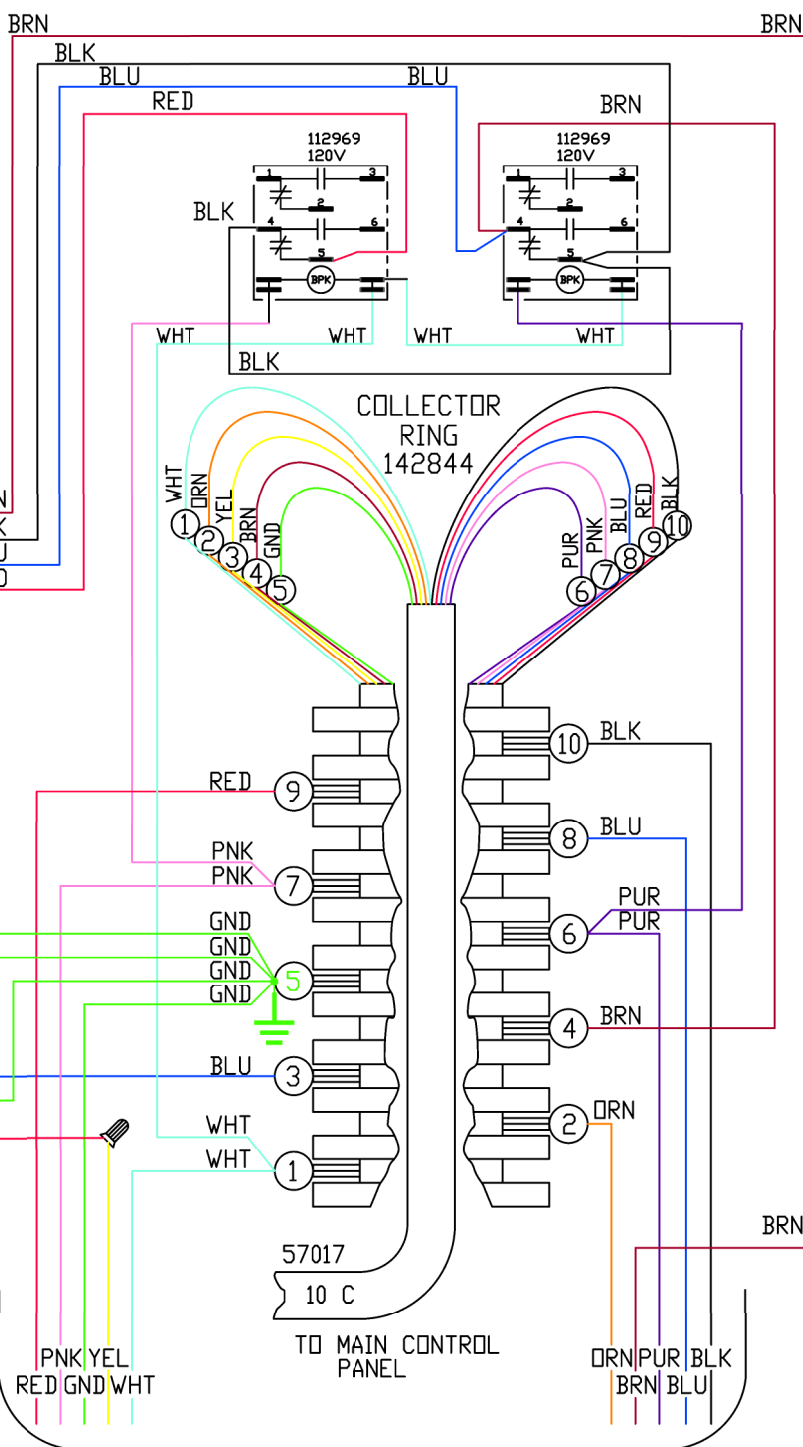
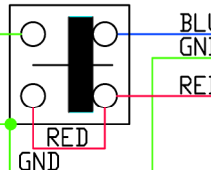
THE BRN WIRE FROM
THE SPAN CABLE IS
CONNECTED TO RED
TERMINAL ON THE
TERMINAL BLOCKS IF
THE AUTO STOP
SWITCH IS NOT USED



AUTO
STOP SWITCH
LSA3K-2C-112750



END
GUN SWITCH
C2-JK528-112746



57028
TO 'A-BOX'-- SPAN CABLE-11C

Copyright, 2007 , Reinke Manufacturing Co., Inc

REINKE MANUFACTURING CO., INC.
DESHLER, NEBRASKA 68340

Title: COLLECTOR REEL A/R PVT CTR RPM

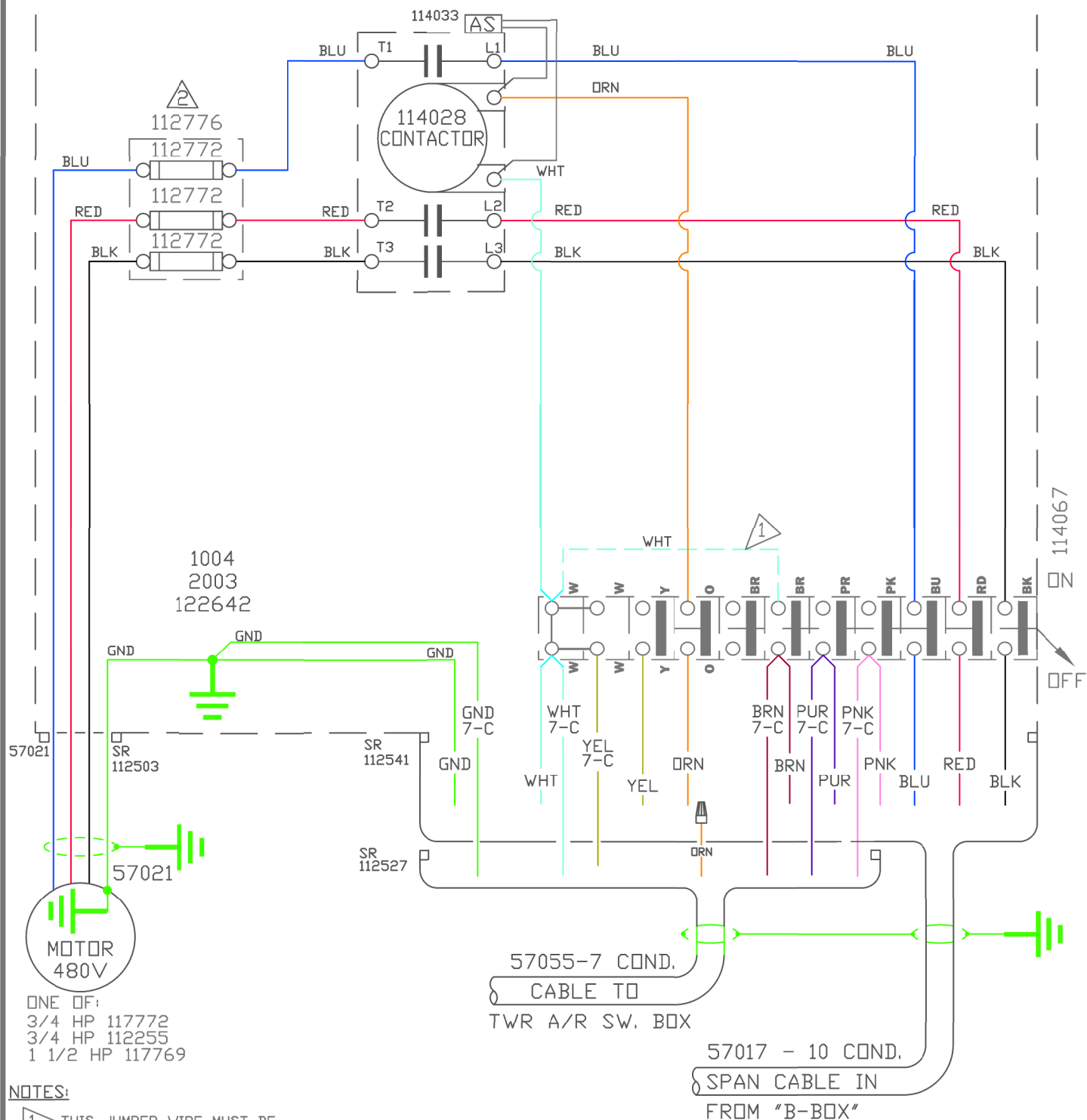
LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY
B	CHANGED NOTE FOR BRN WIRE BACK TO ORIGINAL	01/13	CDR	SLV
A	UPDATE TITLE BLOCK AND DESCRIPTION	02/12	CDR	SLV

Dr. By: SLV	Ck. By: SLV	App. By: YD
Date: 01/10/07	Reference Bill of Materials: 142844	Revision Letter: B

REVISIONS

Reinke Electrical Document Number: 84610434

REVISION
B



NOTES:

1 THIS JUMPER WIRE MUST BE REMOVED IF USING AN AUTO STOP OR TAR ON THIS TOWER.

2 142870 KIT-FUSE 6 AMPS REQD W/5 HP BSTR PMP

F	MODIFICATION DESCRIPTION	DATE	BY	CHK BY
E	UPDATE DIAGRAM TO BE AS BUILT	09/11	DEC	
D	UPDATE DIAGRAM TO BE AS BUILT	10/08		
LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY

REVISIONS

REINKE MANUFACTURING CO., INC.
DESHLER, NEBRASKA 68340

Title: C BOX-RPM-LAST TWR W/TAR

Dr. By:

Ck. By:

App. By:

Date:

Reference Bill of Materials:
142650, 142766, 142650-H, 142766-H
CONNECTOR KIT 142697

Revision Letter:

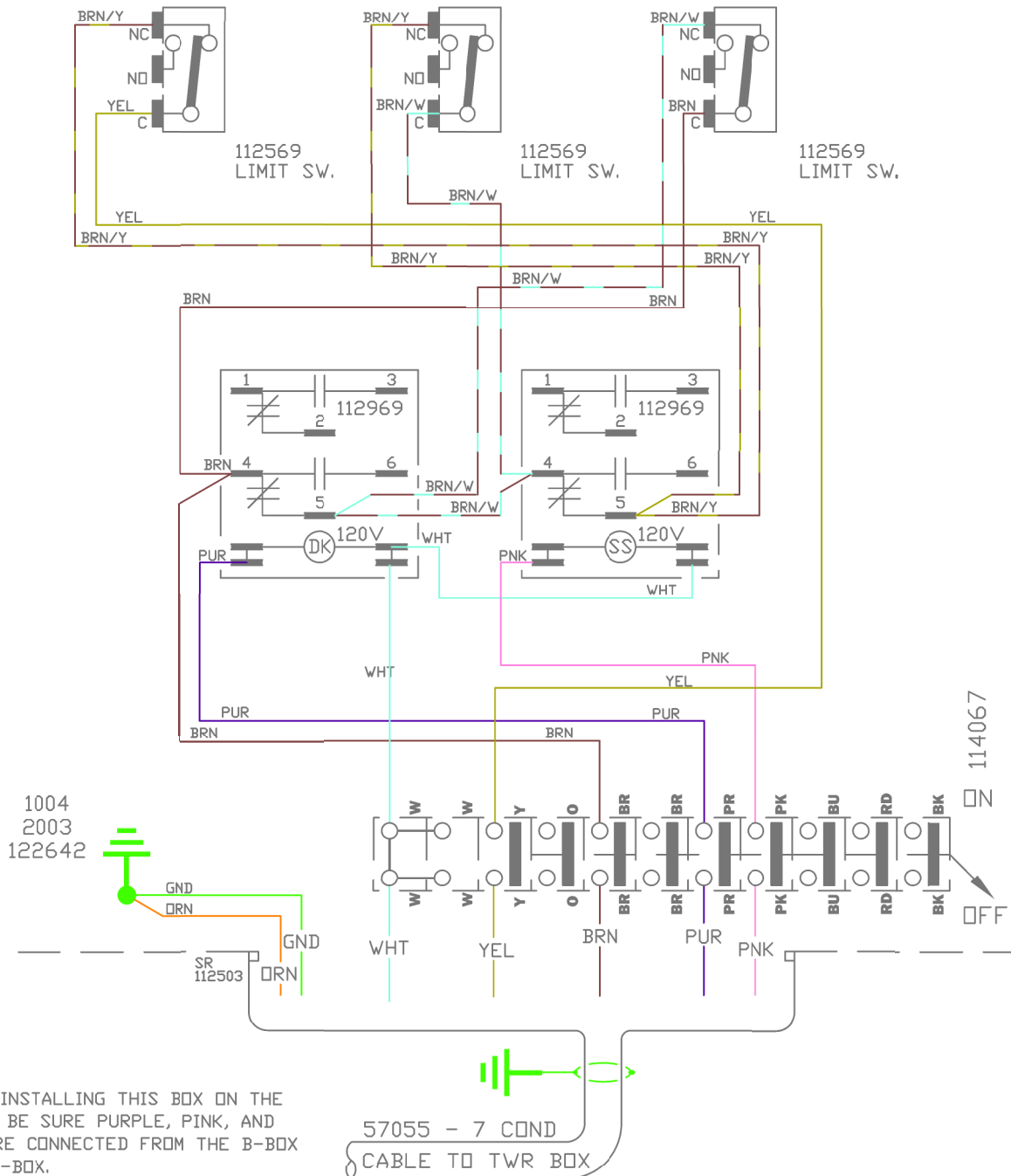
F

Reinke Electrical Document Number: 84610263

BOTTOM SWITCHES
LIMIT
3RD CAM = 8° CLOCKWISE
4TH CAM = 8° C-CLOCKWISE

MIDDLE SW.
TRIPS IN REV
2ND CAM = 6° C-CLOCKWISE

TOP SW.
TRIPS IN FWD
TOP CAM = 6° CLOCKWISE



REINKE MANUFACTURING CO., INC.
DESHLER, NEBRASKA 68340



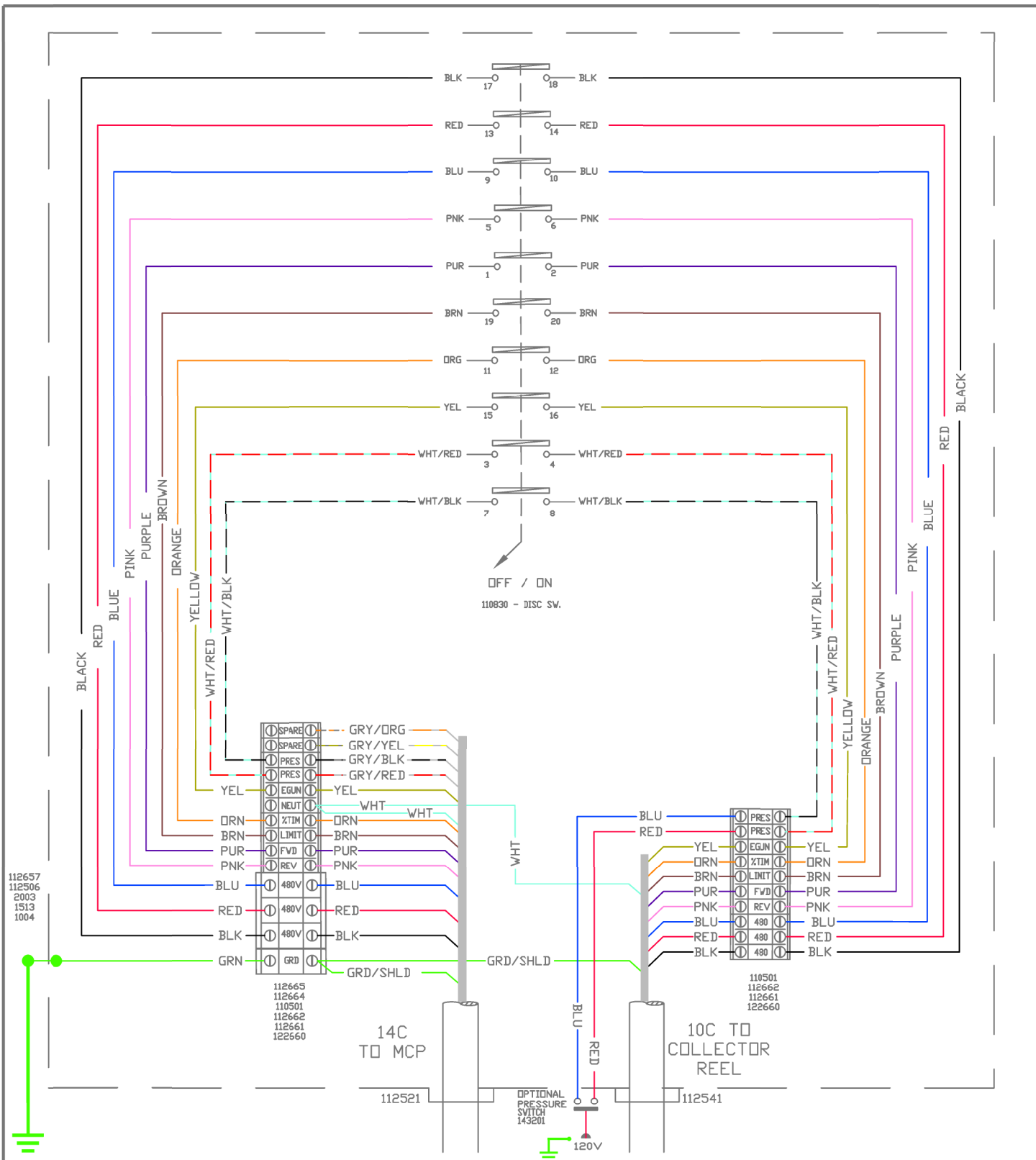
Title: SWITCH BOX RPM TAR UL

D	UPDATE DESCRIPTION TO STANDARDIZE PER ECN 74828	08/11	DOC	RLH
C	DISCONNECT SW. WAS: 114029, IS: 114067	08/11	DOC	RLH
B	REDRAWN, ADDED BR/W & BR/Y, WAS: 112615 IS: 114029	4/11	DOC	RLH
LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY

Dr. By: LRF	Ck. By:	App. By: LRF
Date: 01/23/02	Reference Bill of Materials: 142703	Revision Letter: D

Reinke Electrical Document Number: 84610279

REVISIONS



NOTE:

1. INSTALL ALL TERMINAL BLOCK WITH RELEASE LATCH TO CENTER OF THE BOX.

Copyright, 2011 , Reinke Manufacturing Co., Inc

REINKE MANUFACTURING CO., INC.
DESHLER, NEBRASKA 68340



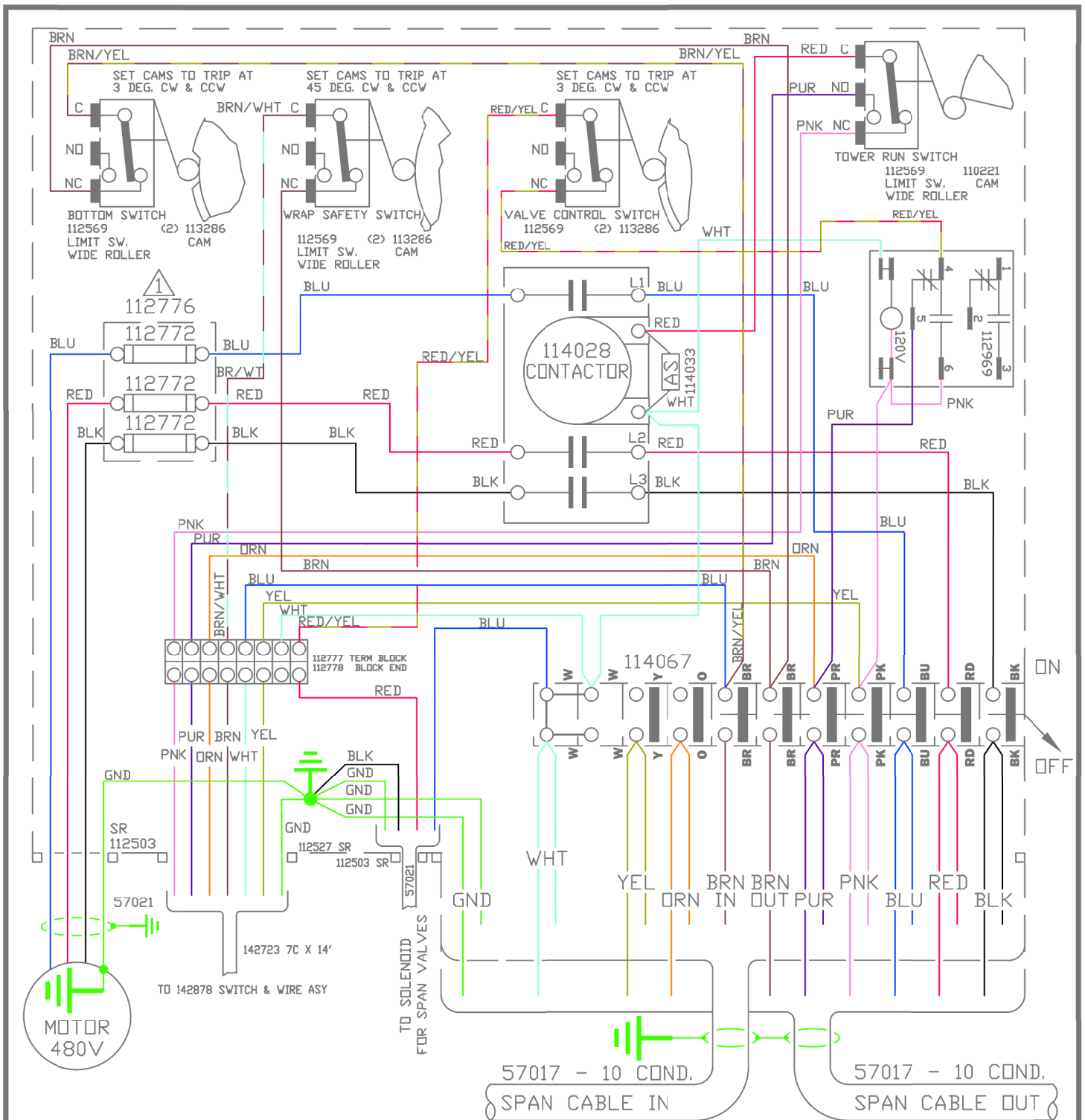
Title: PIVOT JUNCTION BOX DIAGRAM

B	ADDED NOTE ABOUT RELEASE LATCHES TO CENTER OF BOX	05/14	CDR	ELK
A	WIRE COLOR WAS: WHT/%, IS: GRY/%	12/10	DOC	ELK
LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY

Dr. By: DOC	Ck. By: ELK	App. By: YD
Date: 7/23/11	Reference Bill of Materials: P/N 141122	Revision Letter: B

REVISIONS

Reinke Electrical Document Number: 84610666



1 142870 KIT-FUSE 6 AMPS REQD W/5 HP BSTR PMP

* THIS PRINT USED WHEN USING SWITCH & WIRE ASY-PIVOT TWR AND WHEN WRAPPING 10 DEGREES OR LESS *

REINKE MANUFACTURING CO., INC.
DESHLER, NEBRASKA 68340



Title: A BOX 15 DEGREE PVT TWR CSA

K	REMOVED 114051 ADDED 112777/112778 - CHG BRN WIRES	04/14	CSR	RLH
J	REMOVED 112777 & 112778 ADDED (3) 114051	08/13	CSR	RLH
H	UPDATE TO ADD 6 AMP FUSE KIT 142870	05/12	CSR	RLH
LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY

Dr. By: LRF

Ck. By:

App. By:

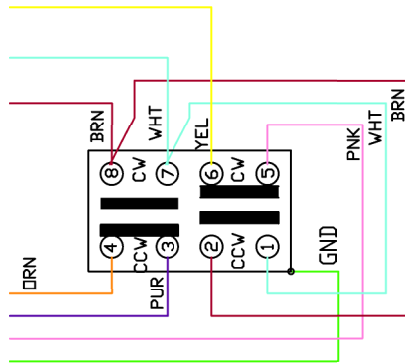
Date: 1/7/04

Reference Bill of Materials: 142877-U

Revision Letter: K

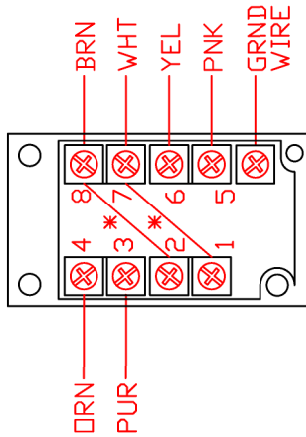
REVISIONS

Reinke Electrical Document Number: 84610339



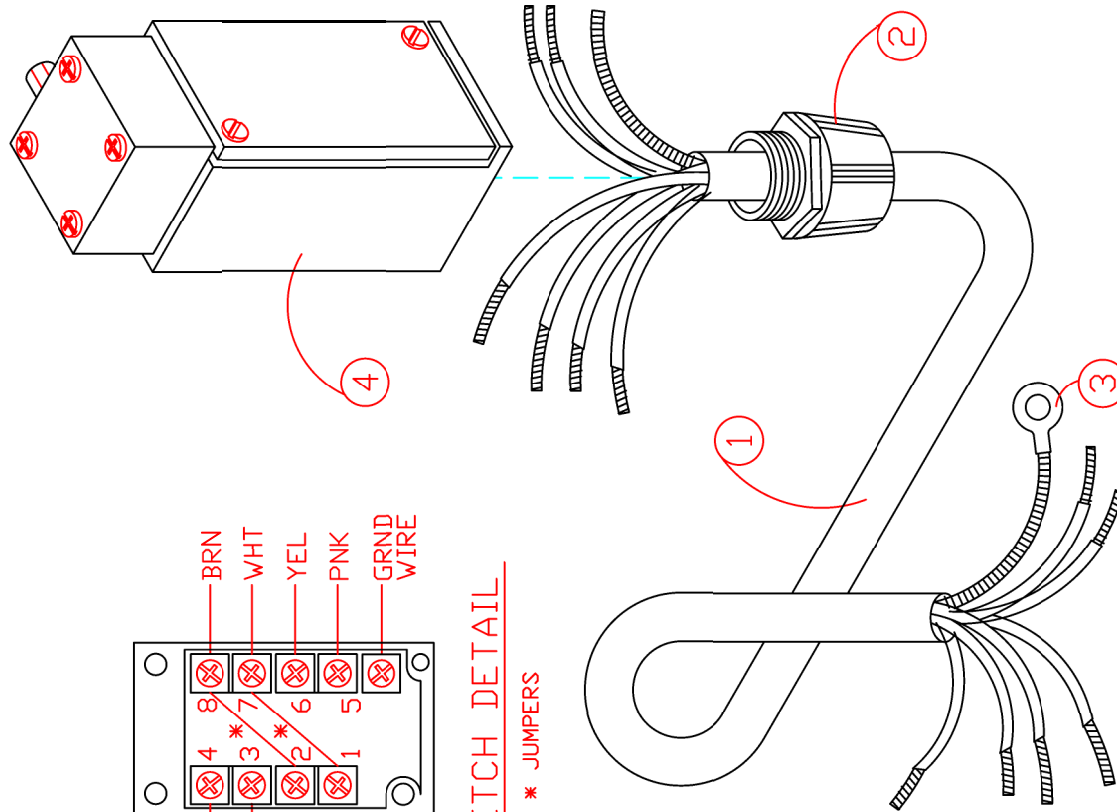
SWITCH CONTACT DETAIL

ASY:
STAND THE SWITCH ON END WITH THE 1/2" FITTING HOLE DOWN AND THE KNOB ON THE HEAD FACING YOU. LOOSEN THE 4 SCREWS FASTENING DOWN THE HEAD AND TURN THE HEAD WITH THE KNOB POINTING TO YOUR RIGHT. FASTEN THE HEAD BACK DOWN WITH THE 4 SCREWS. REMOVE THE FRONT COVER OF THE SWITCH AND PUT SWITCH IN VISE. SCREW THE 1/2" FITTING IN THE BOTTOM OF THE SWITCH, AND PUT 7C/CABLE THROUGH FITTING AND TIGHTEN FITTING. STRIP WIRE BACK 2 1/2" ON SWITCH END AND 6" ON OTHER END.



SWITCH DETAIL

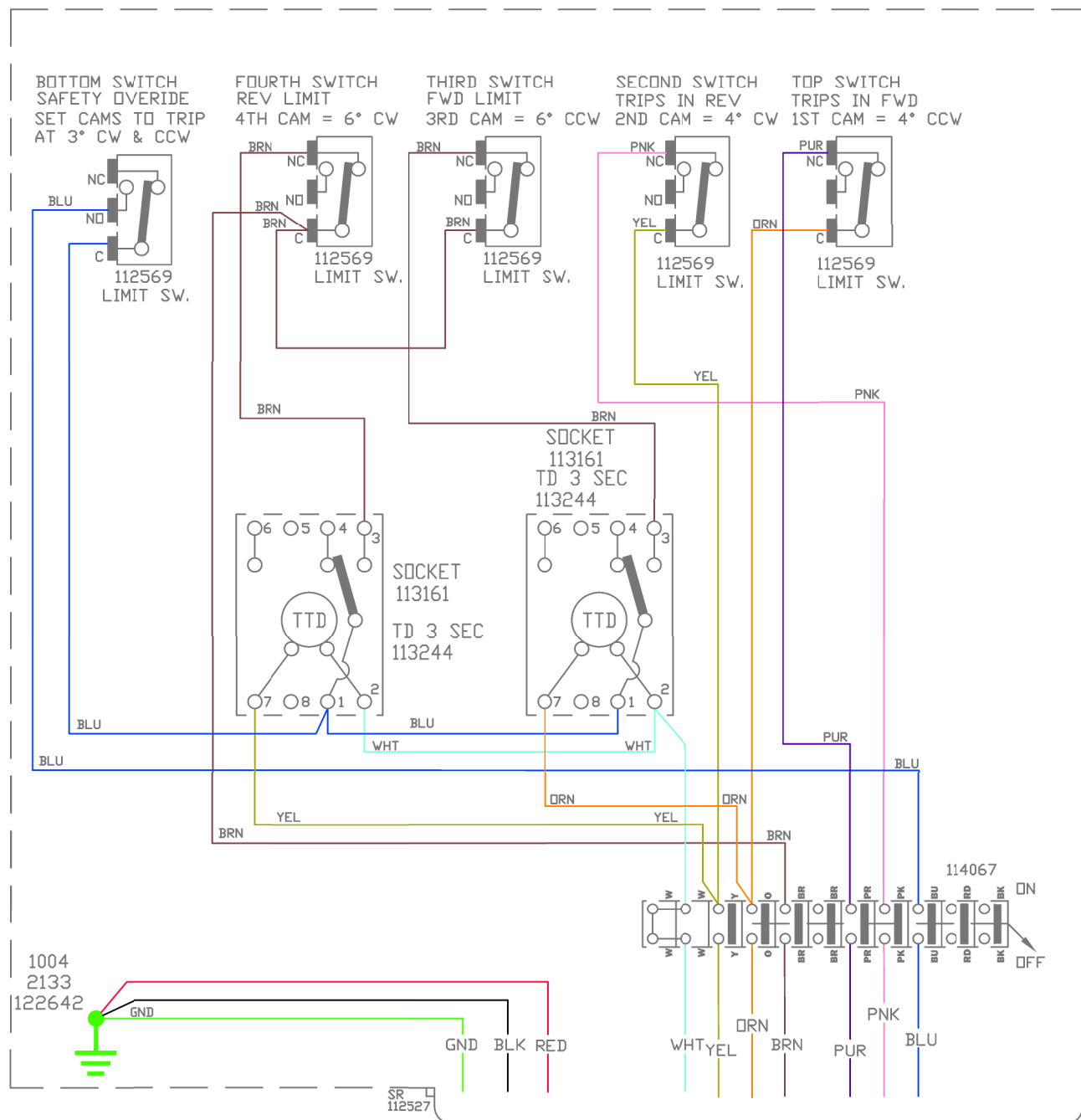
* JUMPERS



Copyright, 2004 , Reinke Manufacturing Co., Inc

TOLERANCES U.D.S.		REINKE MANUFACTURING CO., INC. DESHLER, NEBRASKA 68340			
LINEAR ±1/32 - X ≤ 60" ±1/16 - X > 60" ±.005 - .XXX ANG = ±1/2"		TITLE: SWITCH & WIRE ASY-PIVOT TOWER			
DIAMETERS +005 DIA's = X ≤ 3" -010 DIA's = X ≤ 8" -015 DIA's = 3/8" < X < 1"		Dr. By:	Tr. By:	Chk. By:	App. By:
		Scale: 1/2		Date: 01/07/2004	SHEET of
		Reinke Part No. 142878			
		REVISION A			

LIST OF MATERIALS		REVISIONS						
REF	R. PART NO.	REQ'D	DESCRIPTION	LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY
6	3005	2.00	SCREW #10-32 X 5/16 PAN HD ZC					
5	142689	1.00	ARM-SW W/LOOP TWR A/STOP					
4	113025	1.00	MICROSWITCH-LSYMB7N					
3	112518	1.00	TERMINAL-RING #33457 SOLIS					
2	112527	1.00	STRAIN RELIEF 3/4"-W/DHD#5540					
1	57055	14.00	CABLE-SHIELDED 14/7	A	REPLACE 142691 WITH 142689	01/06	LRF	RLH



NOTE: FIELD ADJUSTMENT OF CAMS IS AS FOLLOWS
 SAFETY OVERRIDE AT BOTTOM OF RAMP
 FWD & REV TRIPS 1/4 TURN OFF OF RAMP
 FWD & REV LIMIT 3/4 TURN OFF OF RAMP

57076 - 10 COND
 CABLE TO TWR BOX

Copyright, 2008 , Reinke Manufacturing Co., Inc

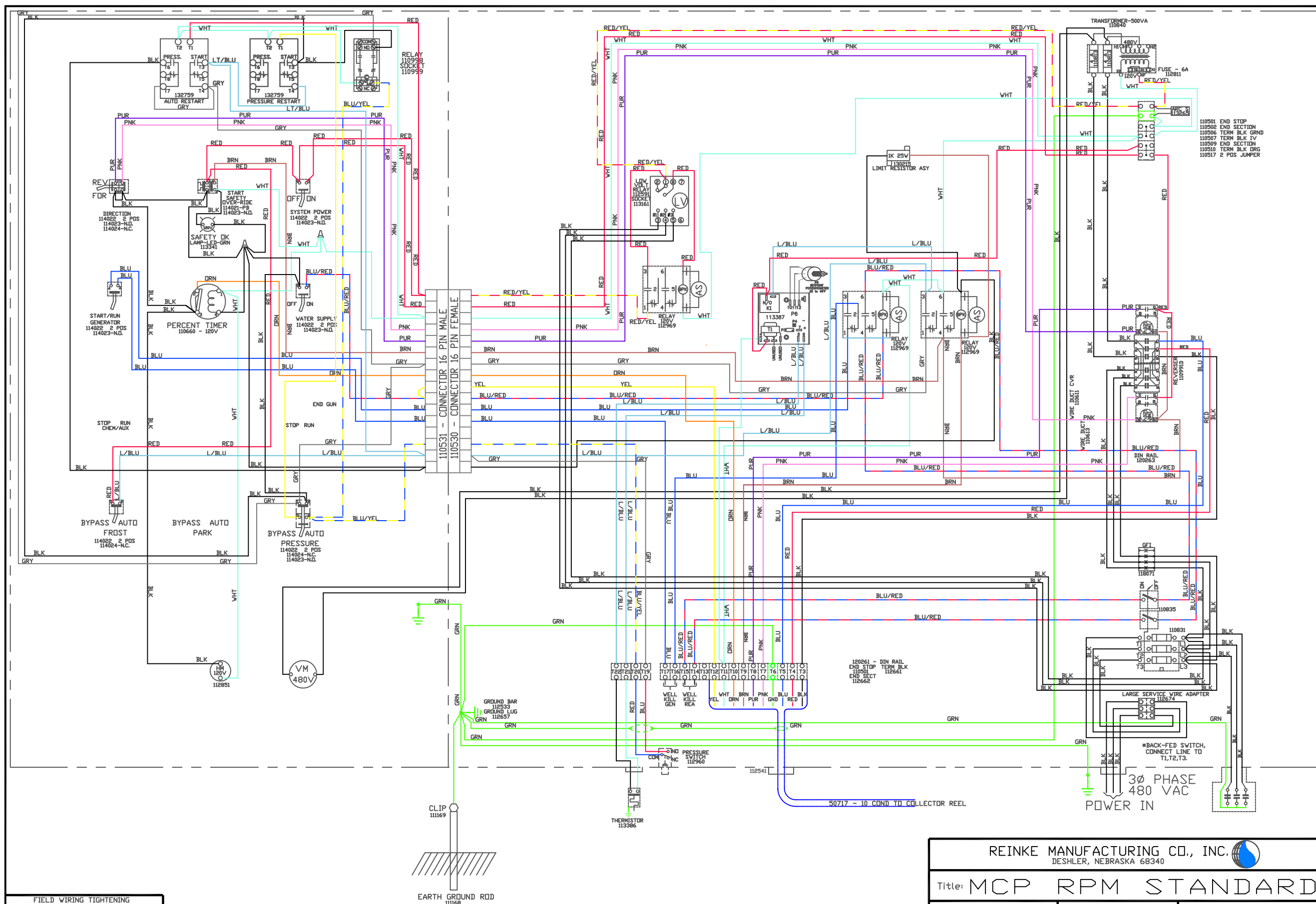
REINKE MANUFACTURING CO., INC.
 DESHLER, NEBRASKA 68340



Title: SWITCH BOX - 90 PVT TWR

B	DISCONNECT SW. WAS: 114029 IS: 114067	07/11	CDR	<i>[Signature]</i>	Dr. By: MKS	Chk. By: <i>[Signature]</i>	App. By:
A	DISCONNECT SW. WAS: 112615 IS: 114029	12/10	DOC	<i>[Signature]</i>	Date: 02/08/10	Reference Bill of Materials: 142869	Revision Letter: B
LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY	Reinke Electrical Document Number: 84610450		

REVISIONS

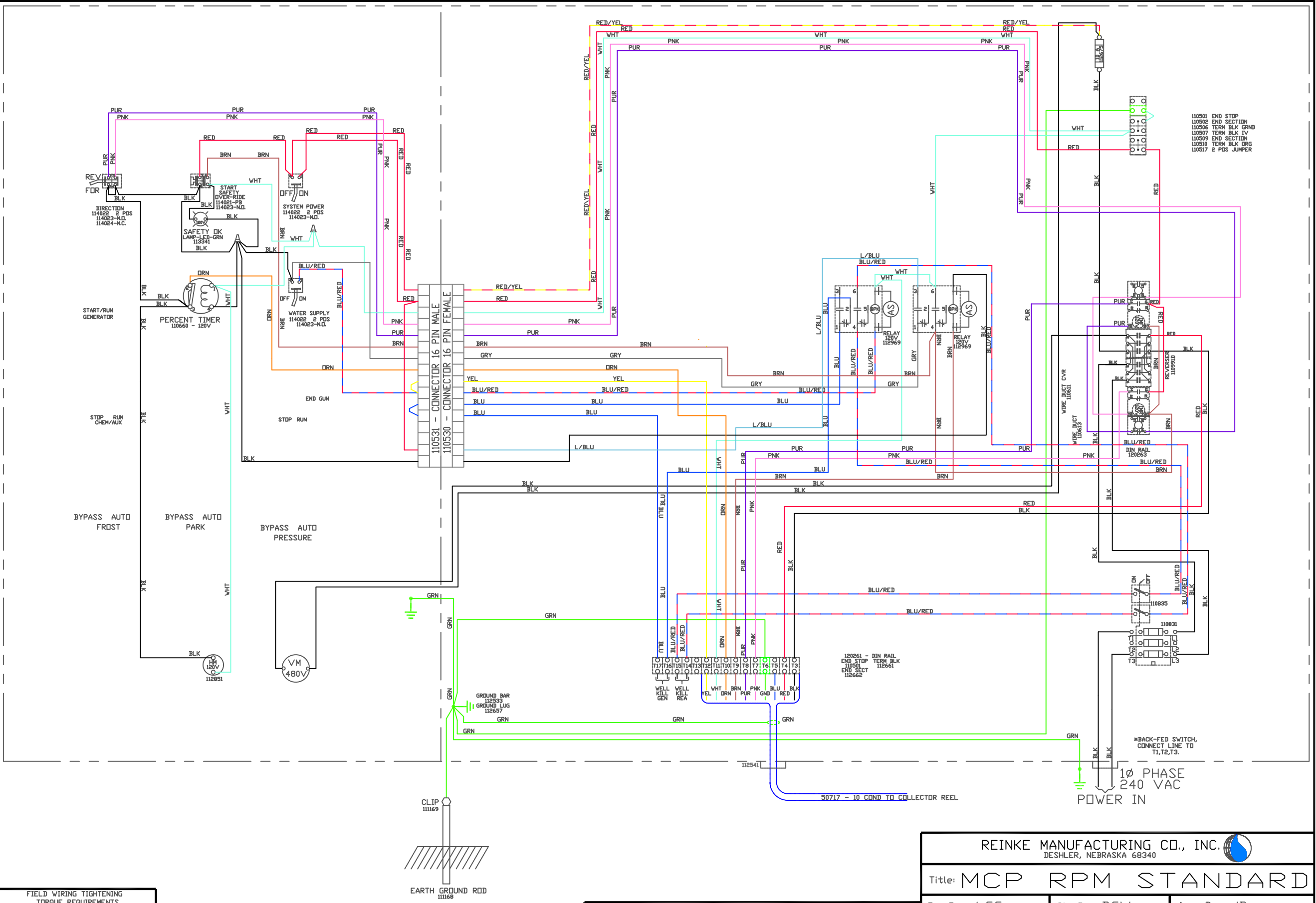


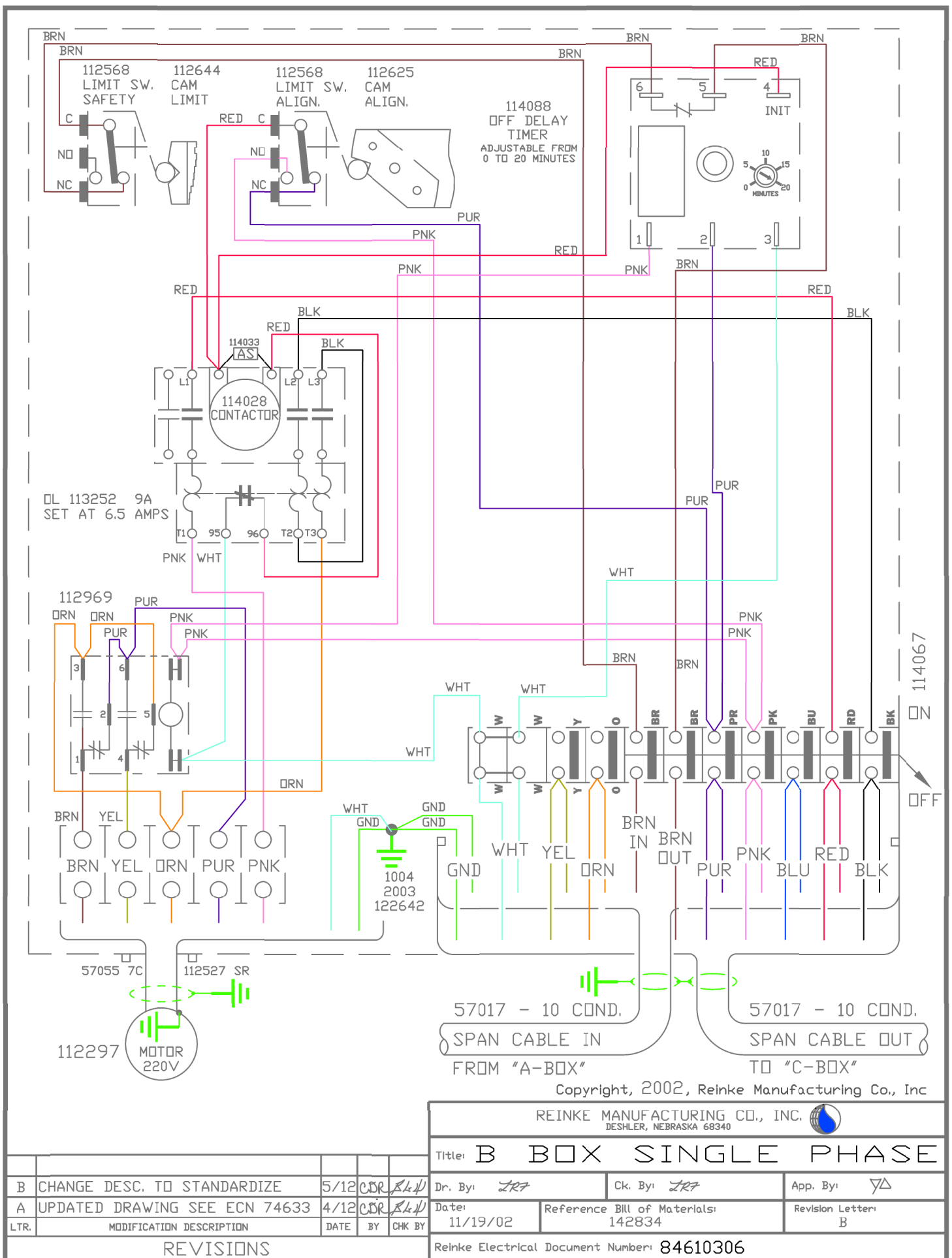
FIELD WIRING TIGHTENING TORQUE REQUIREMENTS	
TERMINAL	TORQUE IN-LBS
MAIN DISCONNECT	12
GRAY TERM. BLOCKS	12.4
GROUND LUG	45
LARGE WIRE ADPT.	45

AA	ADD LARGE SERVICE WIRE TORQUE	08/12	DOC	
Y	PRESSURE RESTART MOVED TO TOP RAIL	01/12	DOC	BGW
X	PERCENT TIMER WAS 112784, IS 110660	11/11	DOC	BGW
LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY

REVISIONS

REINKE MANUFACTURING CO., INC.			
DESHLER, NEBRASKA 68340			
Title: MCP RPM STANDARD			
Dr. By: LGG		Ck. By: BGW	App. By: JD
Date: 2013-2014		REINKE SERVICE SCHOOL	
Reinke Number: RPM STANDARD W/ OPTIONS			REVISION AA





Reinke Electrical Document Number: 84610307

Service Alert - SA13-003



DATE: March 1, 2013

TO: All Reinke Dealers

SUBJECT: Touchscreen Firmware - Problem & Update

The Touchscreen firmware version 12.292 will have problems when running both the VRI and remote communications options simultaneously. This condition may cause the Touchscreen to lockup and display error codes.

The following Touchscreen panels may contain version 12.292 firmware:

- a. Any Touchscreen updated with firmware downloaded from RDSS between November 1, 2012 and January 28, 2013.

- b. Panels or Systems produced by Reinke from January 18, 2013 through January 24, 2013.

All panels currently running 12.292 version firmware should be updated to version 13.23.

Symptoms:

The problem occurs when both VRI and remote communications operate simultaneously. The panel will display either a “NullReferenceException()” or “InvalidCastException” error message and program execution will halt, resulting in Touchscreen lockup.

Resolution:

Update Touchscreen firmware version 12.292 to 13.23 and delete all copies of 12.292 that have been downloaded to local computers or flash drives.

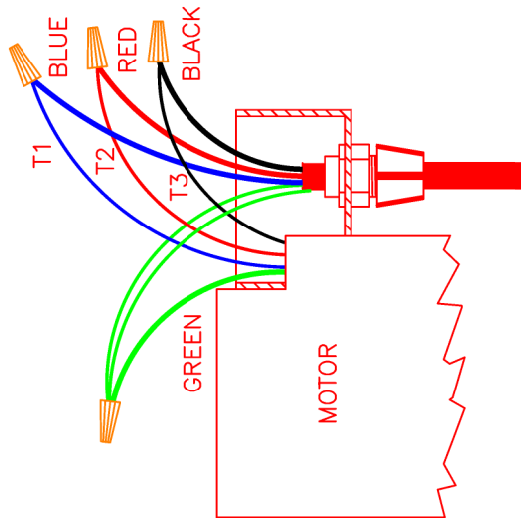
The Touchscreen firmware update can be downloaded from RDSS. You will find the download link in the RDSS Service pull-down menu. Refer to the download instructions on the following pages.

VRI Enabled:

To ensure proper VRI operation both Touchscreen firmware version 13.23 and I/O Board firmware version 008 should be installed. Refer to page 4 for update instructions.

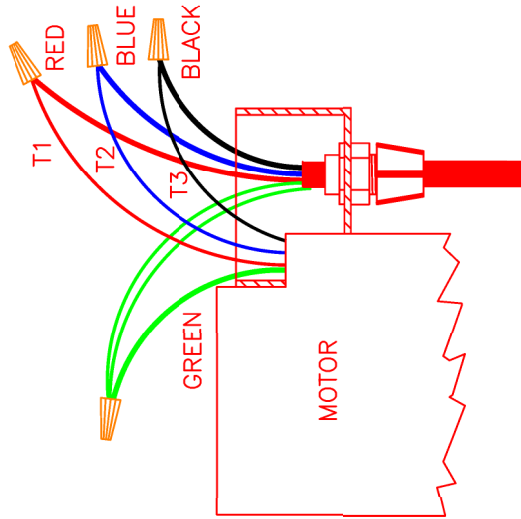
480 VOLT 3 PHASE

US MOTORS
CENTER DRIVES: 112267, 112268,
112269, 112265, 112266
STATOR KITS: 142167-S, 142168-S,
142169-S, 142165-S, 142166-S



UMC
CENTER DRIVES: 117772, 112255,
117769, 117771, 117770
STATOR KITS: 117773, 117774

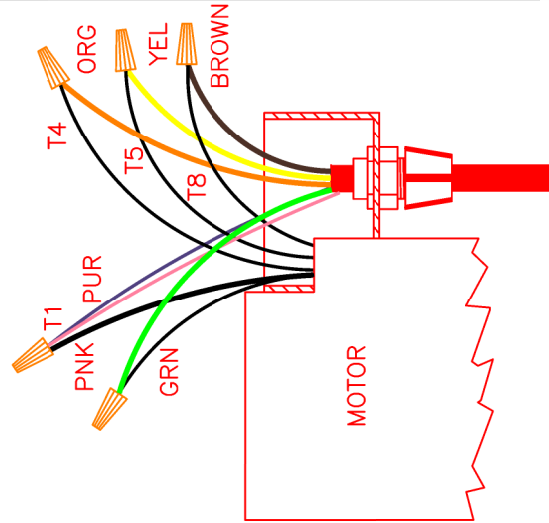
DURST
CENTER DRIVE: 112213



240 VOLT SINGLE PHASE

CENTER DRIVES: 112297
REVERSE MOTOR DIRECTION
USING T5, T8

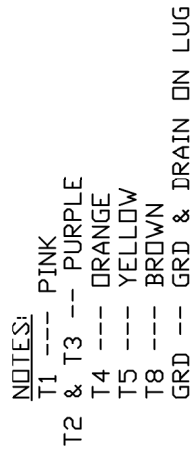
FOR BELT DRIVE UNITS CONNECT:
T5 TO BROWN
T8 TO YELLOW



PART #85152107 (REV B)

Copyright, 2005 , Reinke Manufacturing Co., Inc

[illegible]

[illegible]

To Orange on Term. Strip

To Yellow on Term Strip

To Constant 120VAC

To System Neutral

RELAY 110998
SOCKET 110999

RELAY 110998
SOCKET 110999

Percent Timer Orange from CPU

End Gun Yellow from CPU

Copyright, 2012 , Reinke Manufacturing Co., Inc

REINKE MANUFACTURING CO., INC. 
DESHLER, NEBRASKA 68340

Title: **Isolation Relays for RAMS/Touch**

Dr. By: **MKS**

Ck. By:

App. By:

Date: **10-1-2012**

Reference Bill of Materials:

Revision Letter: **—**

LTR.	MODIFICATION DESCRIPTION	DATE	BY	CHK BY

REVISIONS

Reinke Electrical Document Number: **N/A**

Reinke Part #117722

Rev C 10/21



To exceed our customers' expectations of quality, service, and innovation